Roma Tre

Information Visualization

Perception of color

M. Patrignani

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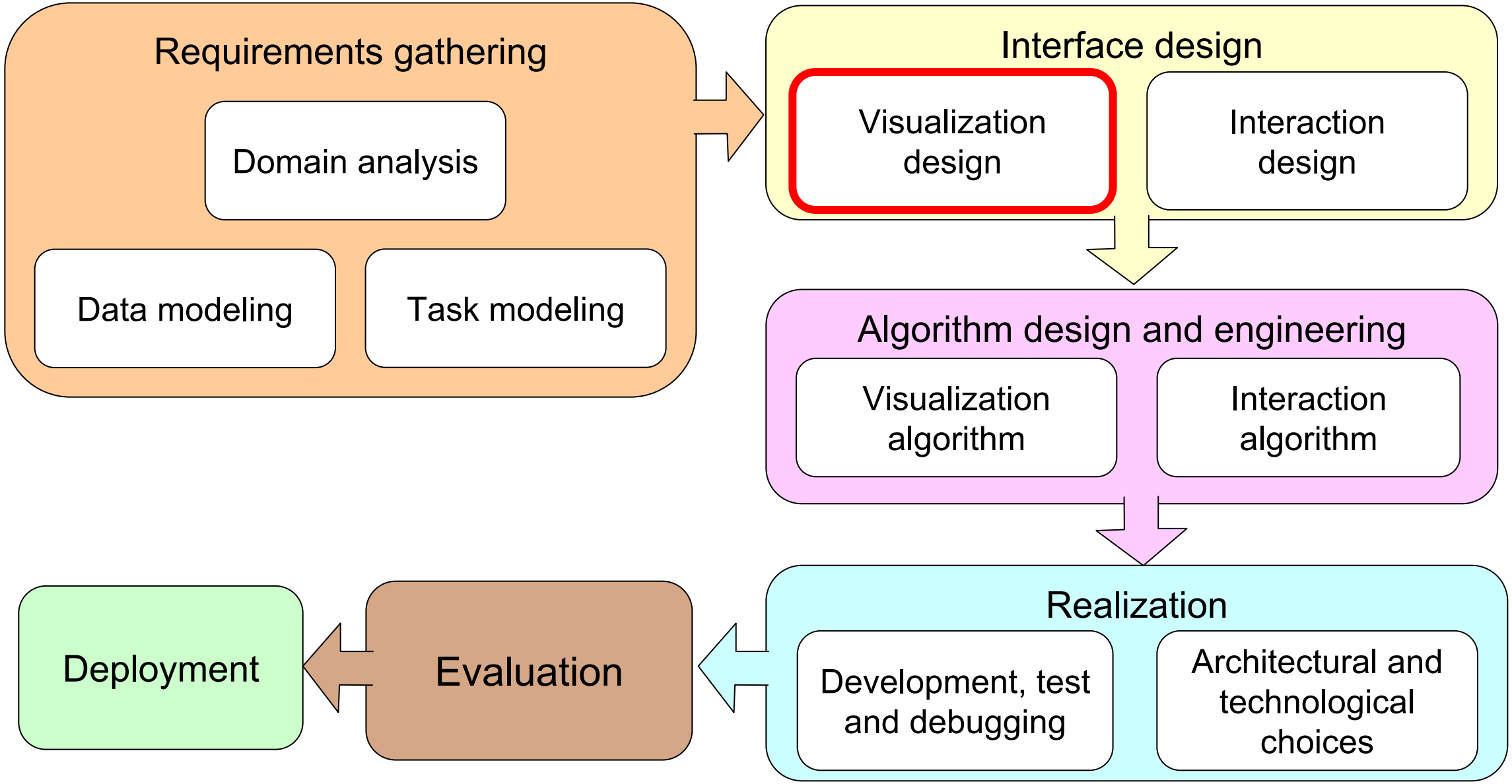
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Summary

- The perception of color
  - chromatic aberration
  - color perception in humans
  - color models
  - color spaces and dialogs windows
  - colormaps

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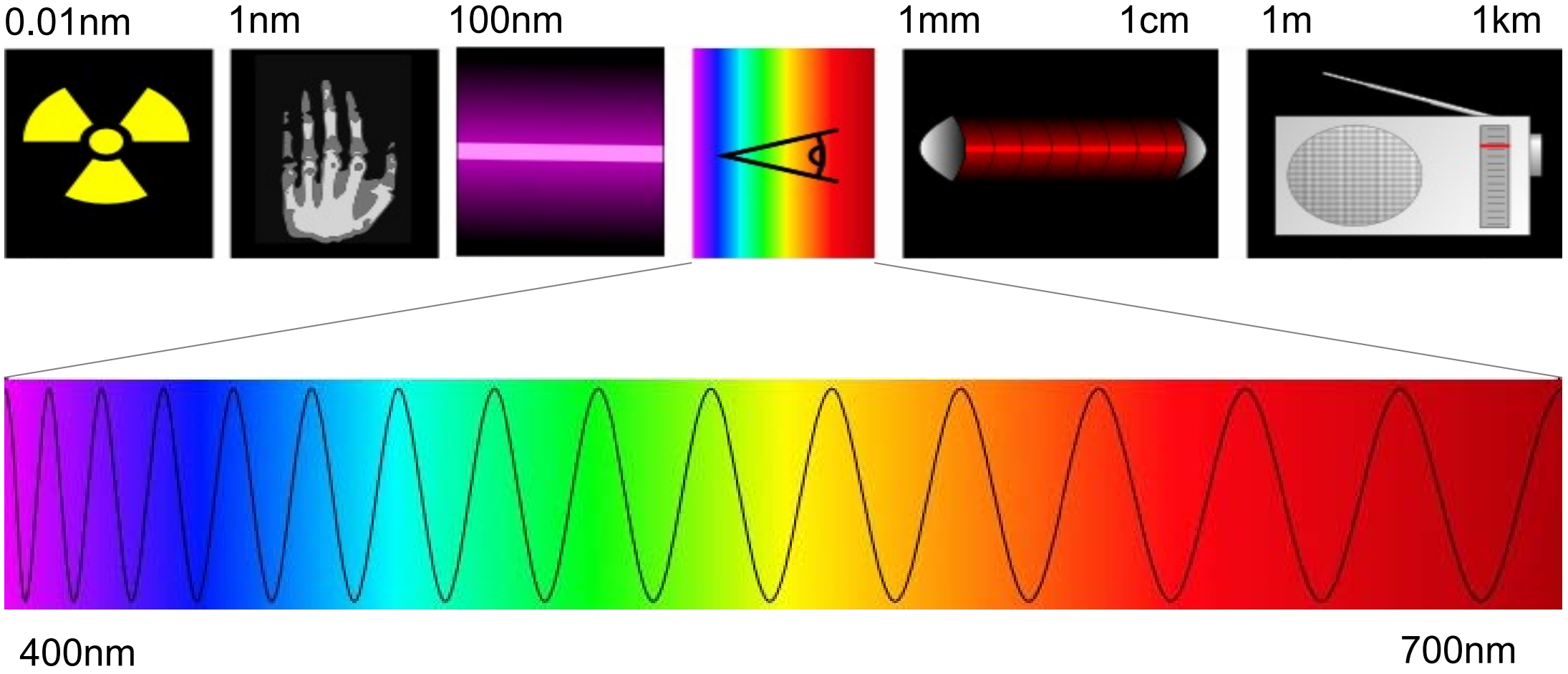
Design pipeline



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Visible spectrum

- Only a portion of the electromagnetic spectrum is visible to the human eye



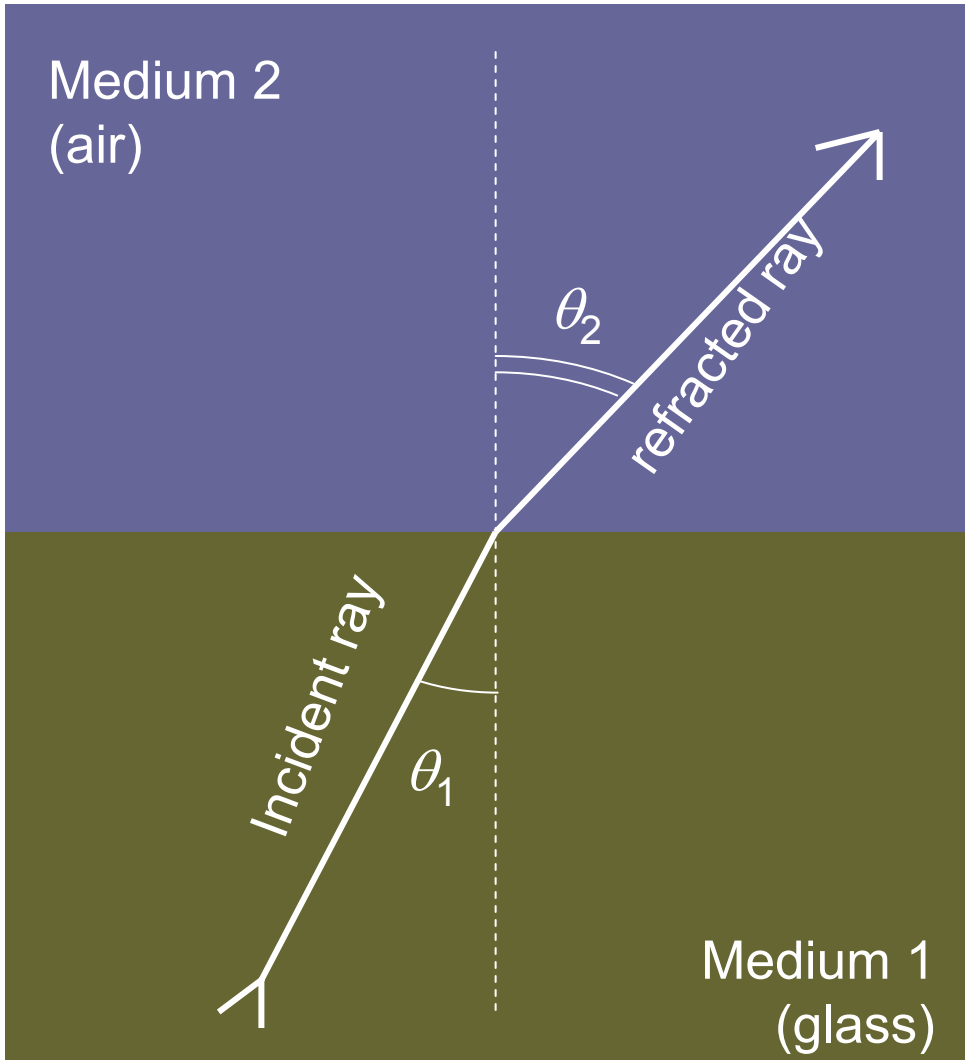
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Snell's law of refraction

- Relationship between the angles of refraction

$$\frac{\sin(\theta_2)}{\sin(\theta_1)} = \frac{v_2}{v_1}$$

- v_1 and v_2 are the phase velocities in the two mediums
- Phase velocity
  - the velocity of any given phase of the wave (for example, the crest) through a medium

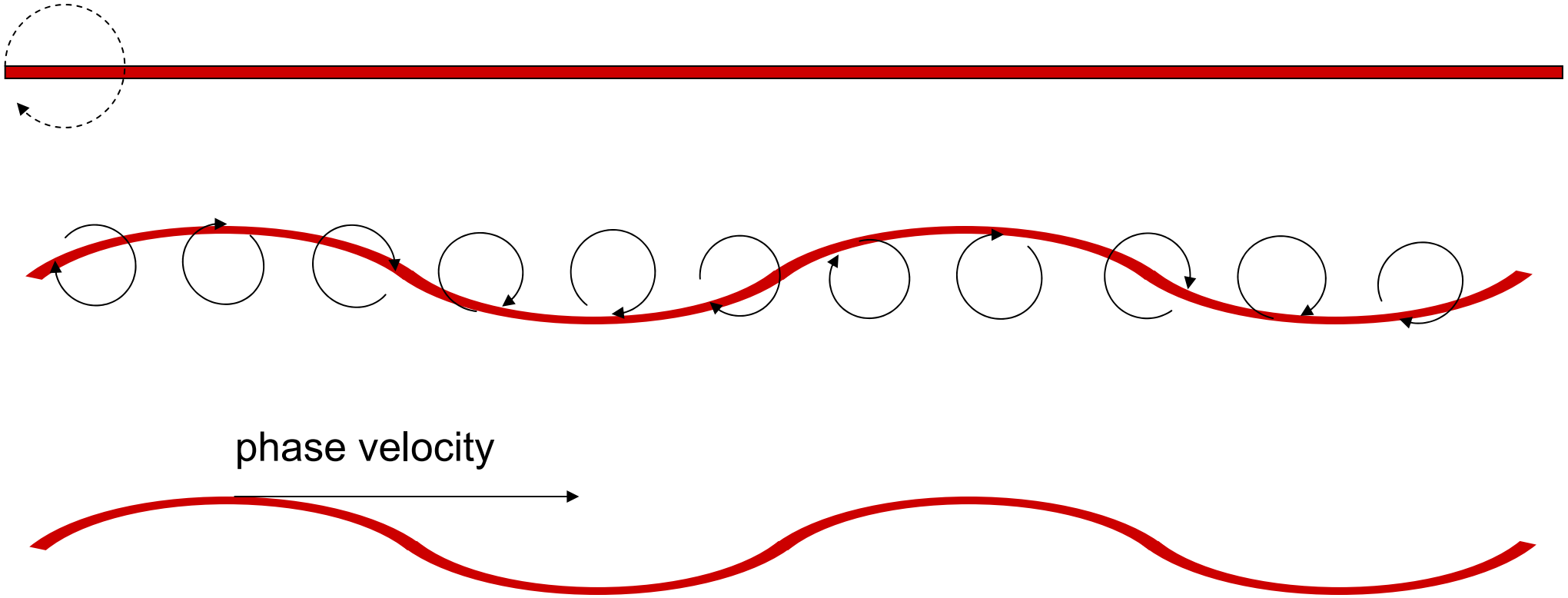


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Phase velocity



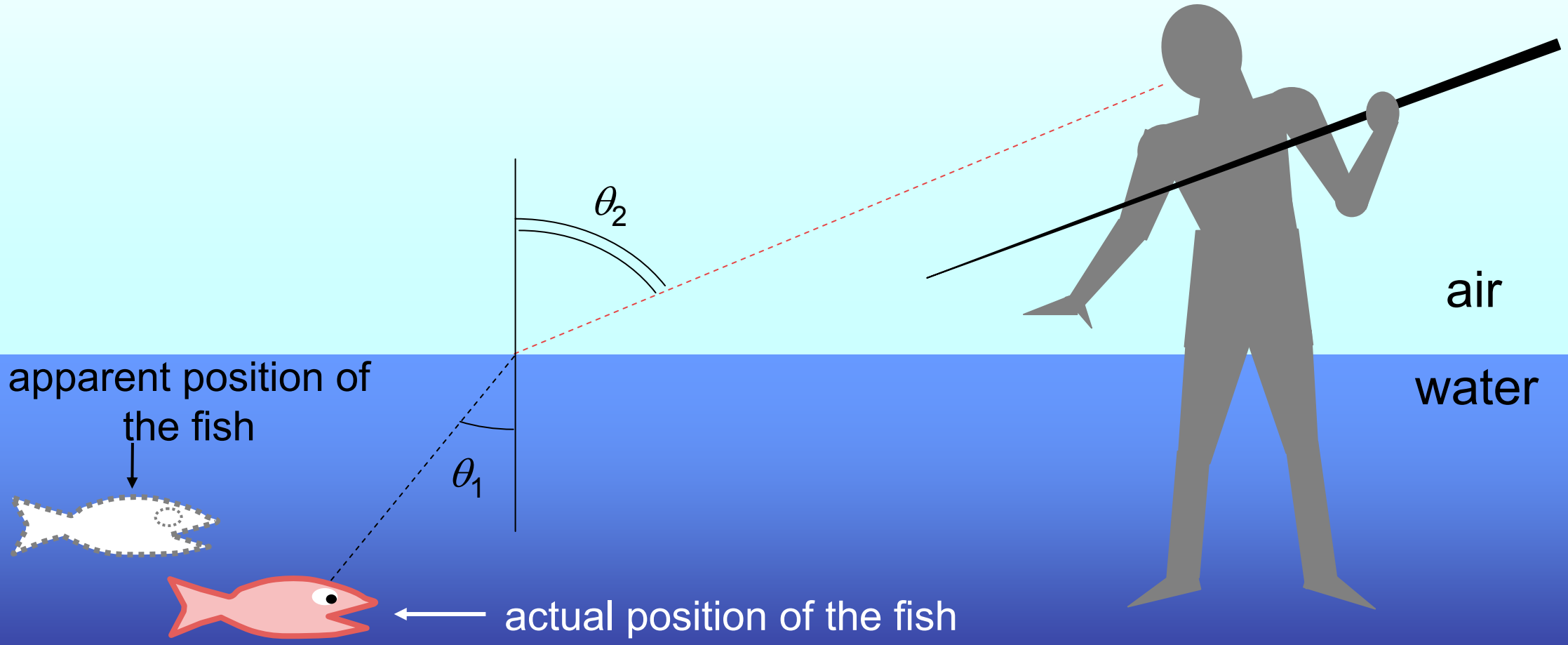
- Imagine to whisk a red carpet



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Snell's law: an example

- The difficulty of fishing



Dispersion

- Phase velocity in a given medium is given by

$$v = \frac{c}{n(\lambda)}$$

  - where c is the speed of light and $n(\lambda) > 1$ is the refractive index of the medium
- For most transparent materias we have

$$1 < n(\lambda_{red}) < n(\lambda_{green}) < n(\lambda_{blue})$$

  which imples $v_1(\lambda_{red}) > v_1(\lambda_{green}) > v_1(\lambda_{blue})$
- While in air or vacuum the refractive index is ~1 and hence $v_2(\lambda_{red}) \cong v_2(\lambda_{green}) \cong v_2(\lambda_{blue})$

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Dispersion

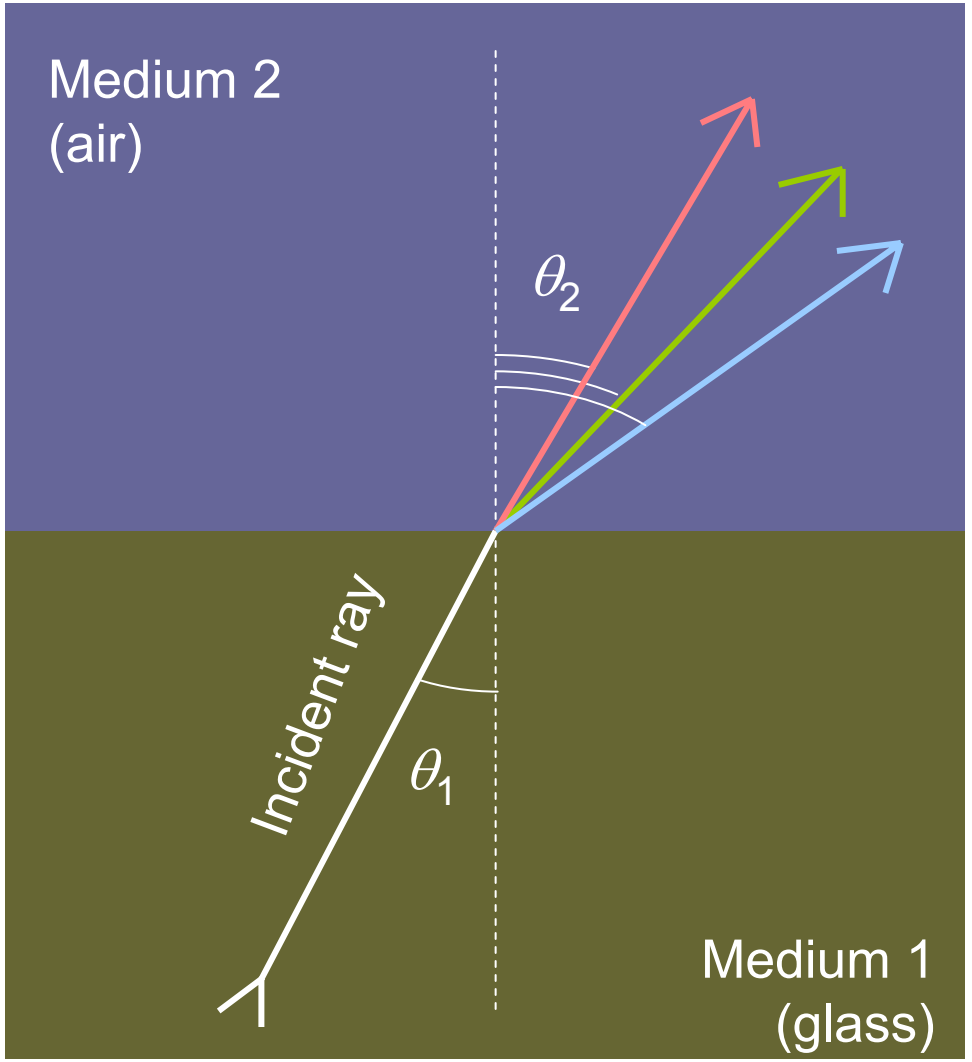
- In conclusion we have

$$v_1(\lambda_{red}) > v_1(\lambda_{green}) > v_1(\lambda_{blue})$$

$$v_2(\lambda_{red}) \cong v_2(\lambda_{green}) \cong v_2(\lambda_{blue})$$

- This implies that different colors have different refraction angles

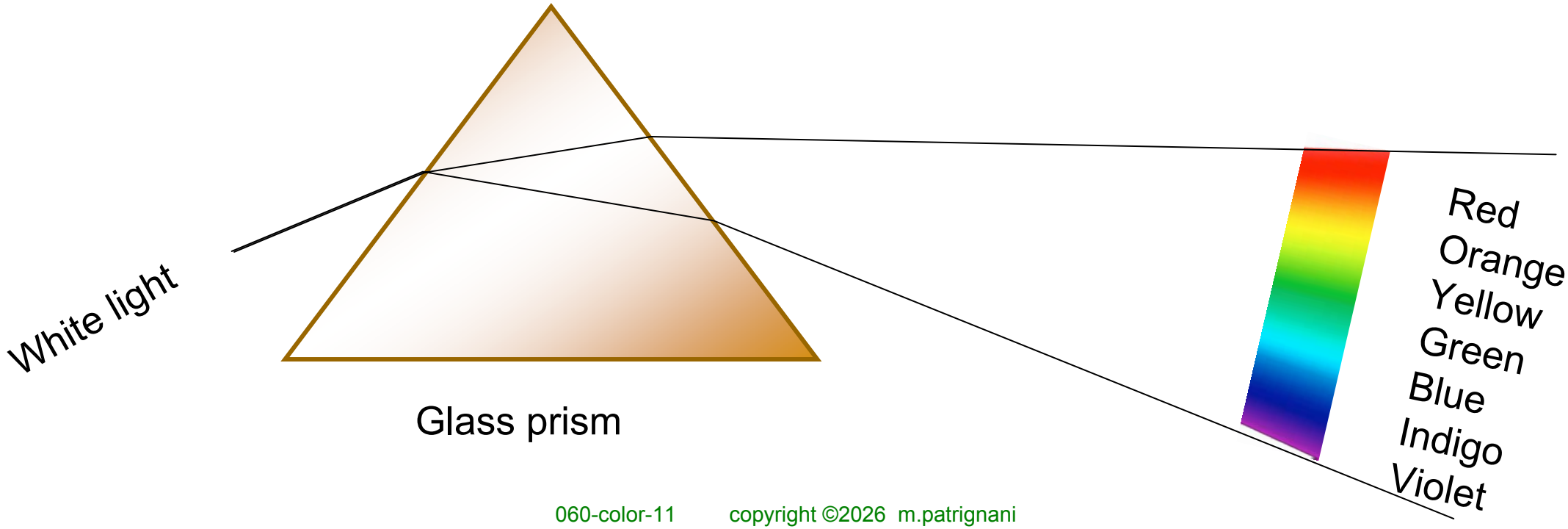
$$\sin(\theta_2) = \frac{v_2}{v_1}\sin(\theta_1)$$



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Prism dispersion

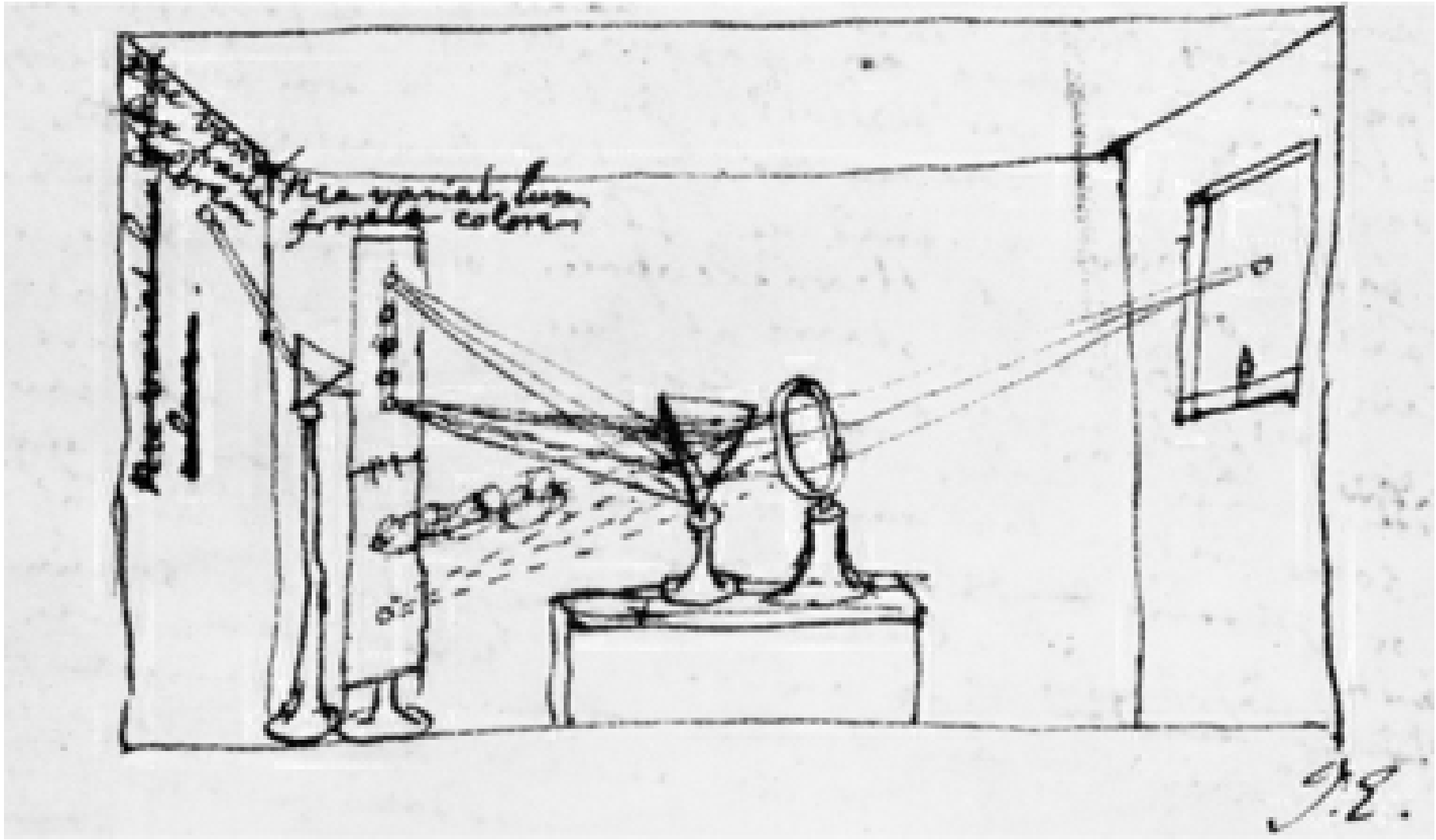
- A triangular prism breaks light up into its constituent spectral colors
  - proves that white light is a composition of colors
  - allows spectral analysis



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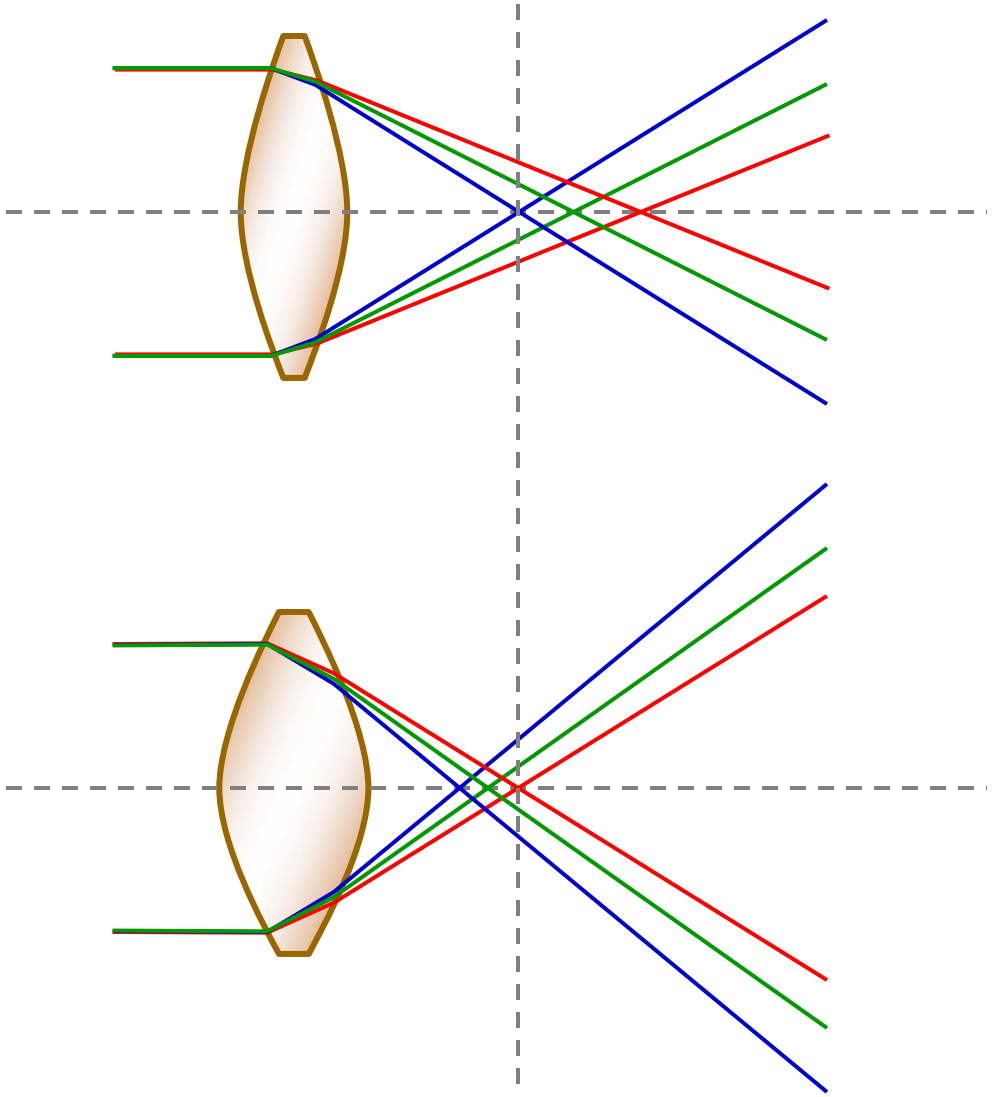
Some credits

- Isaac Newton (1642-1727)
  - correspondence I, p. 107
  - the experimentum crucis



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Chromatic aberration



- In order for the red light to be focused as the blue light, you have to modify your eyes lenses as if the red object was nearer
  - red objects seem to be nearer than blue objects

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Chromatic aberration

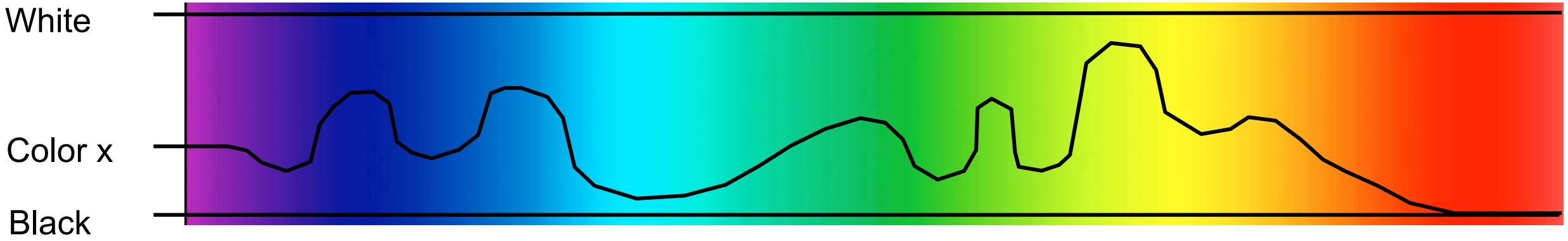
- Chromatic aberration
  - blue color is viewed as in the background
  - red color is viewed as in the foreground
- Need a proof?

Proof Proof Proof Proof Proof Proof
Proof Proof Proof Proof Proof Proof
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Proof Proof Proof Proof Proof

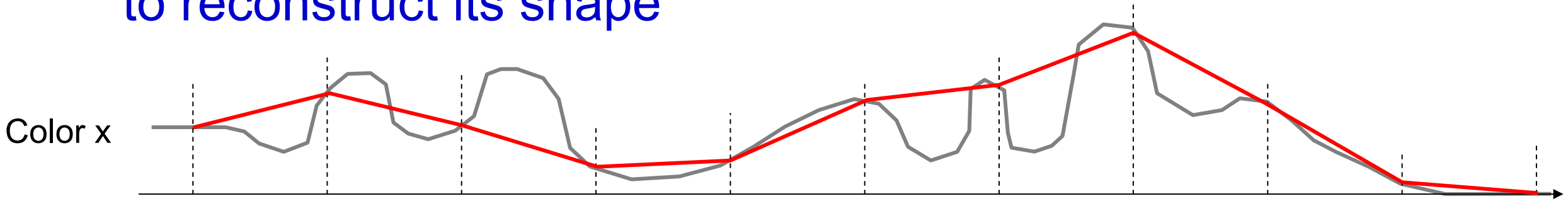
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Spectral analysis

- A color is a combination of wavelengths with different intensities



- Ideally, we would like to have as many samples as possible to reconstruct its shape



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Three distinct color receptors

- Cone sensitivity functions
  - picture adapted from [Ware 13]

S

short wavelength

420-440nm

M

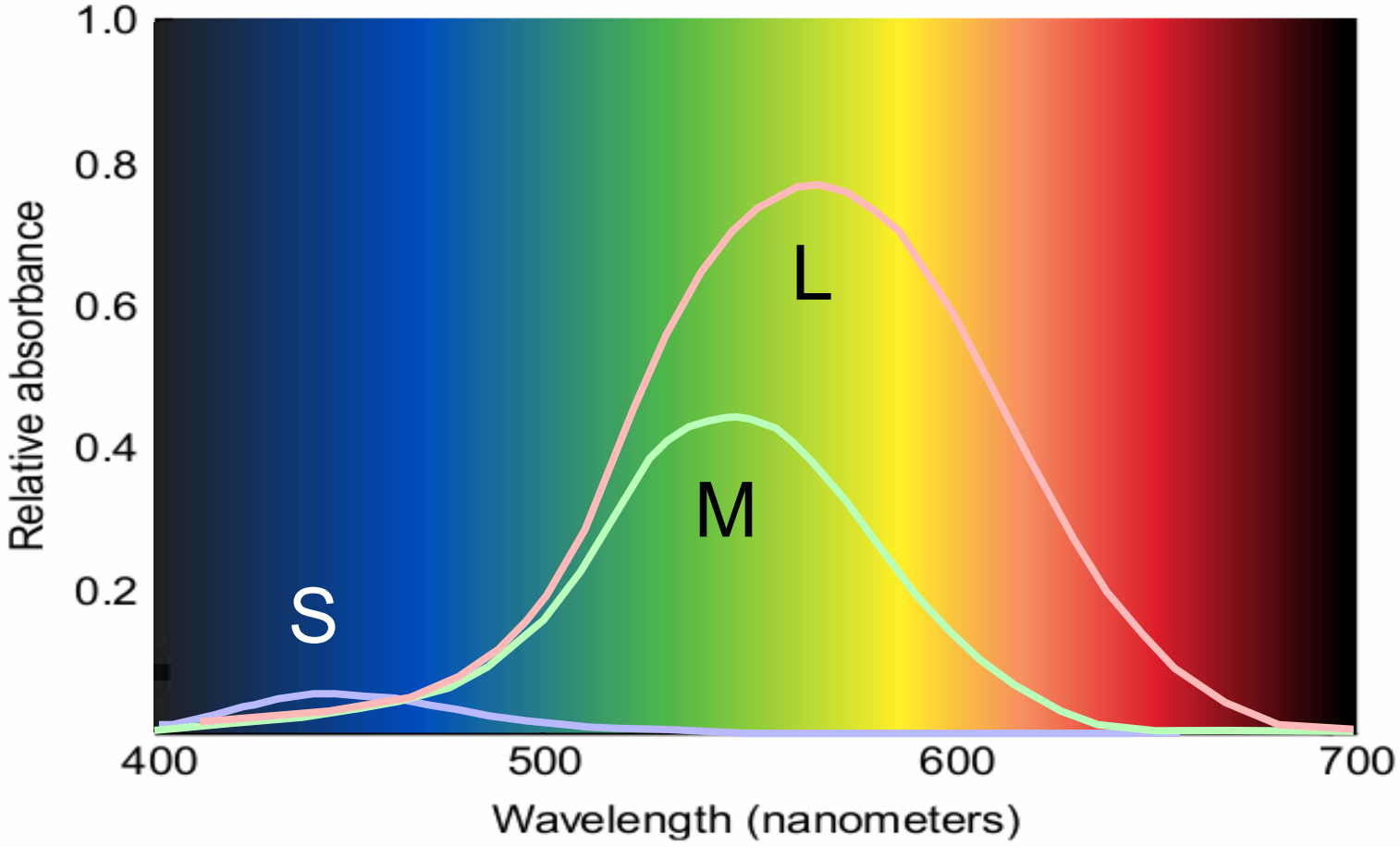
medium wavelength

530-540nm

L

long wavelength

560-580nm



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Three distinct color receptors

- Normalized spectral sensitivity
  - picture adapted from Wikipedia

S

short wavelength

420-440nm

M

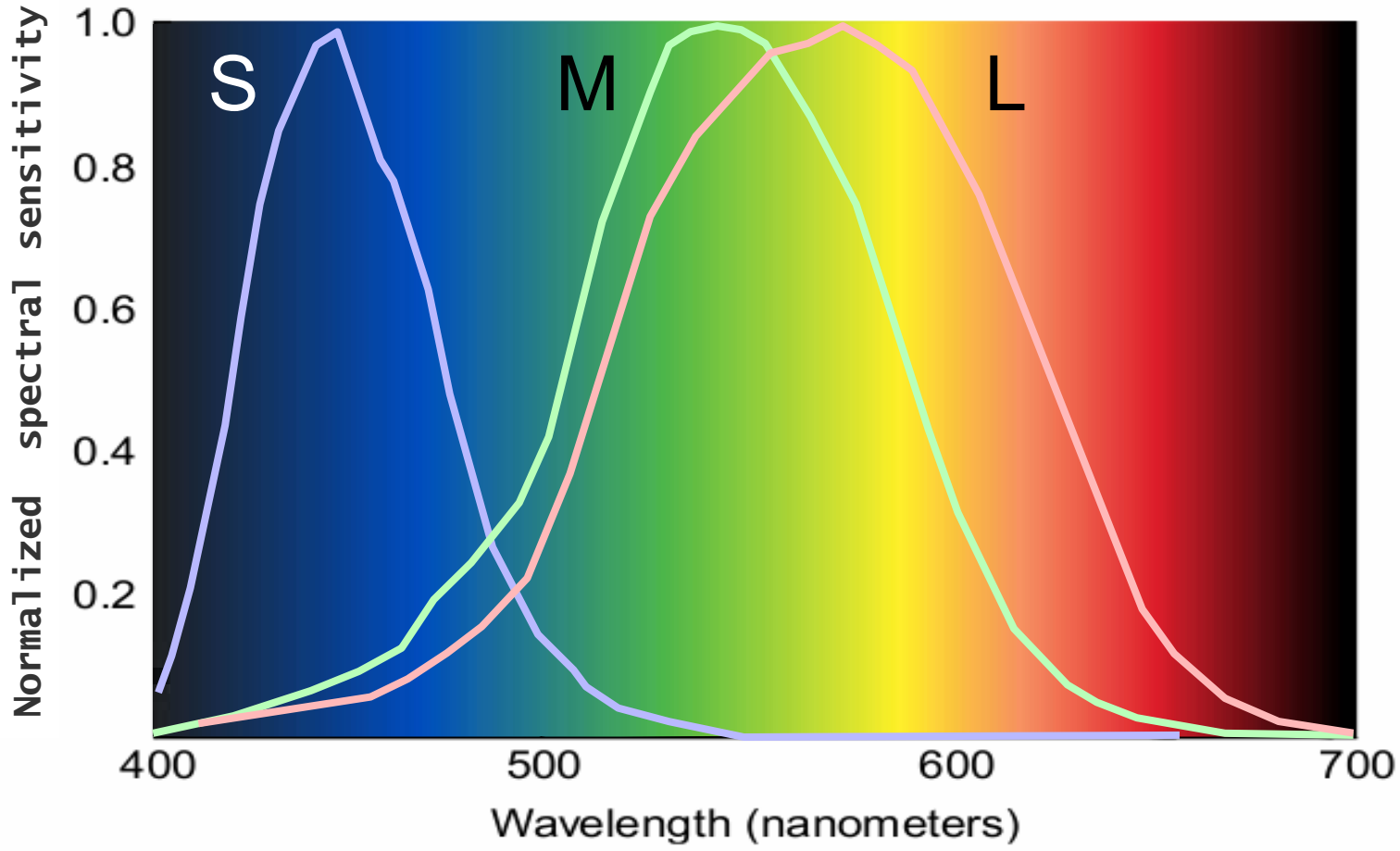
medium wavelength

530-540nm

L

long wavelength

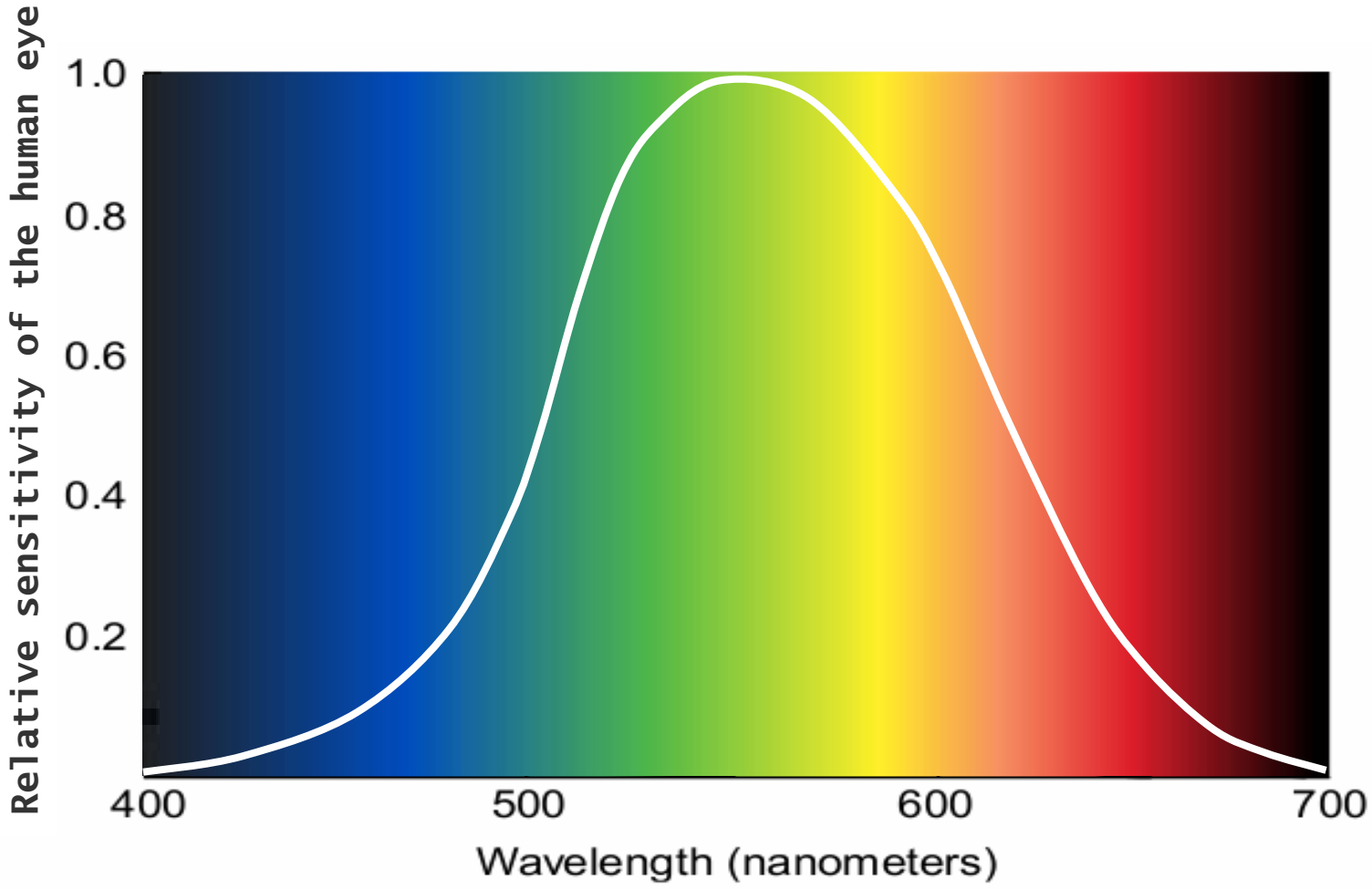
560-580nm



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Relative sensitivity of our eyes

- The luminance of a light measures how we are sensitive to it
  - it is the integral of light spectrum weighted by our sensitivity



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Blue and red lights?

- Very few details can be perceived when a blue or red light is used



From http://www.cdt.ch/ticino/cronaca/3391/inutile-la-luce-blu-nei-wc-pubblici.html

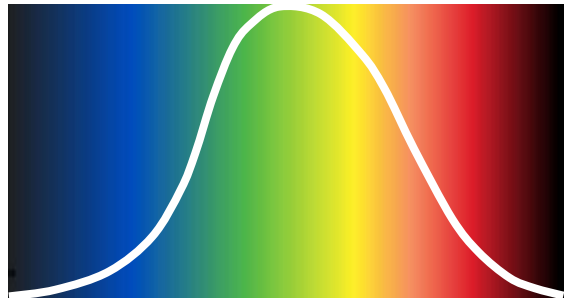


From http://www.scattovisuale.it/blog/

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Luminance and contrast

- When contrast is needed (for example for distinguishing text) luminance is more important than color



Blue text is difficult to view on a black background because of its poor luminance constrast

Blue text is difficult to view on a black background because of its poor luminance constrast

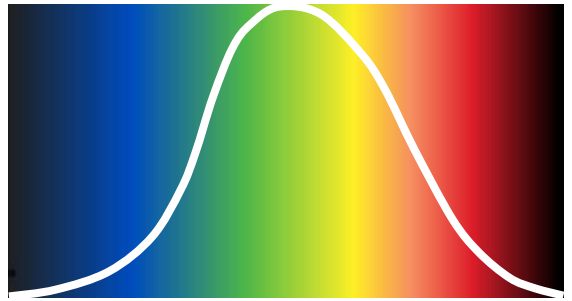
The same holds for yellow text on a white background

The same holds for yellow text on a white background

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Luminance and thumb rules

- Use a minimum 3:1 luminance contrast ratio between a pattern and its background
  - ISO proposes a 10:1 optimal ratio for text (ISO 9241, Part 3)





http://www.pridepainters.ca/article/unique-painting-projects-helicopter-landing-pad.html

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Consequences of trichromacy

- Since we have three distinct color receptors (cones) in our retinas
  - “color spaces” have three dimensions
  - liquid crystal in a television screen have three different colors
    - red, green, and blue
  - we use three primary paint colors
    - red, yellow, and blue
  - printers have a minimum of three colored inks for color printing
    - cyan, magenta, and yellow

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What about other species?

State	Cone cells	# of colors	Carriers
Monochromacy	1	100	marine mammals, owl monkey, Australian sea lion, achromat primates
Dichromacy	2	10 K	most terrestrial non-primate mammals, color blind primates
Trichromacy	3	10 M	most primates, especially great apes (such as humans), marsupials, some insects (such as honeybees)
Tetrachromacy	4	100 M	most reptiles, amphibians, birds and insects, rarely humans
Pentachromacy	5	10 G	some insects (specific species of butterflies), some birds (pigeons for instance)

From Wikipedia (“Color vision”)

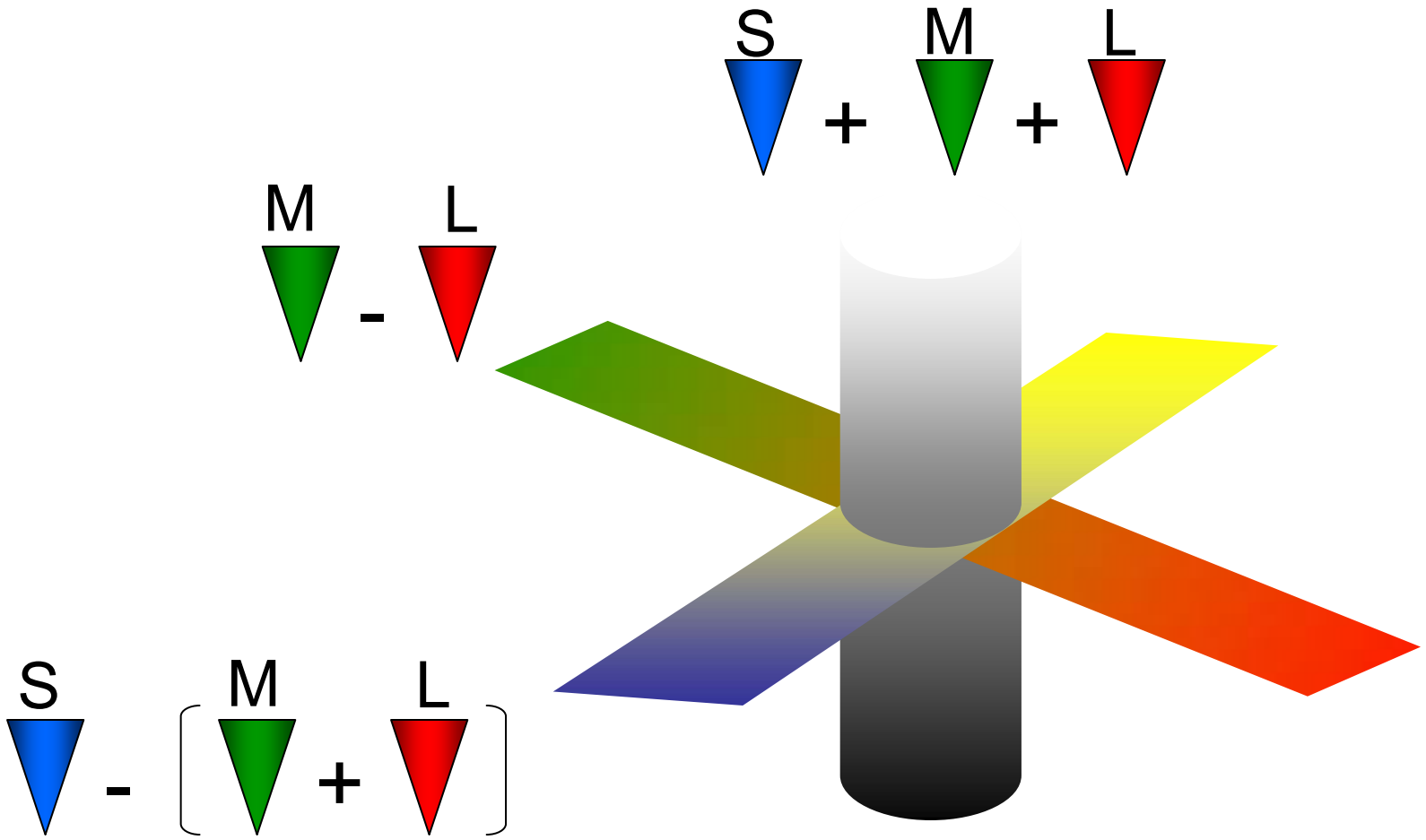
Color models

- What strategies can be used to recognize, generate, identify, name a color
- Color percetive models
  - opponent process theory
- Color generative models
  - RGB: Red Green Blue
  - CMYK: Cyan, Magenta, Yellow, blacK
- Color descritive models
  - HSL: Hue, Saturation, Lightness
  - HSV: Hue, Saturation, Value

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Color perceptive model

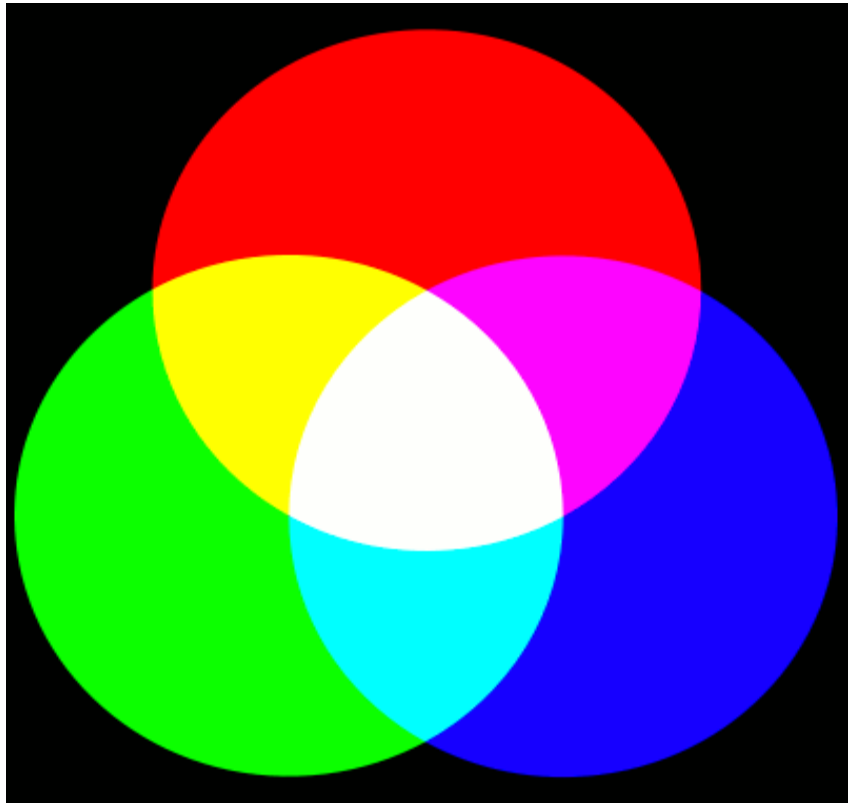
- Opponent process theory
  - introduced by Ewald Hering in 19th century
  - explains how our S M and L cones allows us to perceive all colors



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Color generative models

- We will see two strategies
  - additive model (RGB: Red Green Blue)
  - subtractive model (CMYK: Cyan, Magenta, Yellow, blacK)

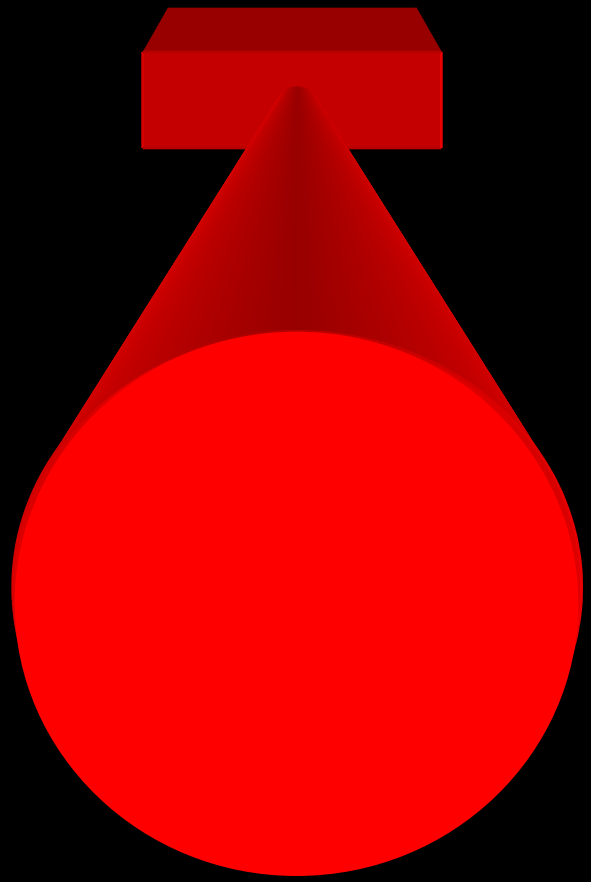




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Additive model

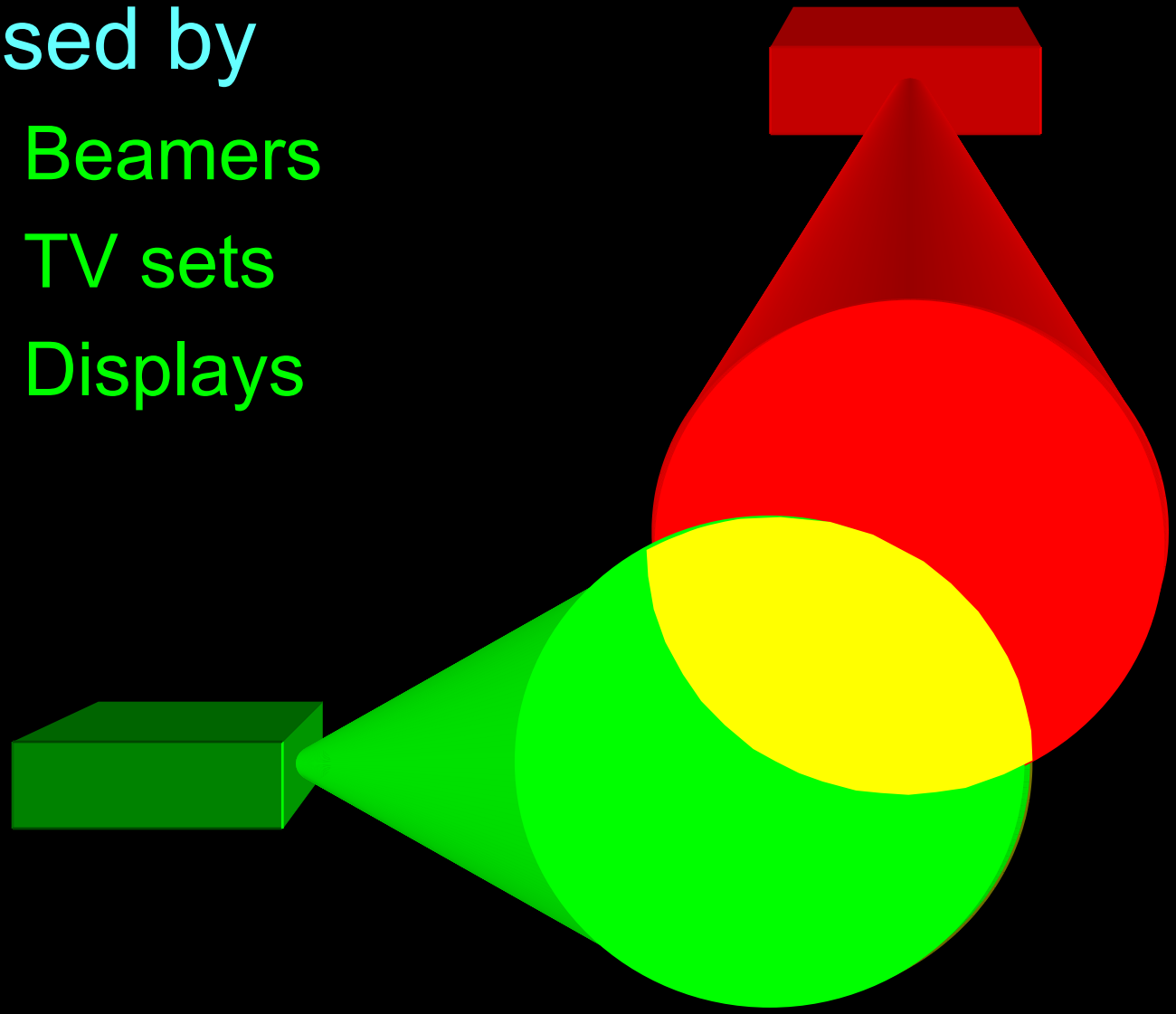
- Used by
  - Beamers
  - TV sets
  - Displays



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Additive model

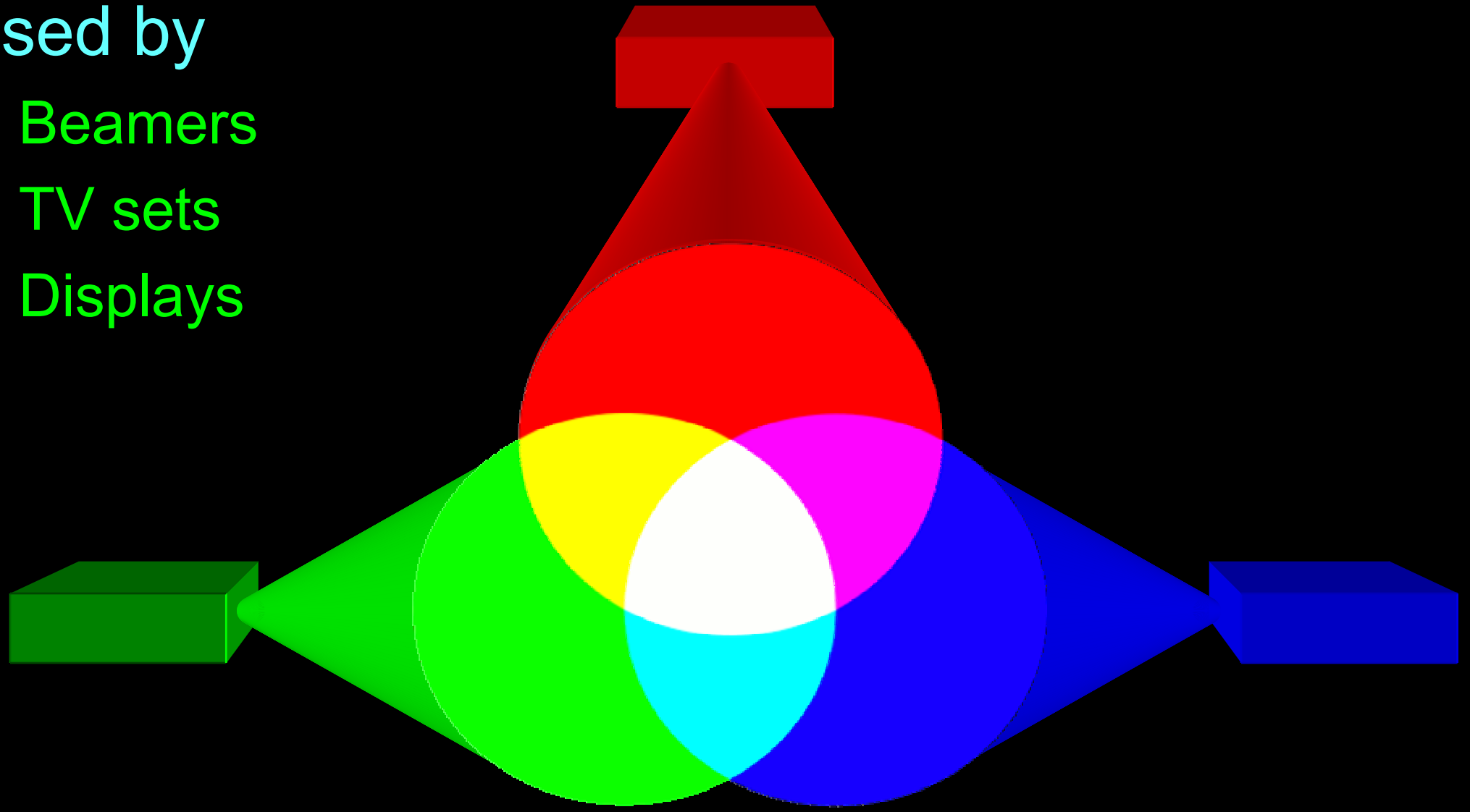
- Used by
  - Beamers
  - TV sets
  - Displays



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Additive model

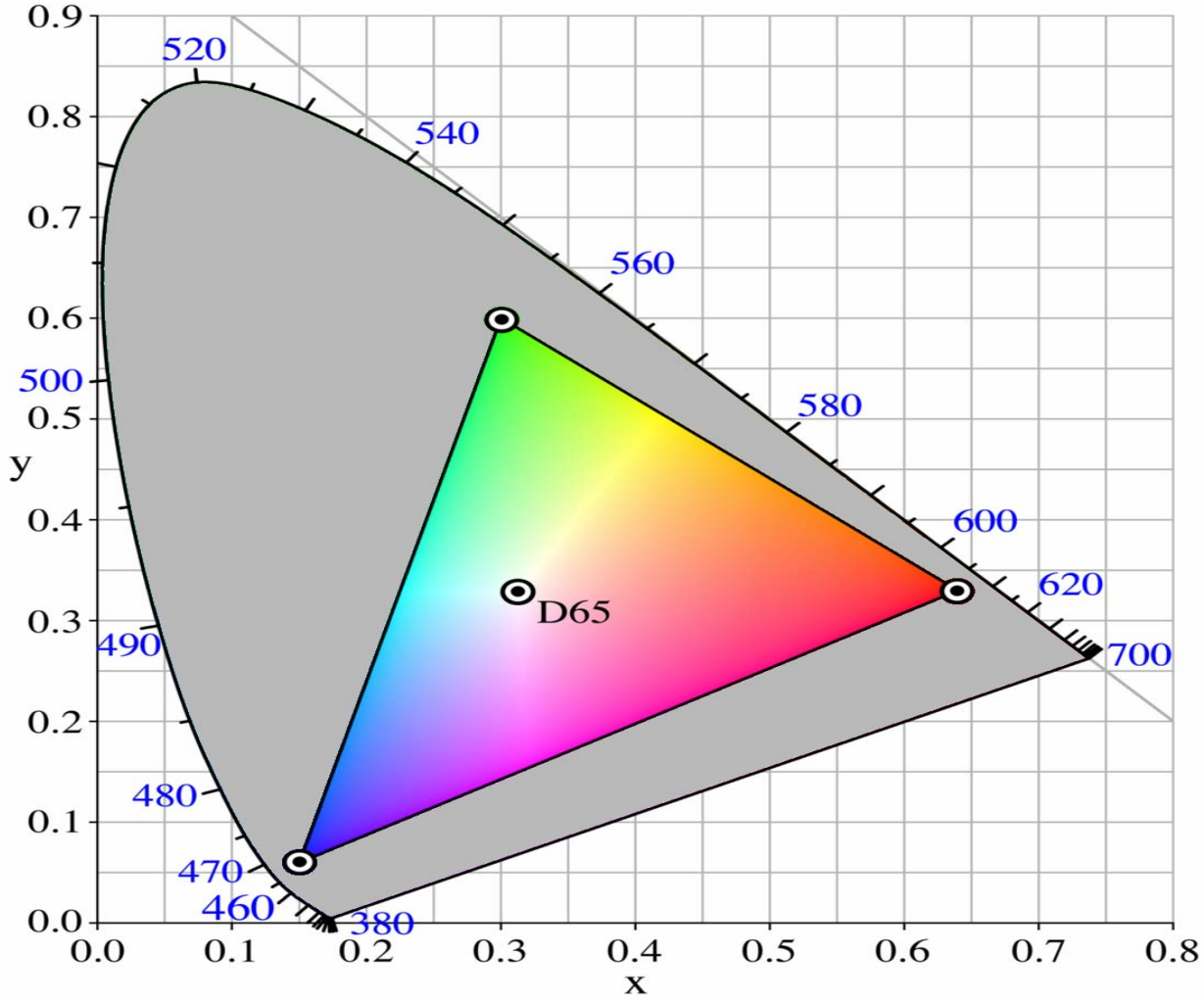
- Used by
  - Beamers
  - TV sets
  - Displays



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RGB color space

- We can describe a color as a combination of red green and blue
  - C = rR + gG + bB
- Only the colors in the “gamut” of the three colors can be obtained
- The effect of a combination is rather unpredictable

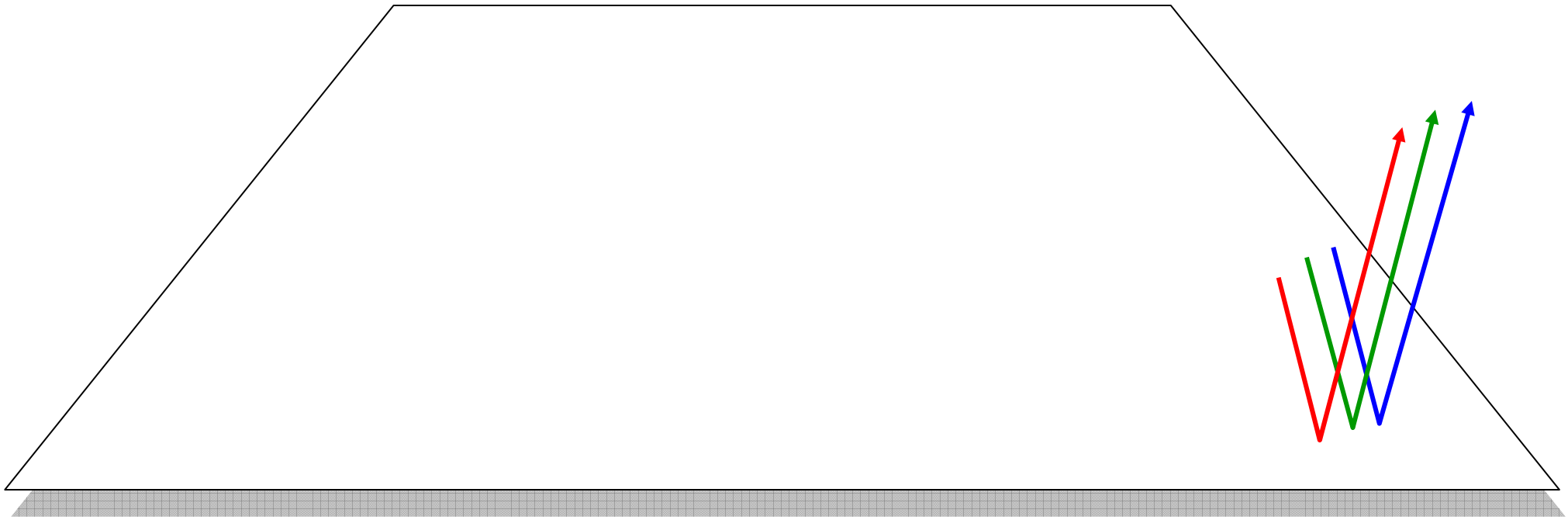


From Wikipedia

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CMYK subtractive model

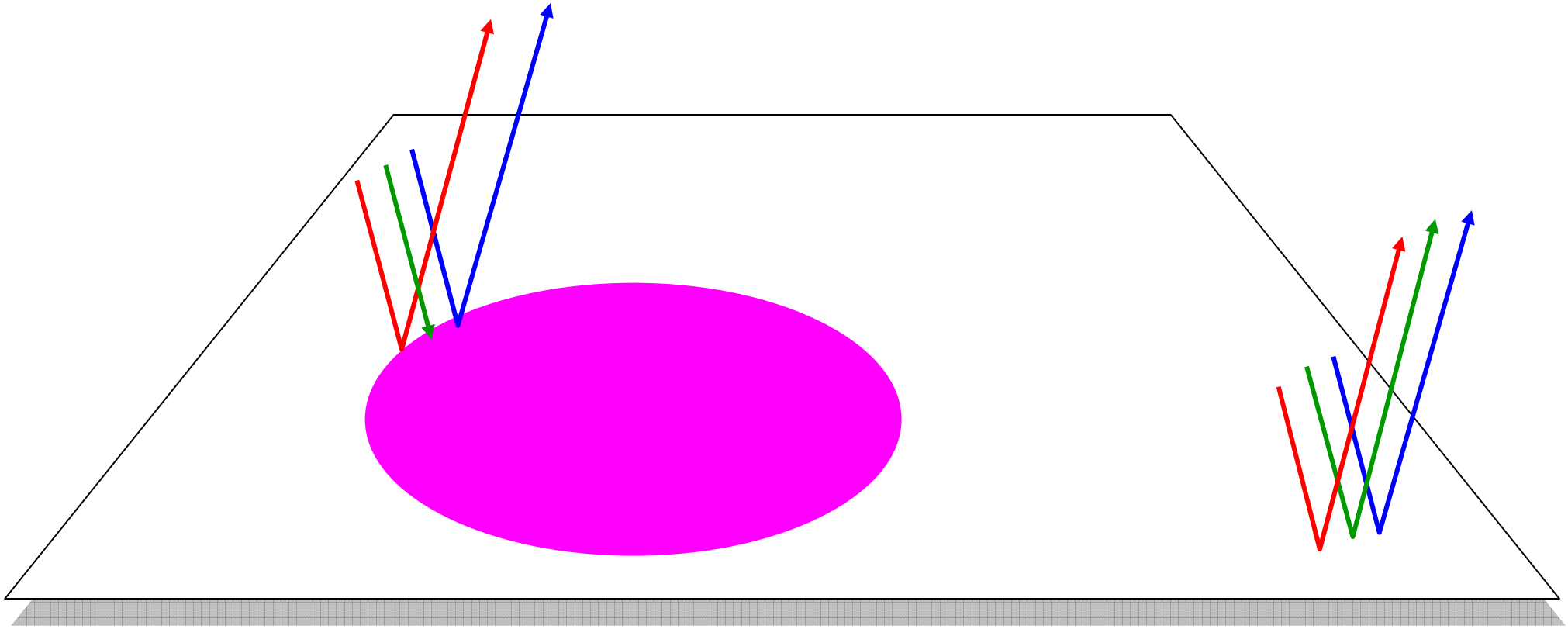
- Each color ink is actually a filter, absorbing part of the spectrum



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CMYK subtractive model

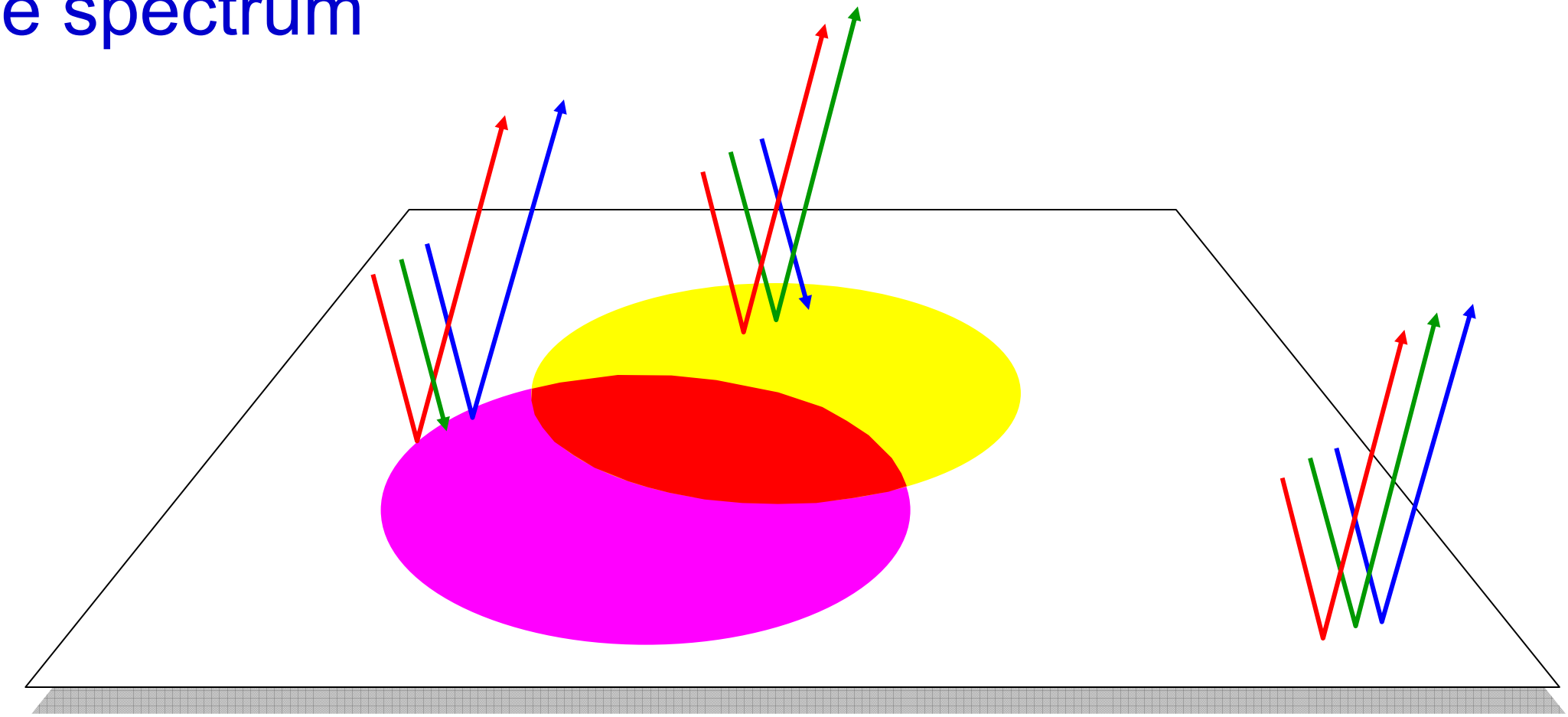
- Each color ink is actually a filter, absorbing part of the spectrum



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CMYK subtractive model

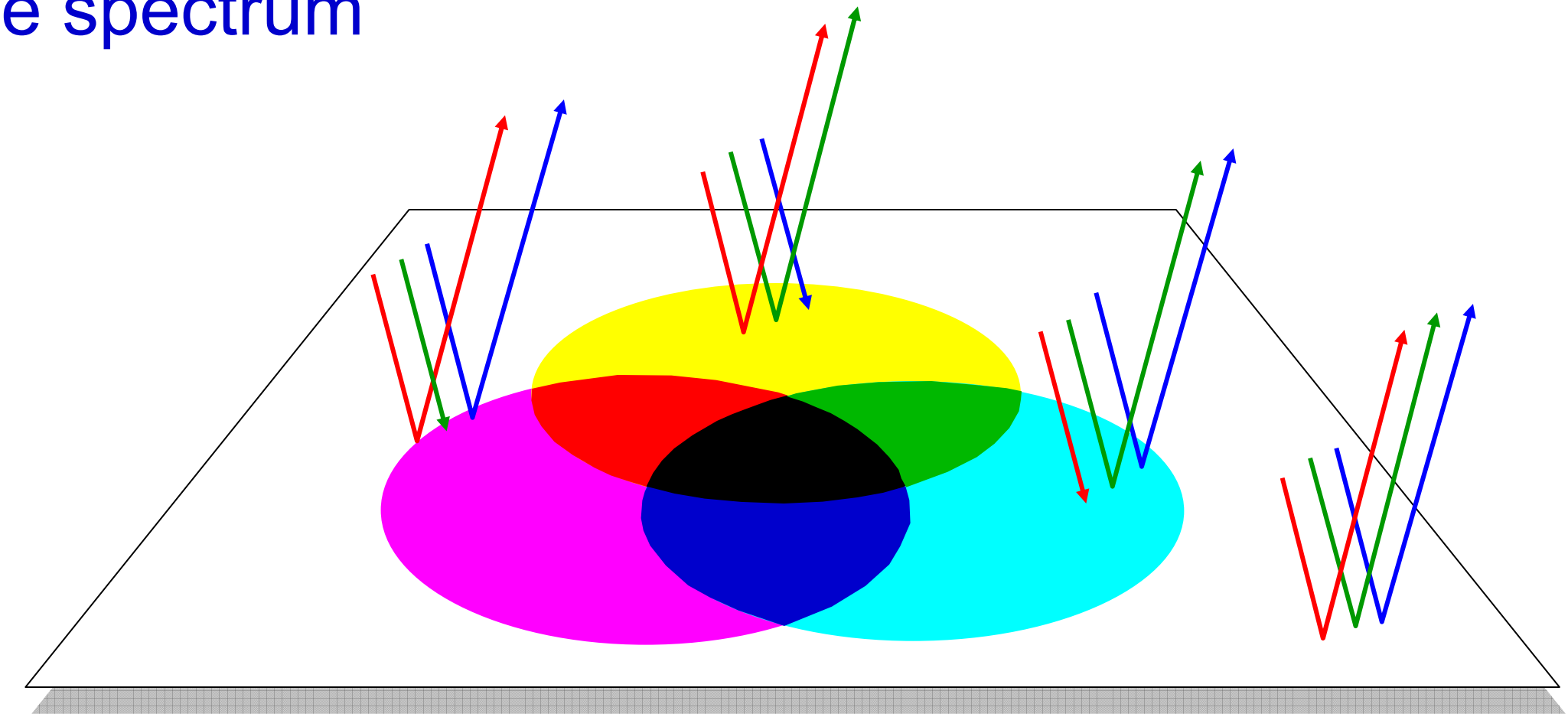
- Each color ink is actually a filter, absorbing part of the spectrum



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CMYK subtractive model

- Each color ink is actually a filter, absorbing part of the spectrum



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HSL color space

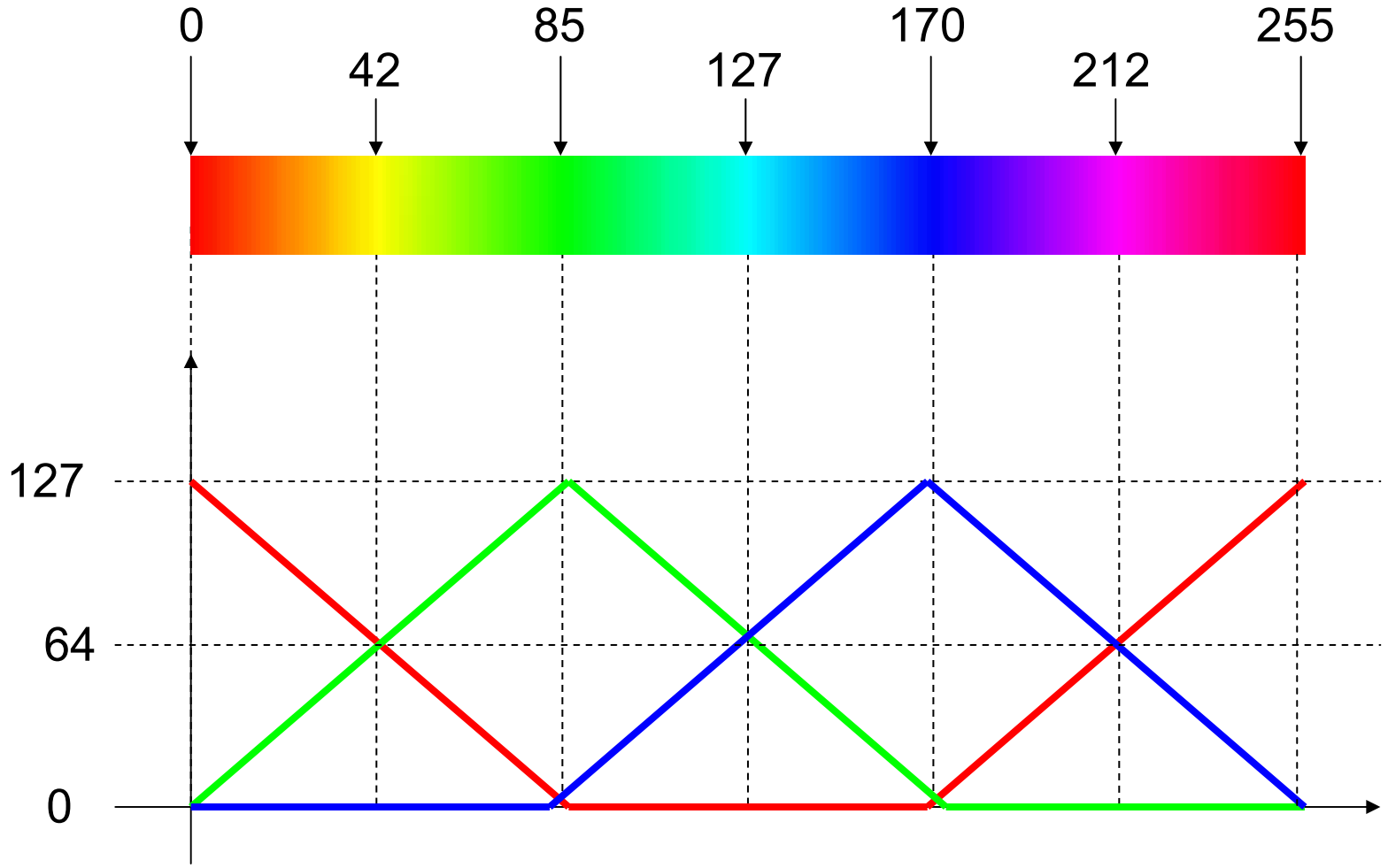
- Hue
  - an approximation to the visible spectrum
    - from red to green
    - from green to blue
    - from blue to red
- Saturation
  - how much of the color is replaced with gray
- Lightness
  - how much light
    - (MaxColor+MinColor)/2



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HSL hue

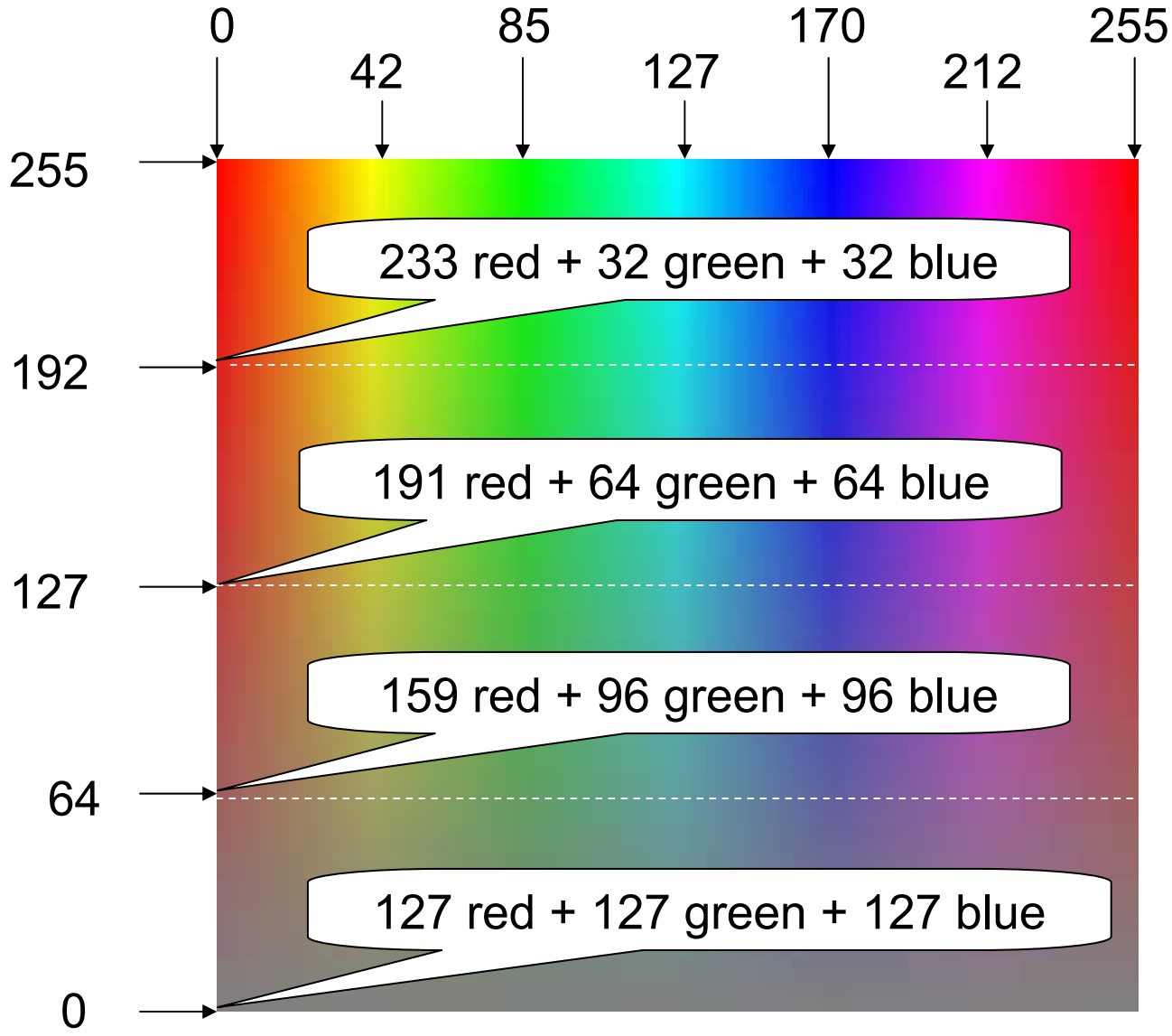
- Assume all values are in [0,255]
- Hue interpolates a color composition from red to green, to blue, to red again
  - cylindrical coordinates



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HSL saturation

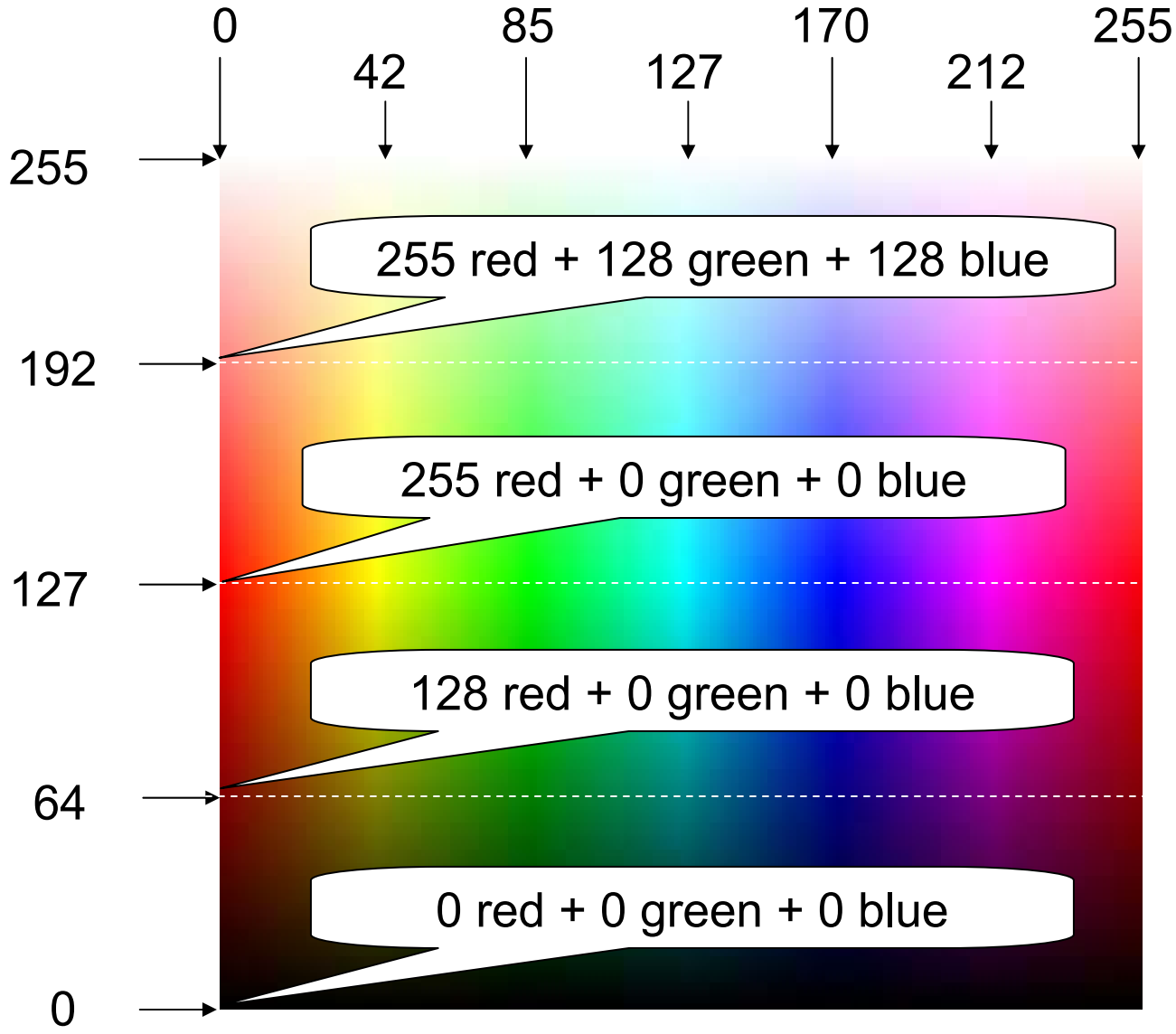
- Saturation vs Hue
- Keeping the same Lightness
  - 127 in this example
- Pointed out values
  - 192+32=233, 32, 32
  - 127+64=191, 64, 64
  - 63+96=159, 96, 96
  - 0+127=127,127,127



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HSL lightness

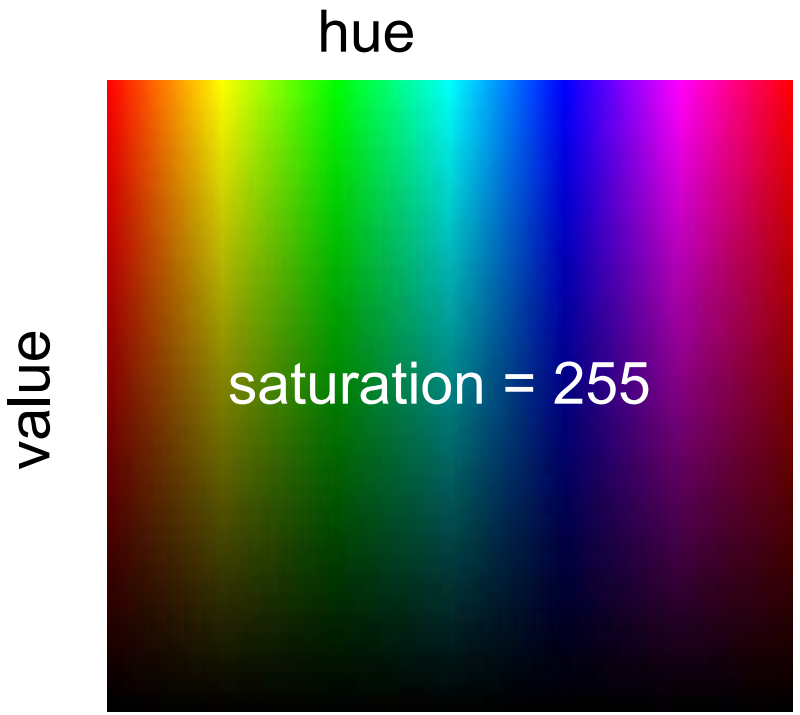
- Lightness vs Hue
- Keeping the same Saturation
  - 255 in this example



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HSV color space

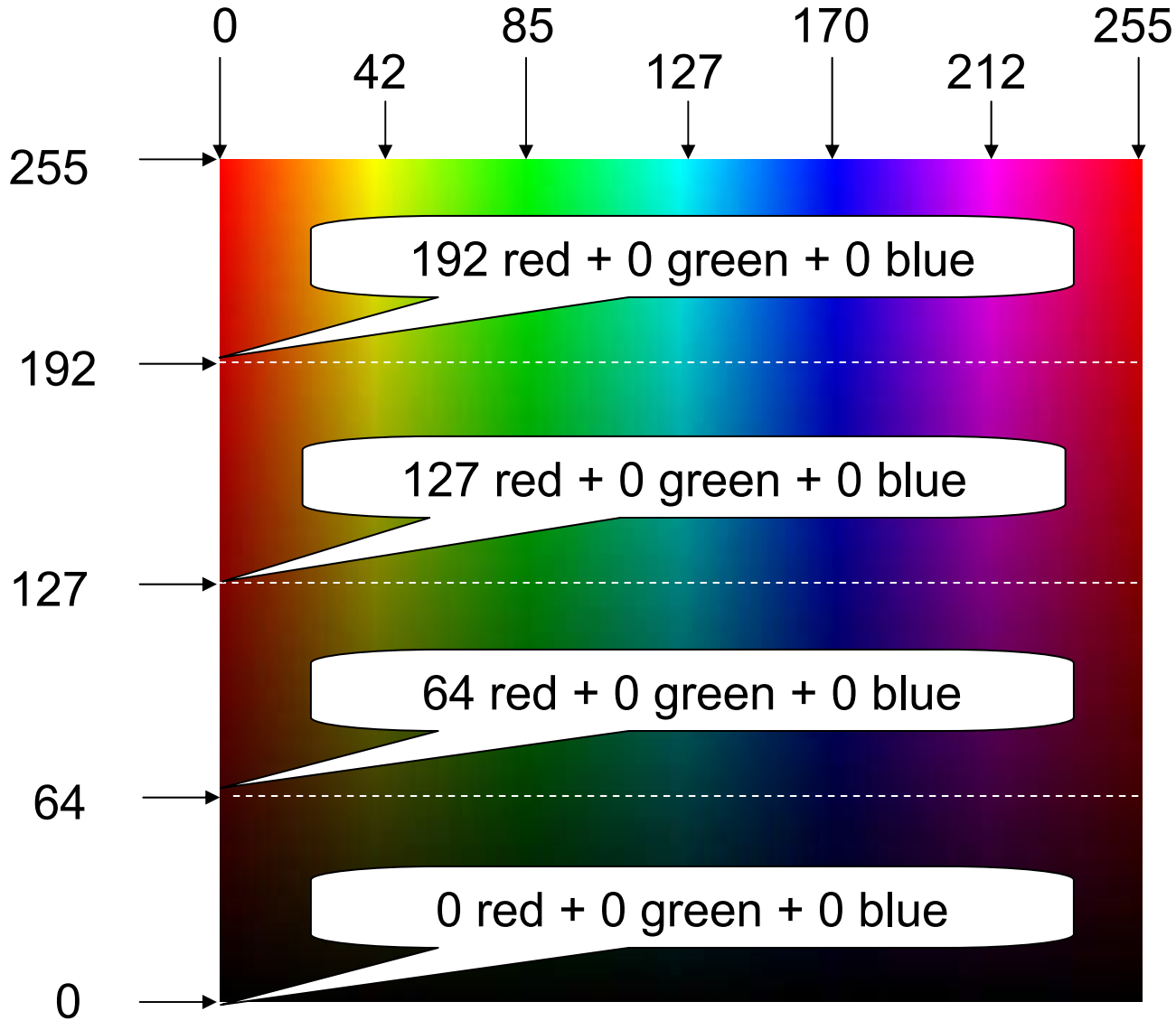
- Hue
  - the same as HSL
- Saturation
  - different from HSL
  - how much of the color is mixed with white
    - 255 = only colors
    - 0 = white
- Value
  - the higher value of the RGB composition
  - how much of the color is mixed with black



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HSV value

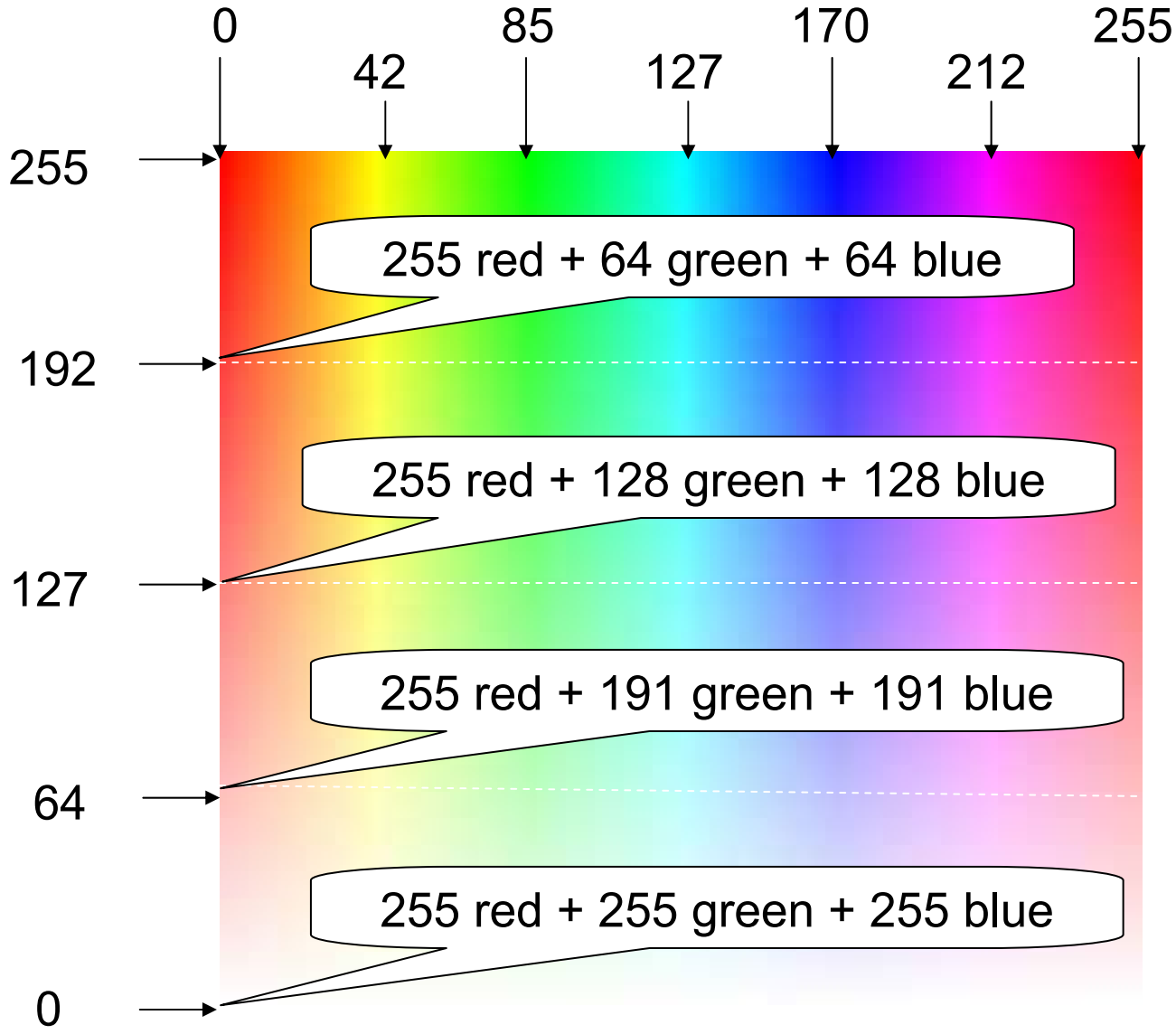
- Value vs Hue
- Keeping the same Saturation
  - 255 in this example



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HSV saturation

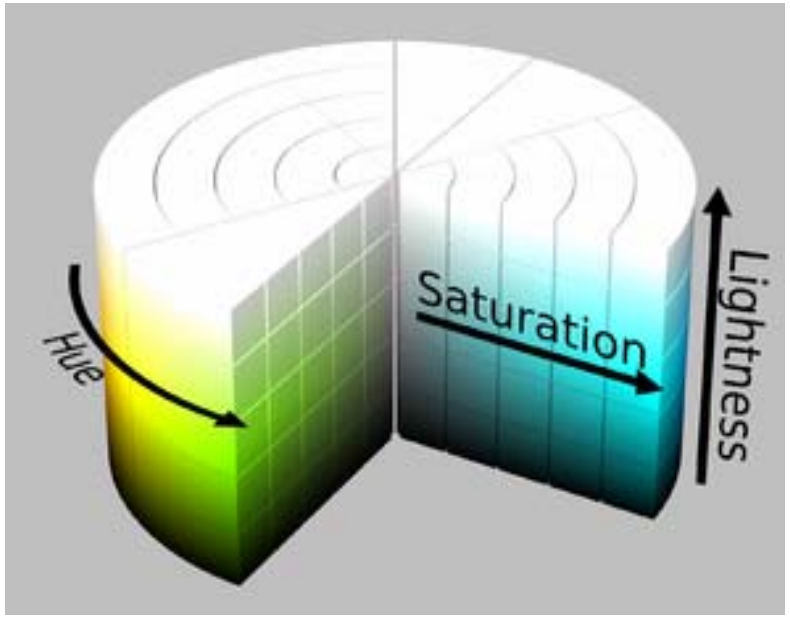
- Saturation vs Hue
- Keeping the same Value
  - 255 in this example



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HSL cylinder

- Gray shades can be found along the vertical axis
- Pure colors (i.e., red = 255,0,0) can be found on the external border at lightness 127
  - that is: at half-height

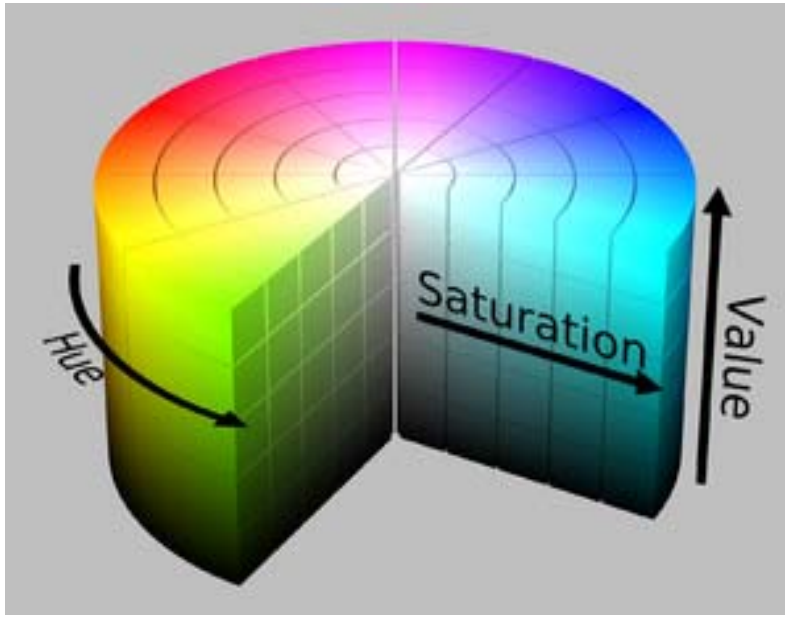


From Wikipedia (“HSL and HSV”)

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HSV cylinder

- Gray shades can be found along the vertical axis
  - the same as HSL
- Pure colors can be found on the external border at value 1
  - on top of the cylinder



From Wikipedia (“HSL and HSV”)

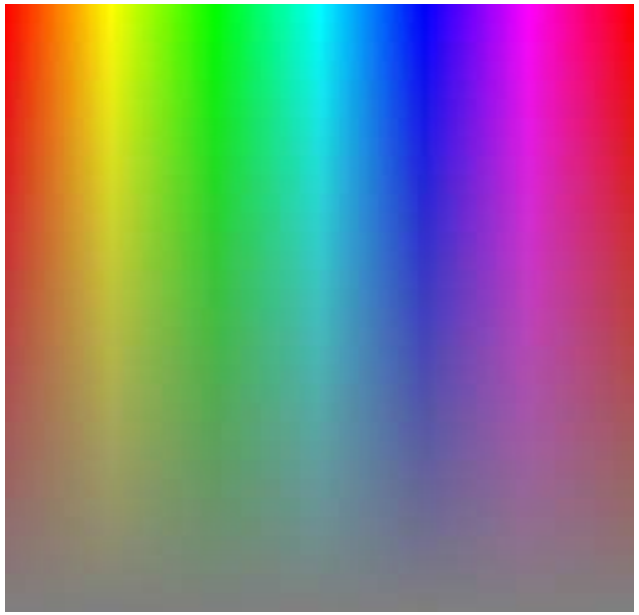
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HSL and HSV color spaces

HSL
(lightness = 0.5)

hue

saturation

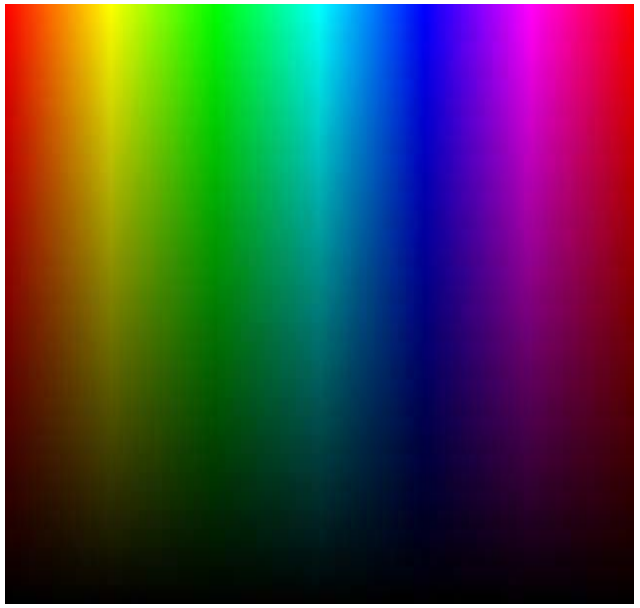


increasing or decreasing lightness mixes with white or black

HSV
(saturation = 1)

hue

value

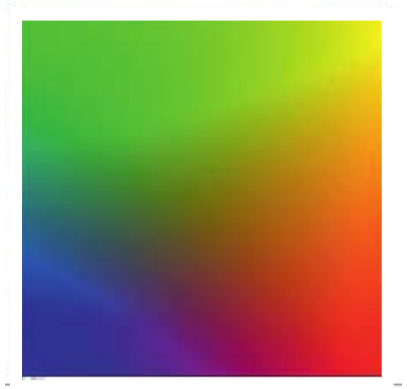


decreasing saturation mixes with white

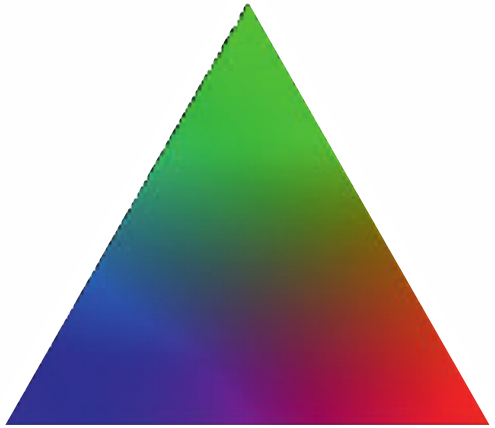
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Different geometric color layouts

- There are several ways of arranging colors on a plane



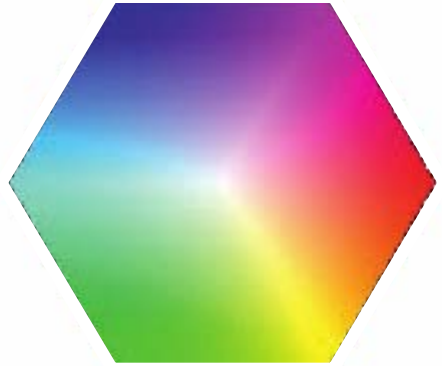
square



triangle



wheel



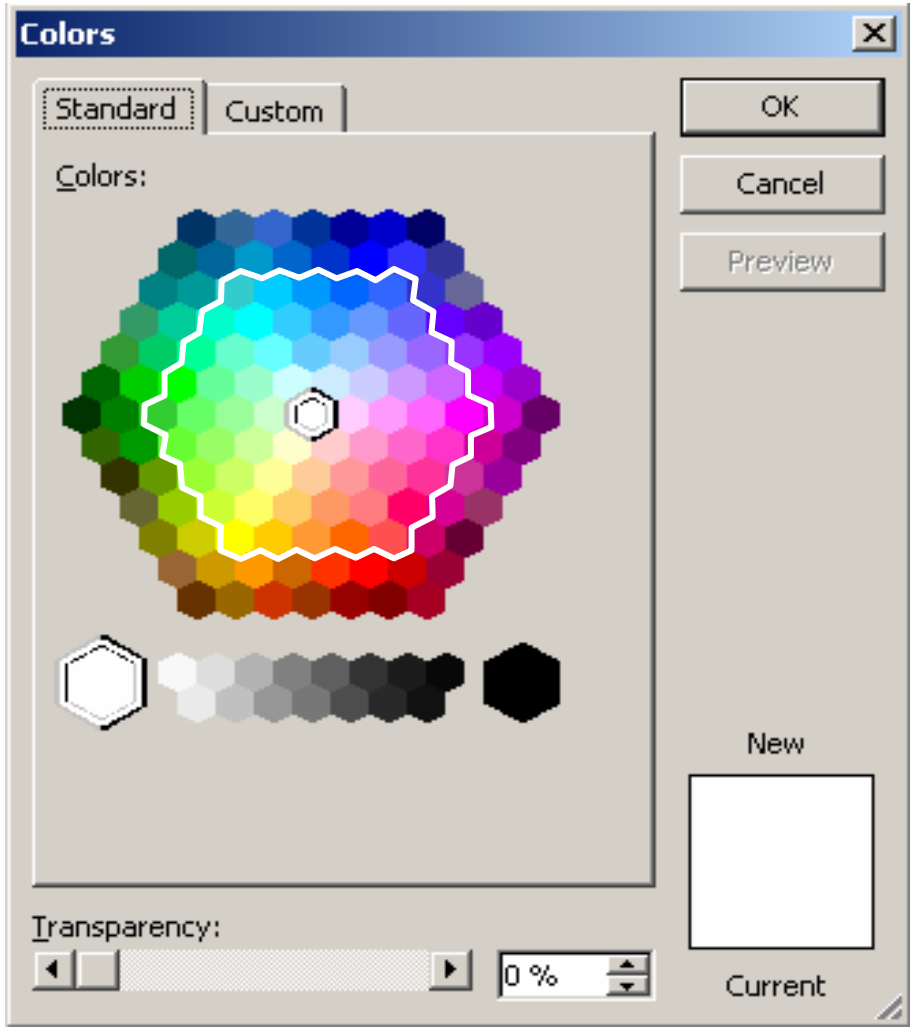
hexagon

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MS Office standard color dialog

- A modification of the hexagonal color layout
  - standard hexagonal color layout



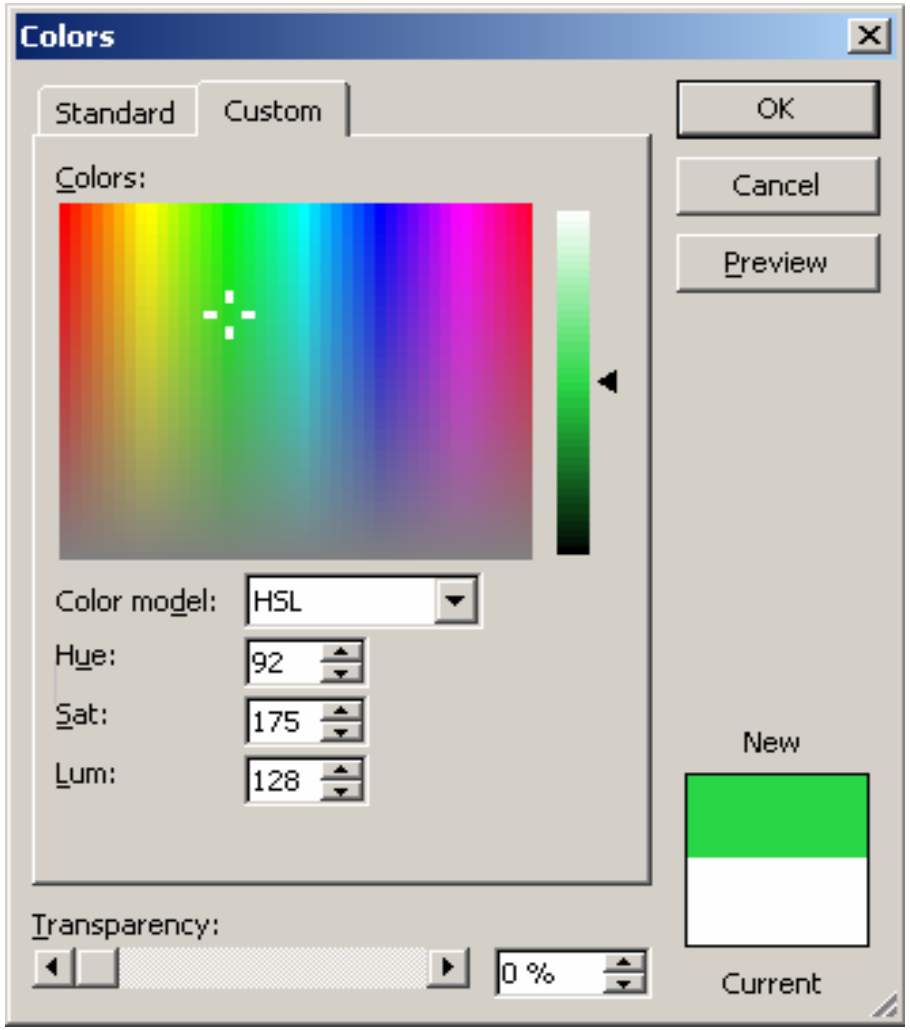


From http://www.devcomponents.com/dotnetbar/colorpicker.aspx

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MS Office custom color dialog

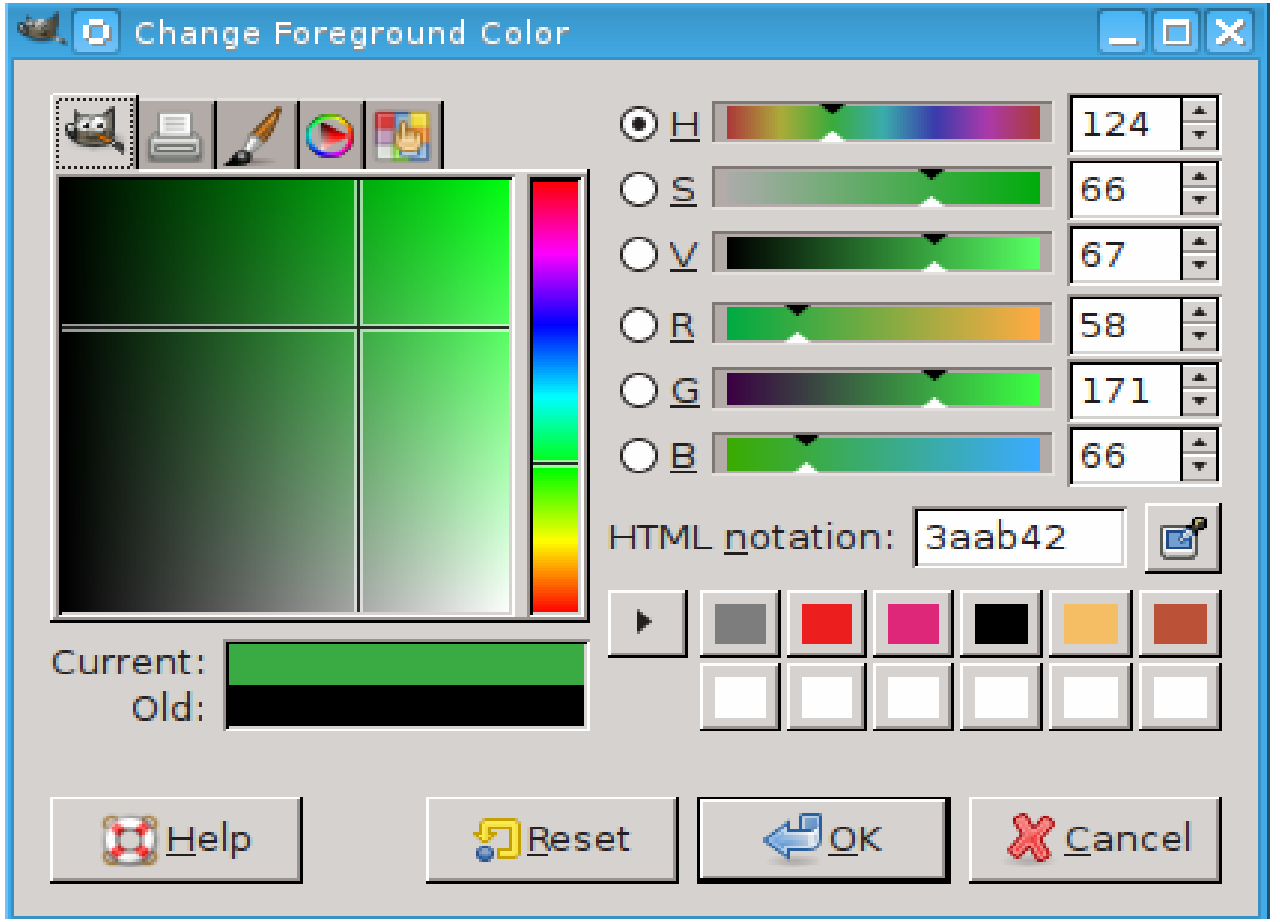
- Offers an HSL color layout
  - the rectangular color space always shows colors for lightness 128
- Use
  - select a hue/saturation combination
  - change lightness and see result in the “new” box



From http://www.devcomponents.com/dotnetbar/colorpicker.aspx

Gimp standard color dialog

- Offers an HSV color chooser
- Use
  - select the desired hue in the vertical bar
  - the rectangular region shows the value (x) and saturation (y) space for the given hue

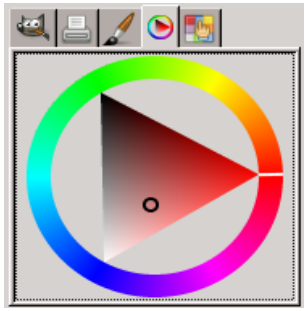


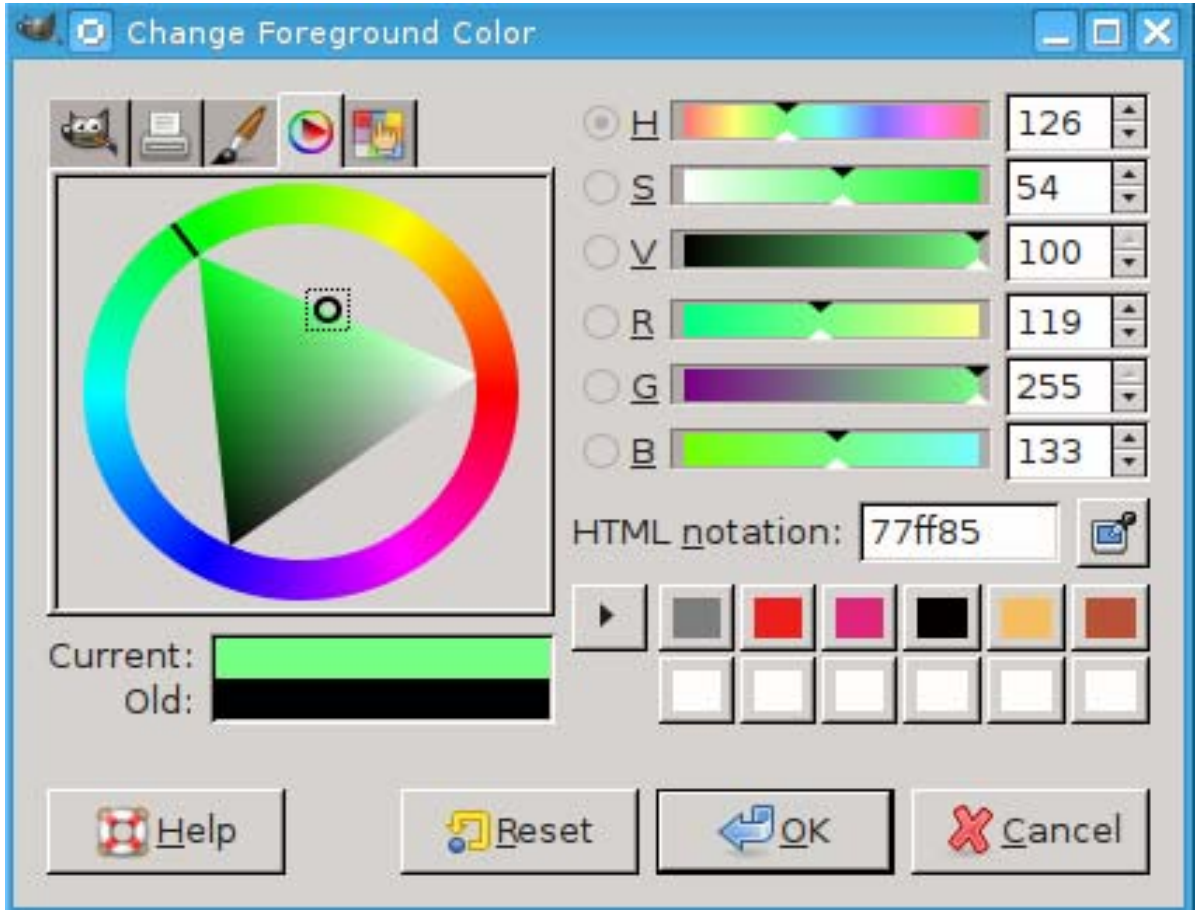
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Gimp triangle color dialog

- Offers an HSV color chooser
- Use
  - select the desired hue by clicking the circular ring

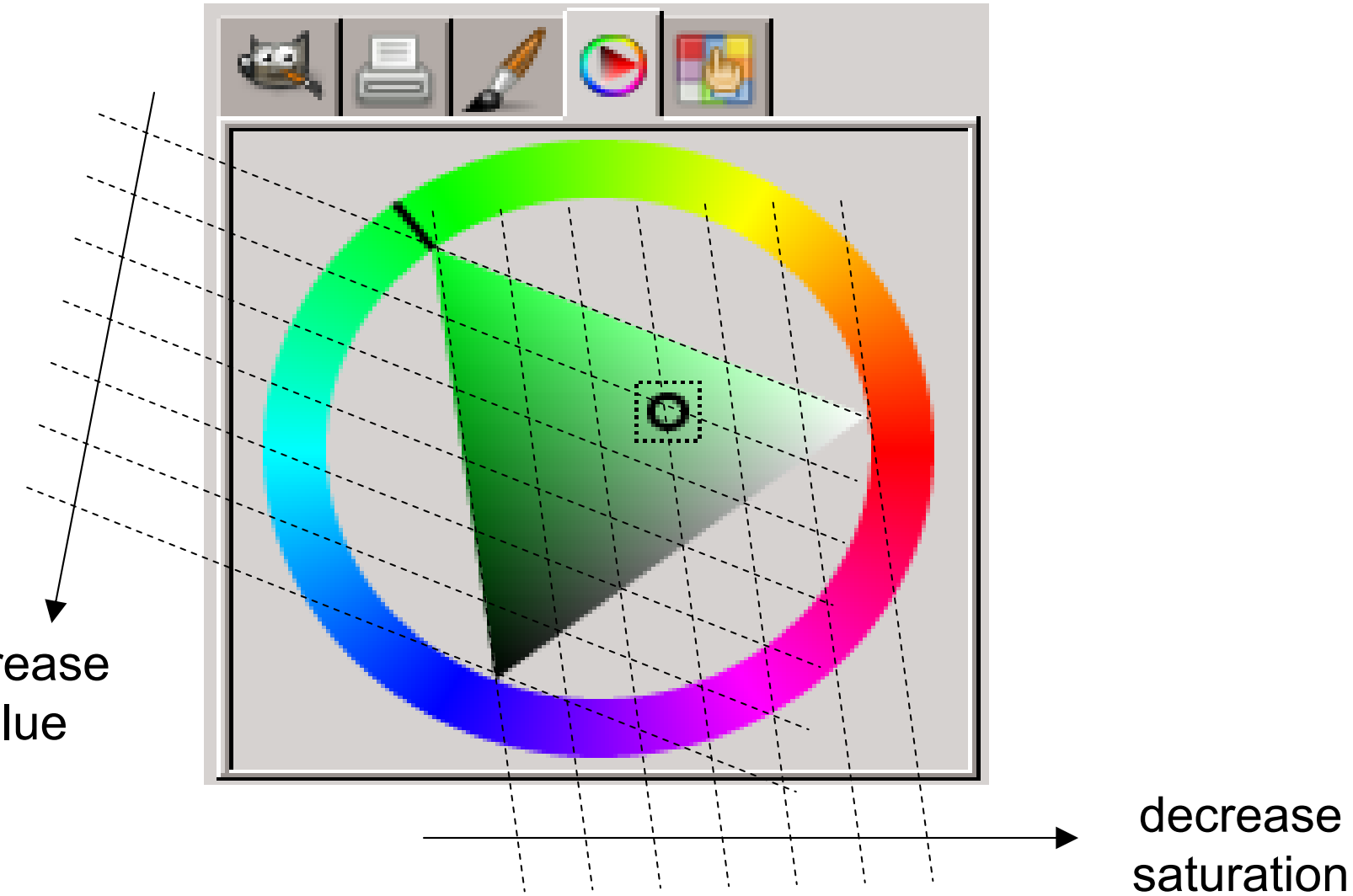






copyright ©2026 m.patrignani

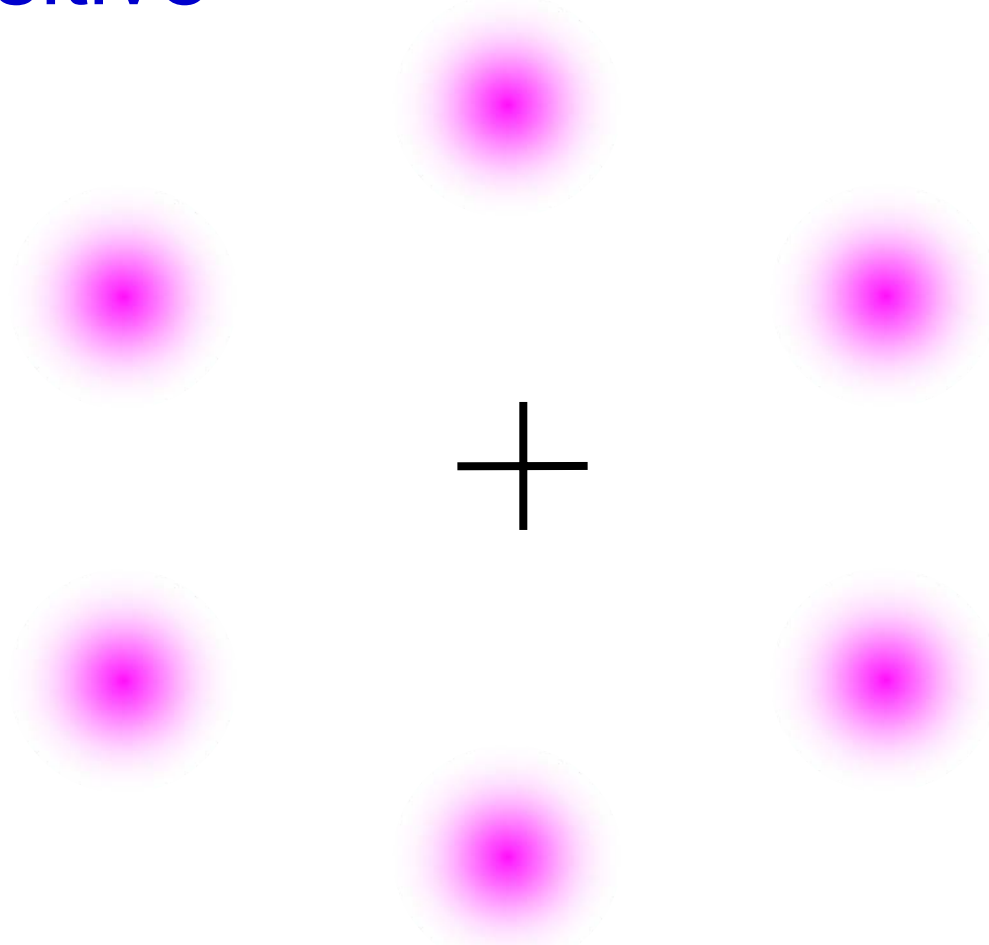
Gimp triangle color dialog



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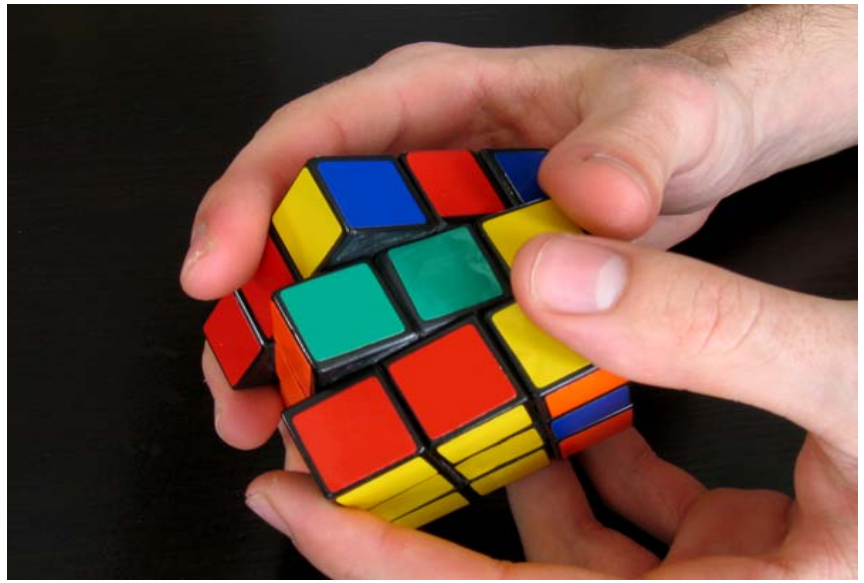
Exhaustion of cones

- Stressing your cones with colors tend to make them less sensitive



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Is color perception absolute?



- We recognize colors in different light conditions



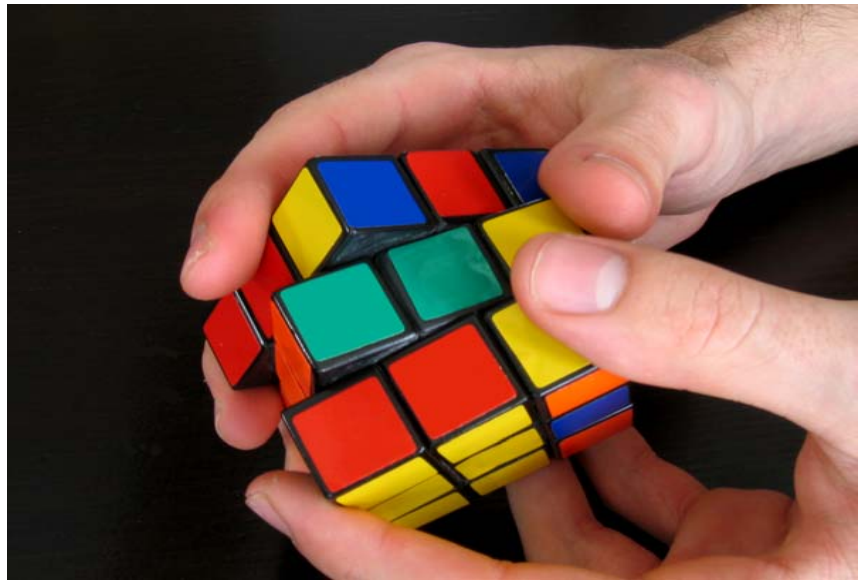
under a yellow light



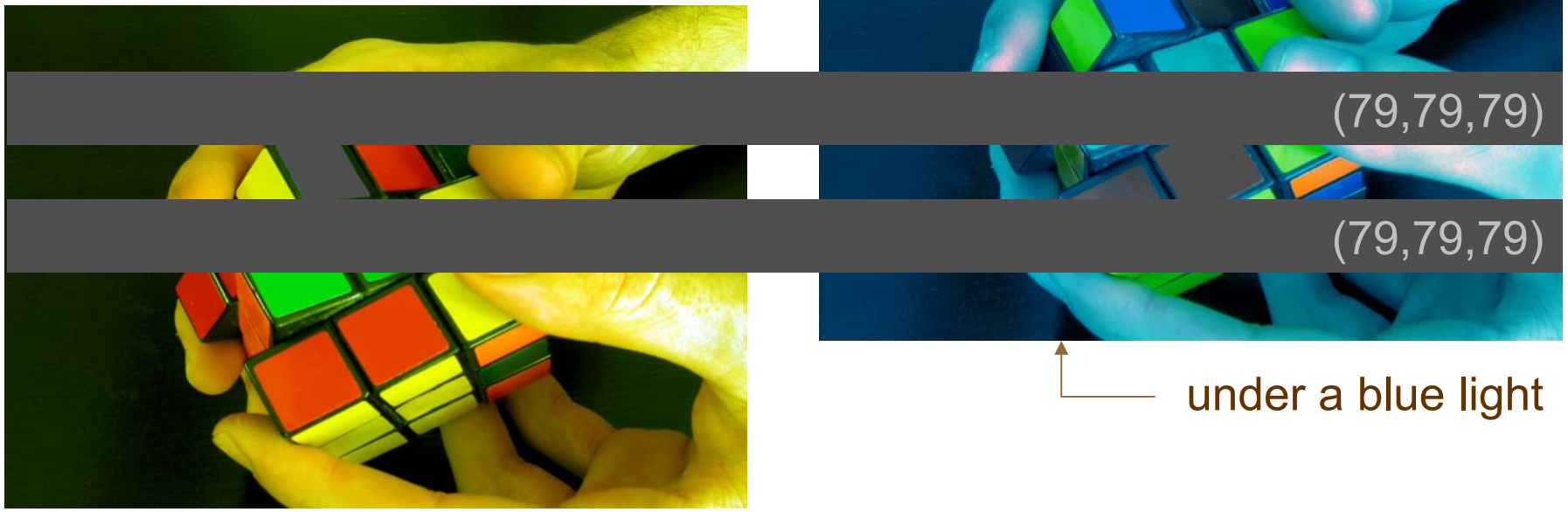
under a blue light

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Is color perception absolute?



- We recognize colors in different light conditions
  - color perception is also based on relative, not absolute, judgements



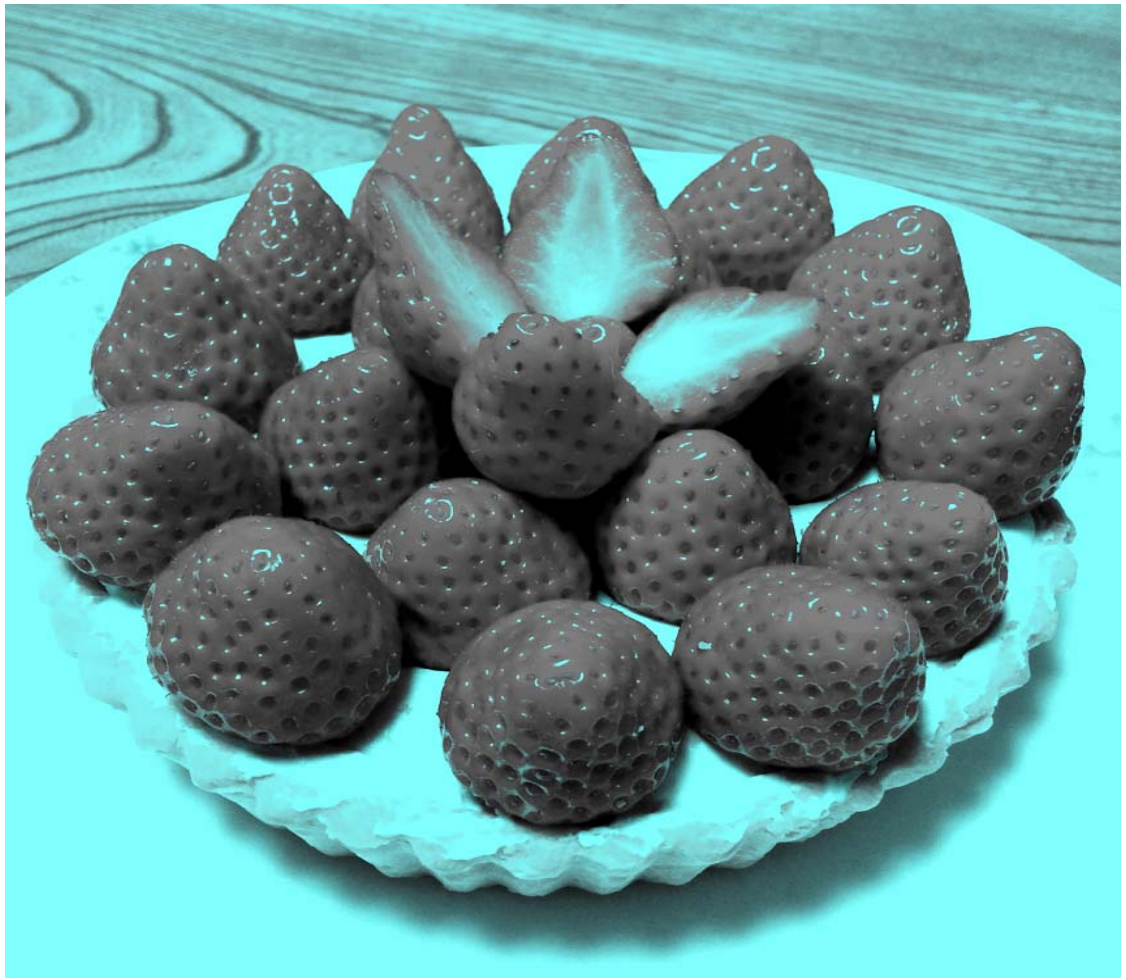
under a yellow light

under a blue light

 copyright ©2026 m.patrignani

Is color perception absolute?

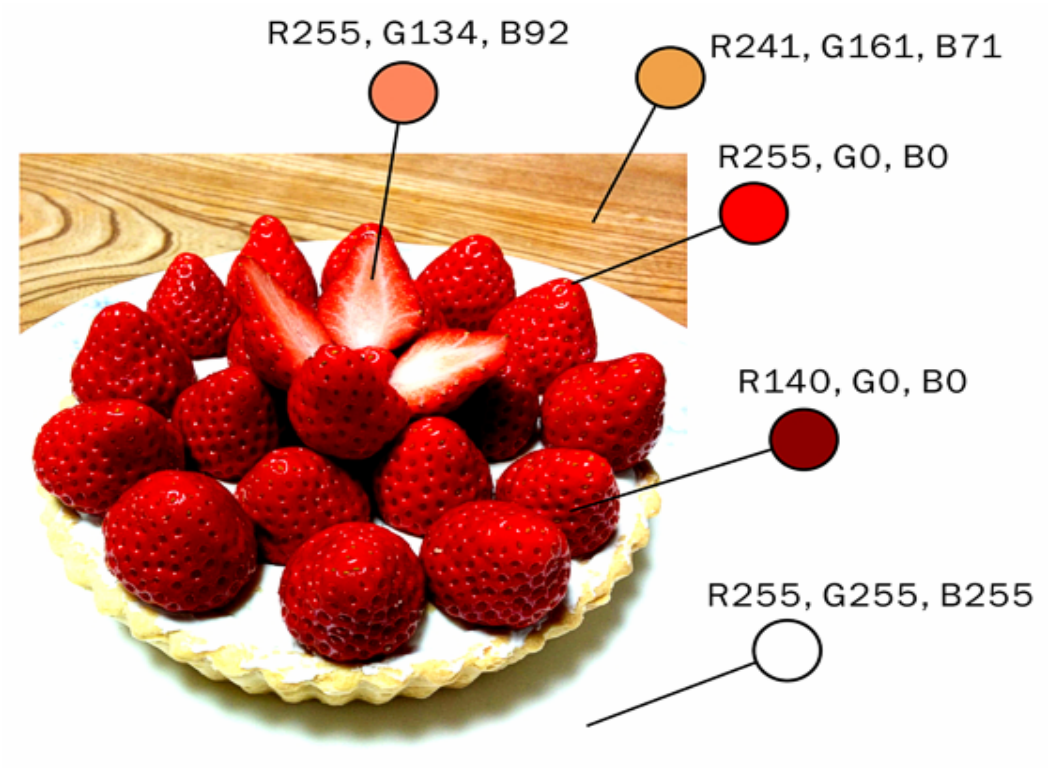
- Red strawberry illusion
  - strawberries appear to be reddish although the pixels are cyan or gray
  - from [Kitaoka 16]



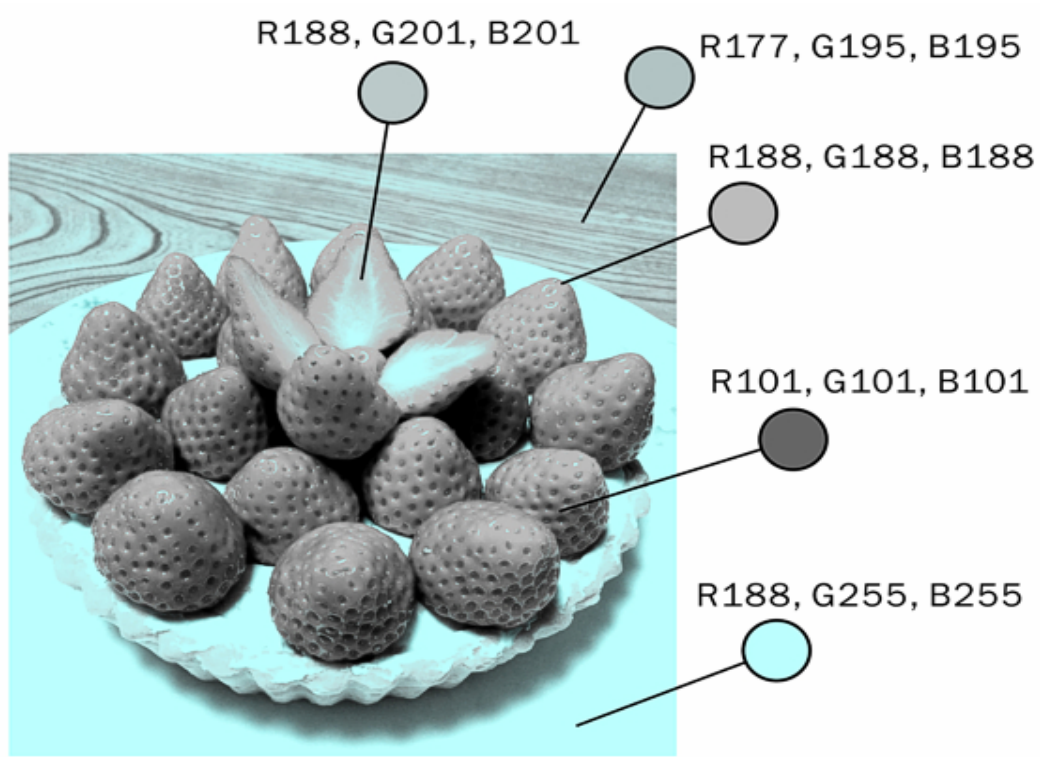
 copyright ©2026 m.patrignani

Is color perception absolute?

- Picture from [Kitaoka 16]



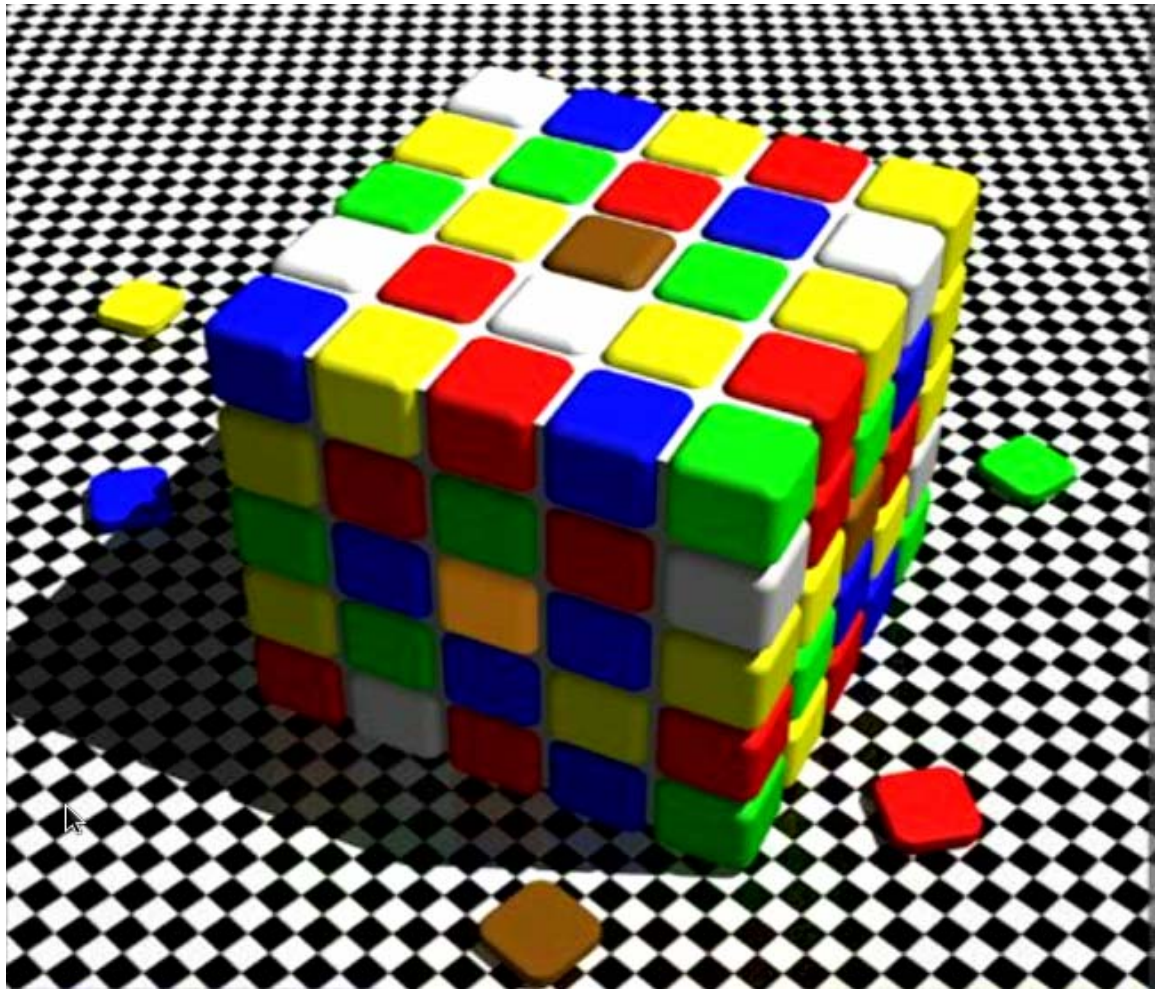
original image



altered image

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Is color perception absolute?

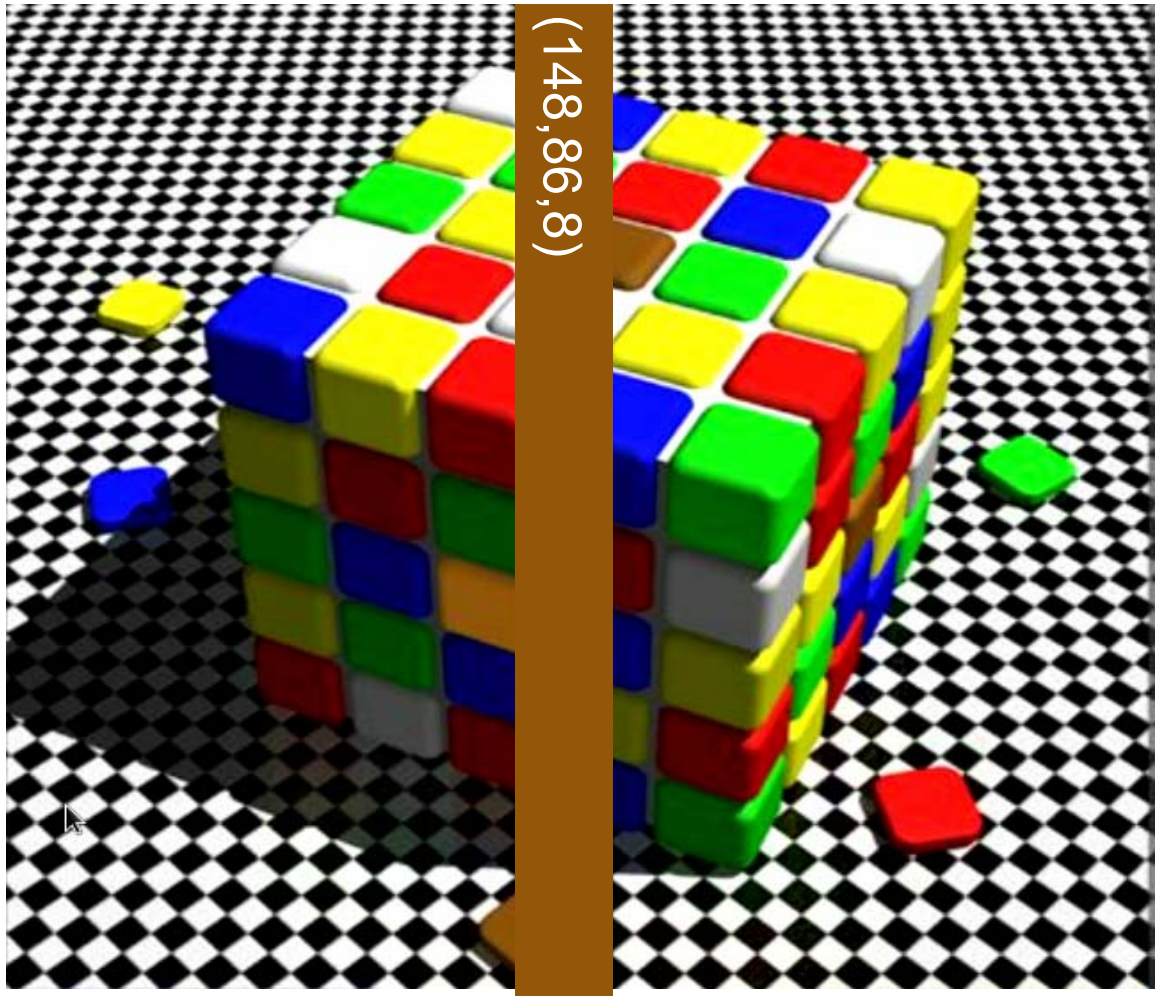


- Color perception is biased by shadows and environment conditions

- From [Lotto ‘14]

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Is color perception absolute?



- Color perception is biased by shadows and environment conditions

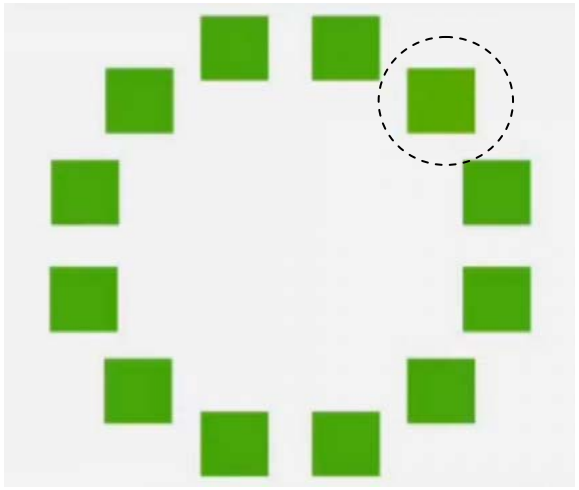
- From [Lotto ‘14]

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Is color perception “cultural”?

- Egiptian are the only ancient culture to know the existence of blue
  - all ancient languages and poems (Odyssey, early Bibles, Koran, Chinese classics..) do not mention blue
- Contemporary Namibians can not spot a blue square among green squares
  - but they immediately spot a different hue of green in the picture on the right: can you?



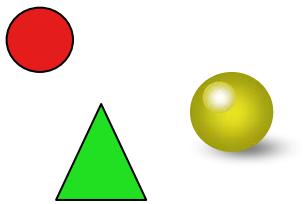
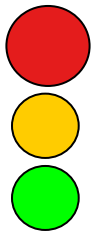
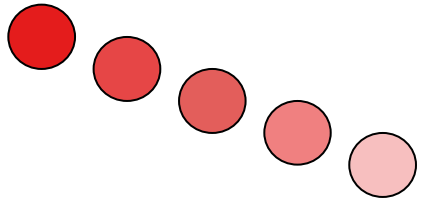
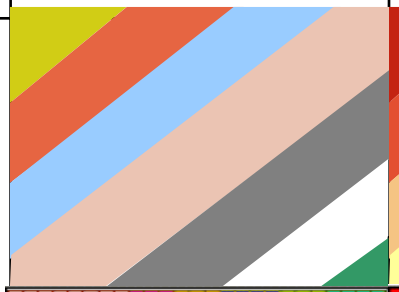
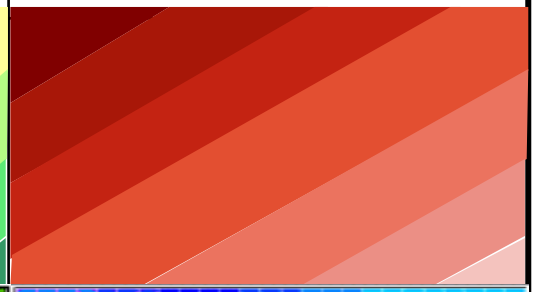
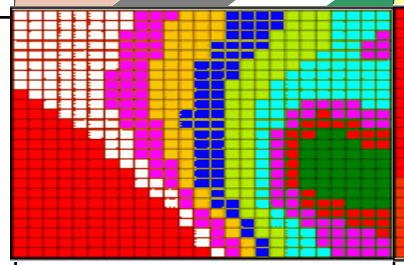
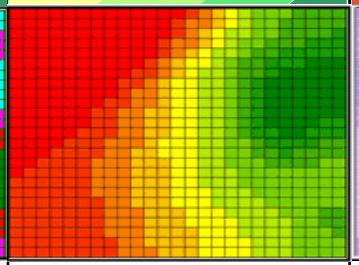
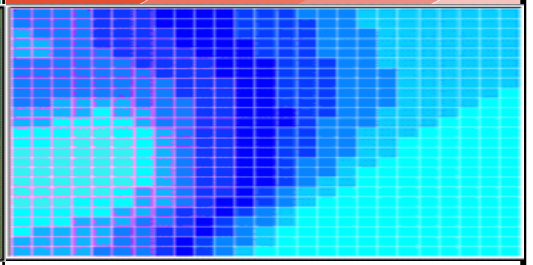


From www.businessinsider.com/what-is-blue-and-how-do-we-see-color-2015-2

Use of Color

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What can we do with colors?

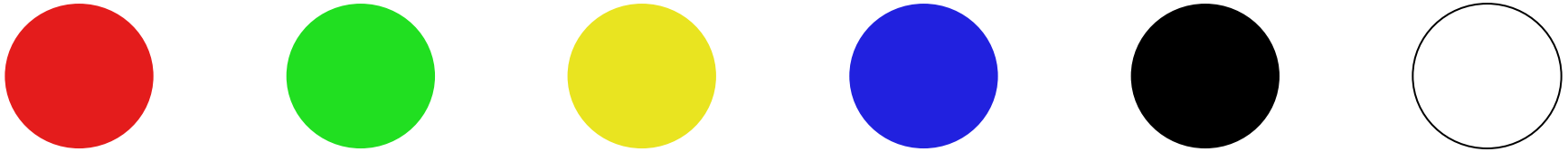
what we color	what values we encode: nominal	ordinal	quantitative
objects (icons, shapes, 3D items, etc)			
regions of the plane			
points (pixels)			

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How to use chromatic coding

- When colors are used for coding nominal information we need colors that have an agreed upon name
  - use these six [Berlin & Kay 69]

red green yellow blue black white



  - use these second six only if needed

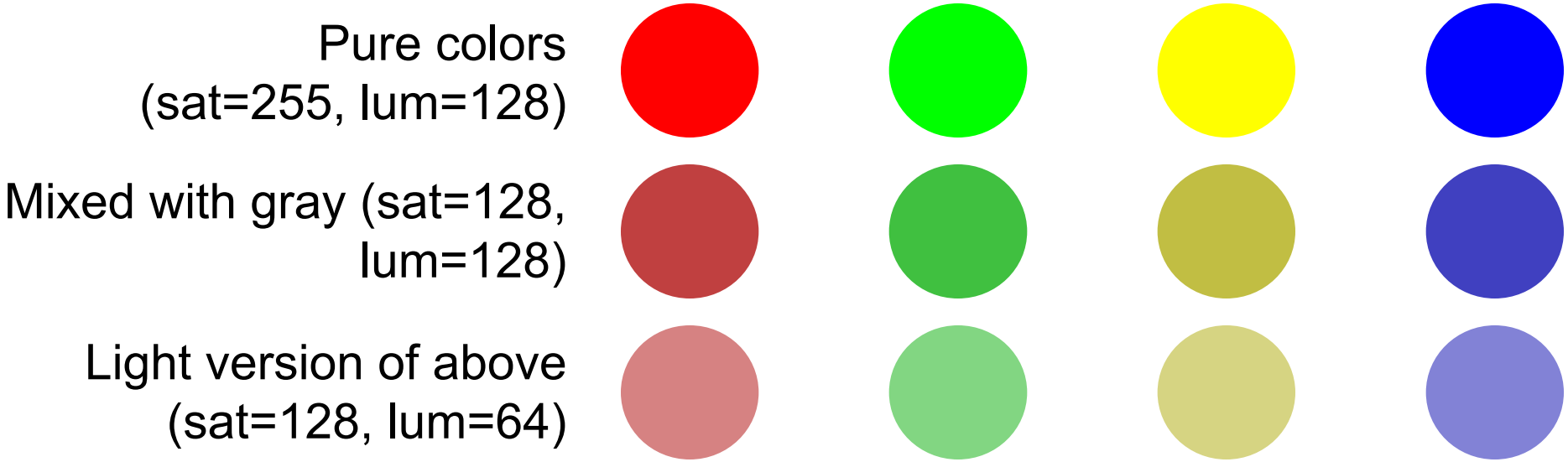
pink cyan gray orange brown purple



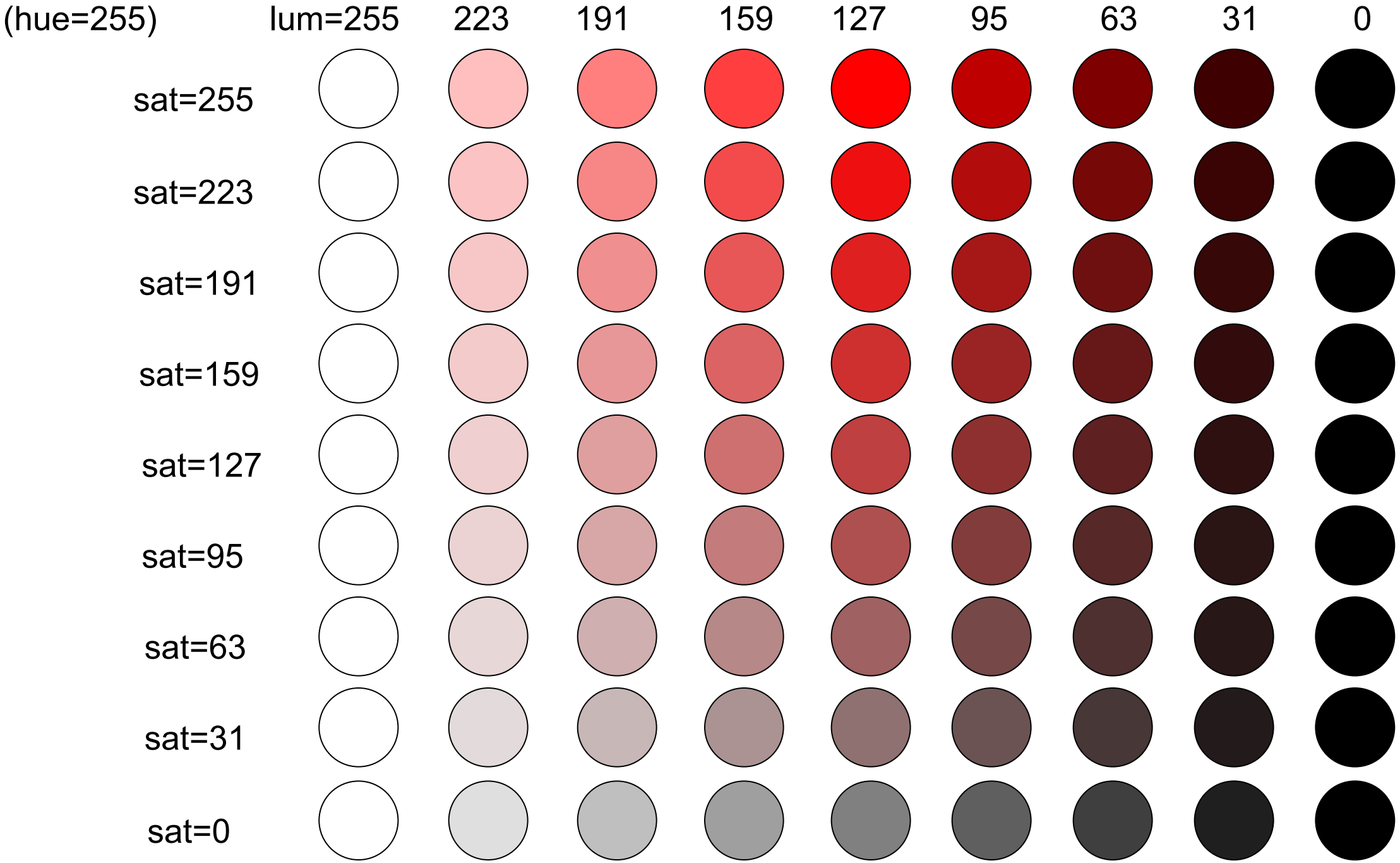
copyright ©2026 m.patrignani

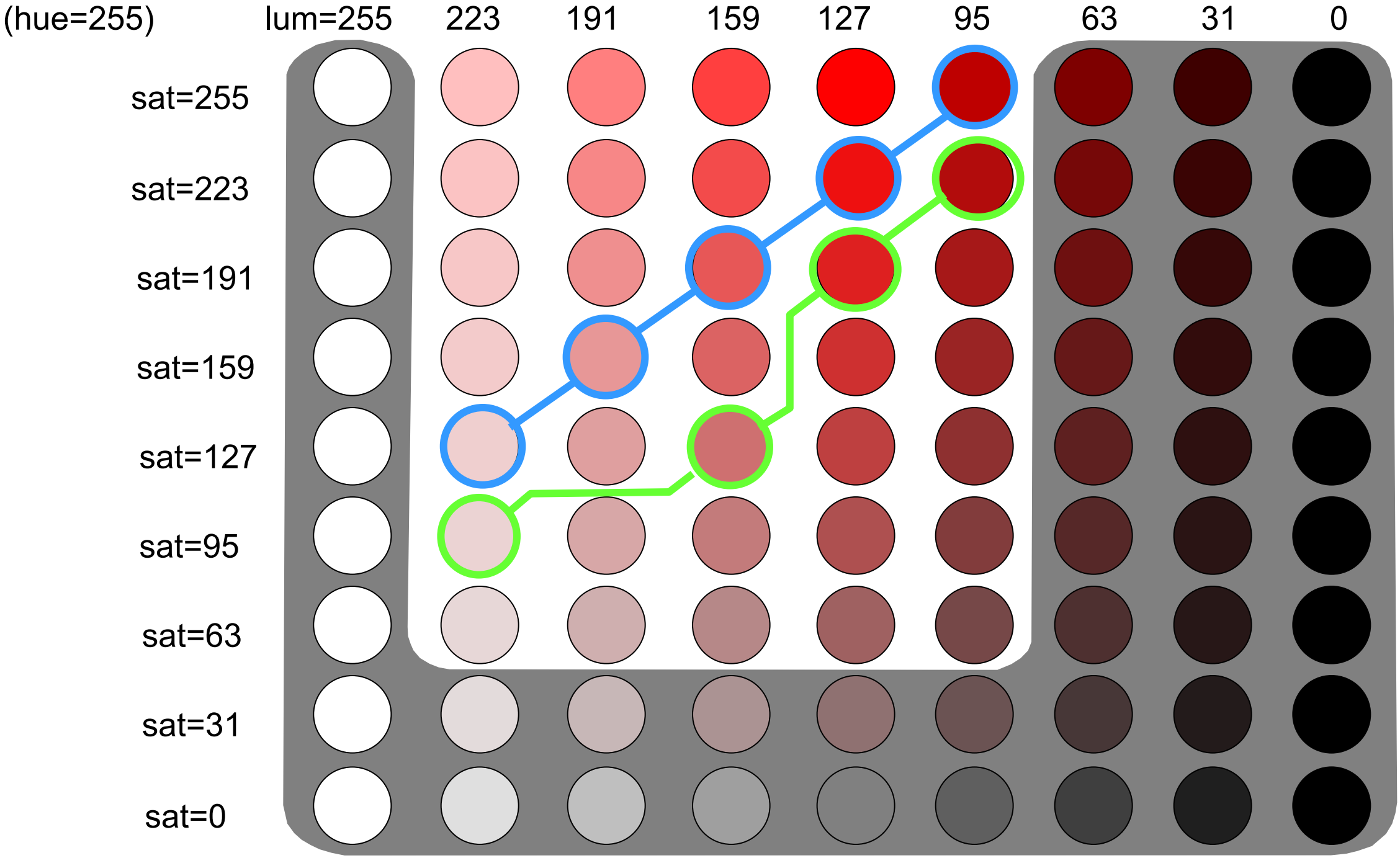
How to find color families

- When colors are used to code ordinal variables it is convenient to generate color families
  - start from a set of colors (one for each variable) different by hue
  - vary their saturation and/or luminance



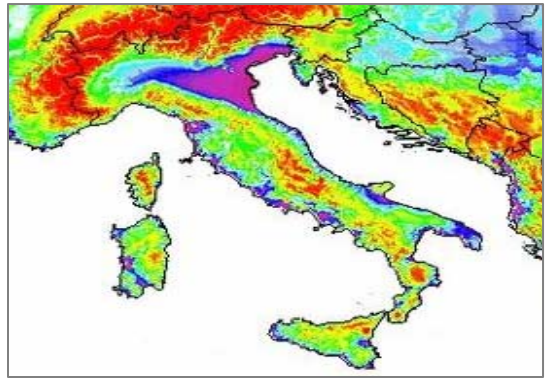
 copyright ©2026 m.patrignani


(hue=255)
lum=255
223
191
159
127
95
63
31
0
sat=255
sat=223
sat=191
sat=159
sat=127
sat=95
sat=63
sat=31
sat=0


(hue=255)
lum=255
223
191
159
127
95
63
31
0
sat=255
sat=223
sat=191
sat=159
sat=127
sat=95
sat=63
sat=31
sat=0

Colormaps

- Colormaps are needed whenever quantitative variables need to be mapped to colors
- Several types are known





rainbow colormaps



grayscale colormaps



isoluminant colormaps



sequential colormaps



diverging colormaps

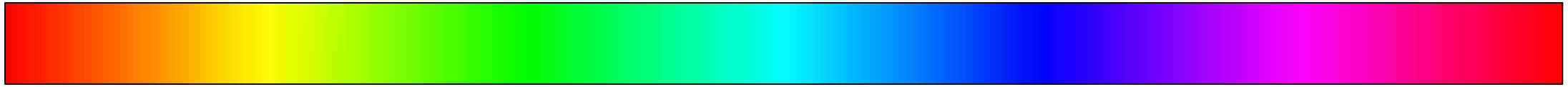
copyright ©2026 m.patrignani

Rainbows colormaps

- The most (in)famous type of colormap
  - changes the hue, maintaining constant lightness and constant saturation
- Different ways of obtaining them
  - one can be obtained from the color spectrum

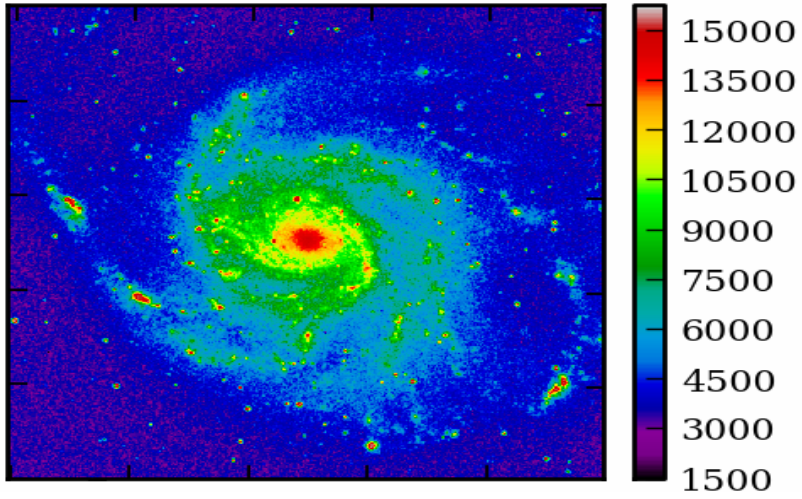


  - one can be obtained from an HSL section

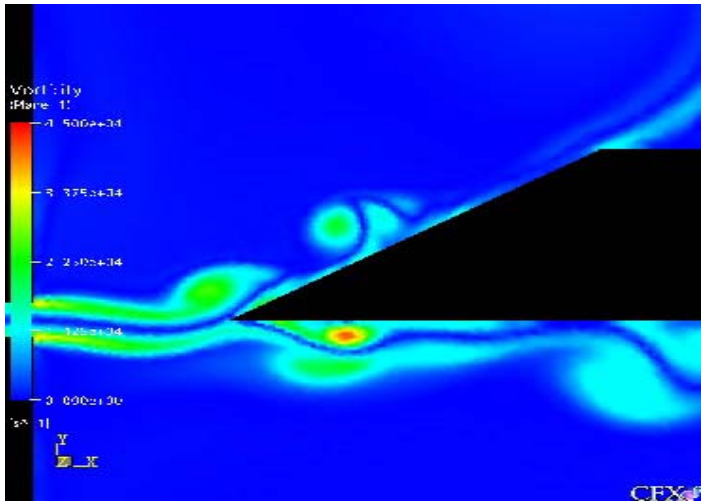


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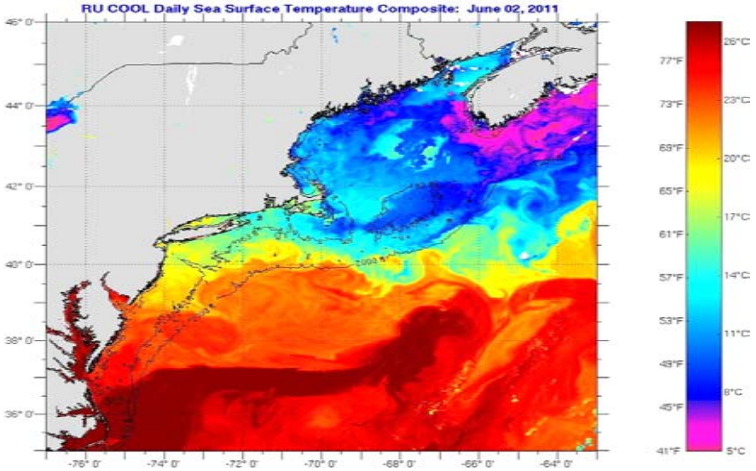
Rainbow colormaps are popular



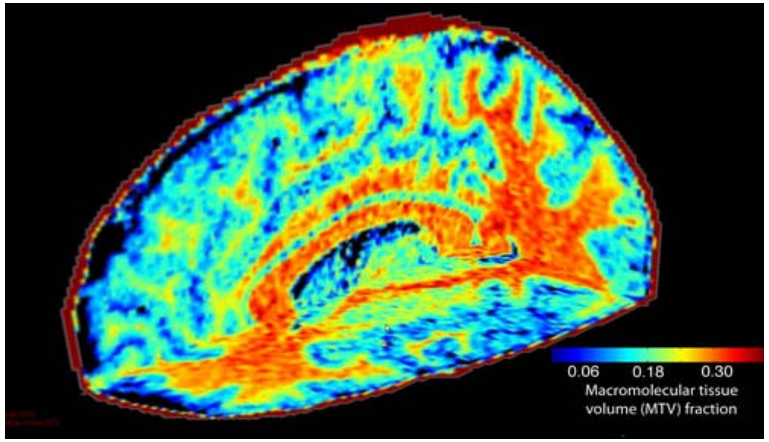
Astrophysics images



Turbulent flows simulations



Weather forecasts

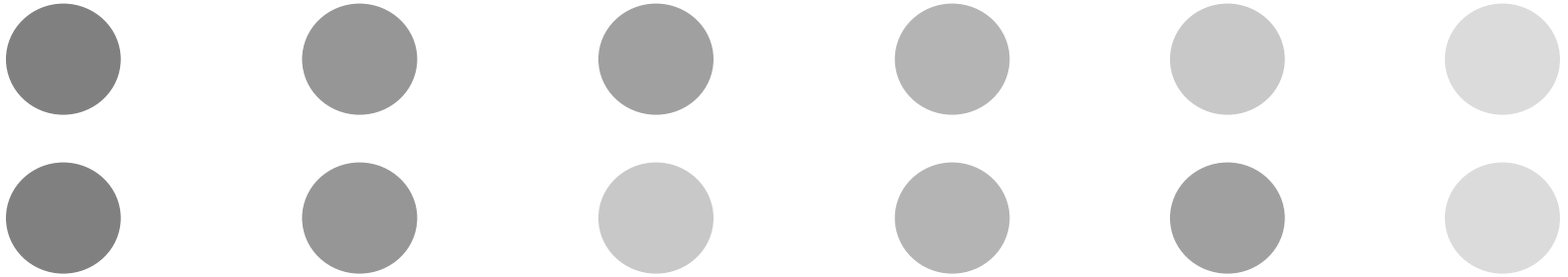


Magnetic resonance imaging

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Known problems of rainbow colormaps

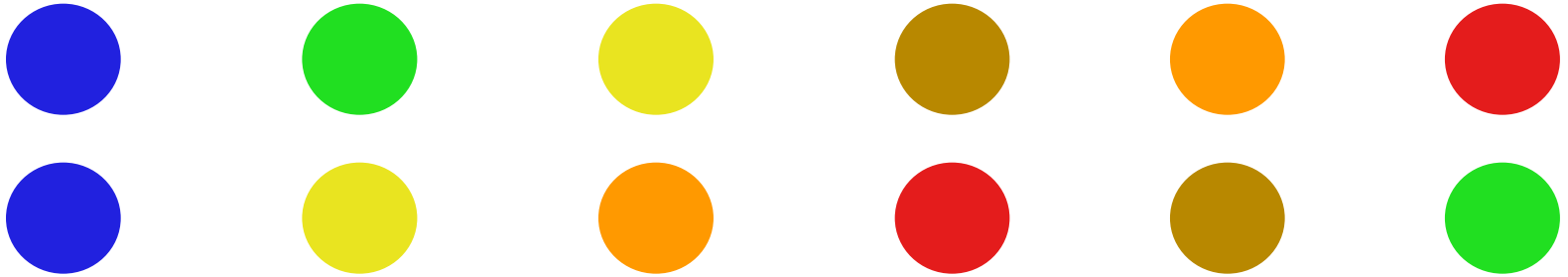
- Perceptual ordering is unclear
  - you can easily place in order grayscale objects based on perception



Okay

Wrong

  - it is difficult to do the same with colors



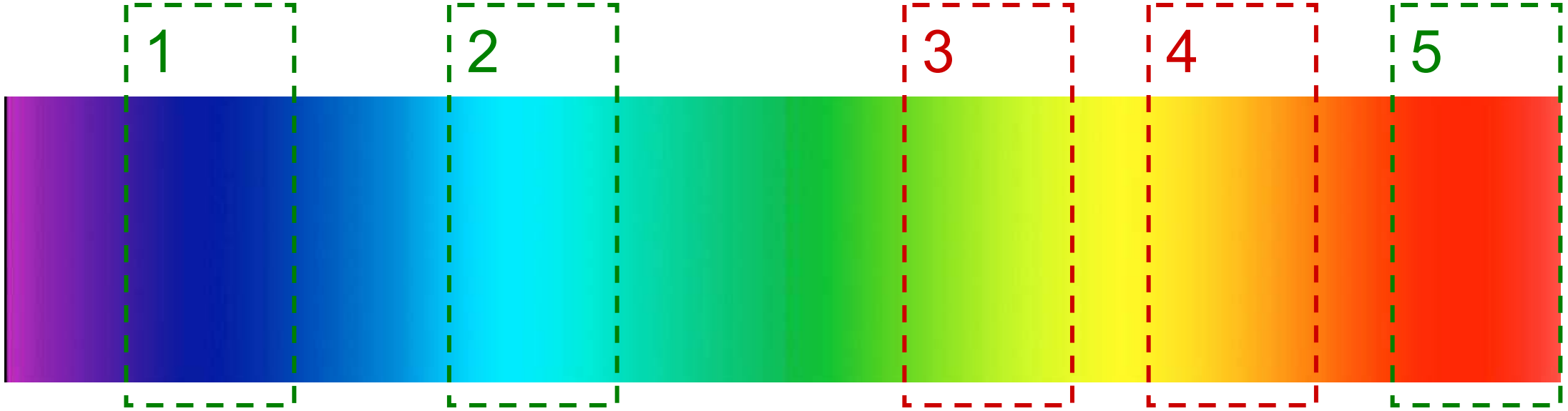
??

??

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Known problems of rainbow colormaps

- Rainbow colormaps are not perceptually uniform
  - intervals 1, 2, 5 seem monocromous
  - intervals 3 and 4 seem to contain color transitions
    - from https://courses.washington.edu/engageuw/why-you-should-dump-the-rainbow/



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Known problems of rainbow colormaps

- Rainbow colormaps have different perceived luminance
  - HSV colormap
    - lightness = 0.5, saturation = 1



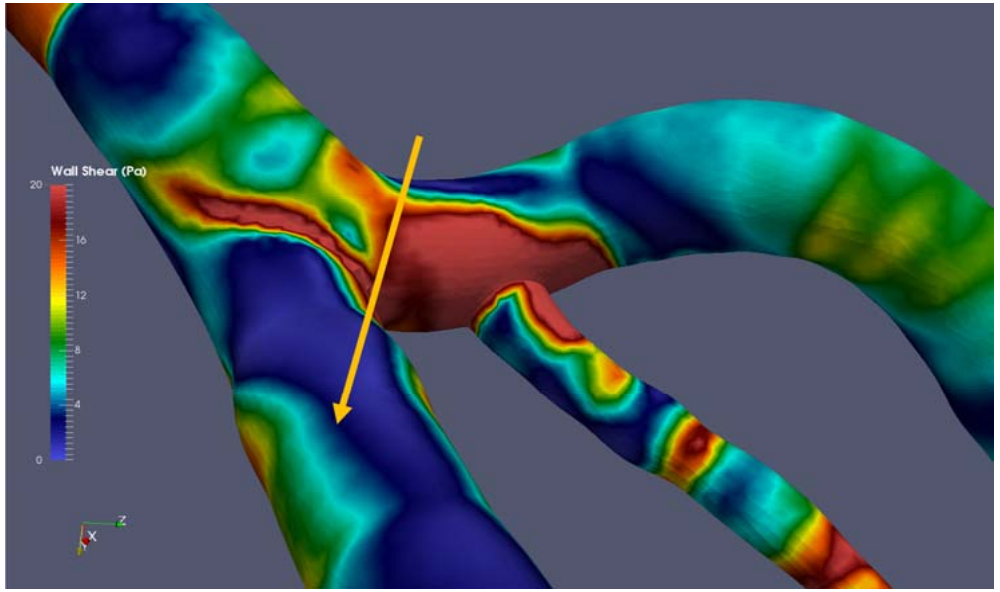
  - perceived luminance
    - yellow, green, and cyan are perceived significantly brighter than other colors



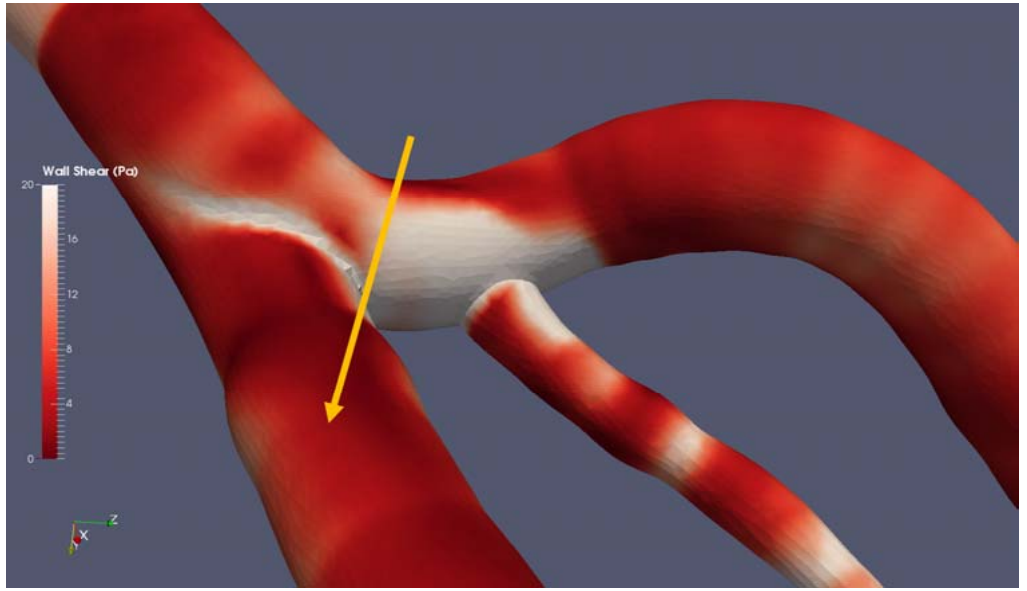
 copyright ©2026 m.patrignani

Artifacts of rainbow colormaps

- Smooth changes result in dramatic color transitions
  - example from https://figureoneblog.wordpress.com/2014/03/12/misleading-with-pictures-the-pitfalls-of-data-visualization/



rainbow colormap

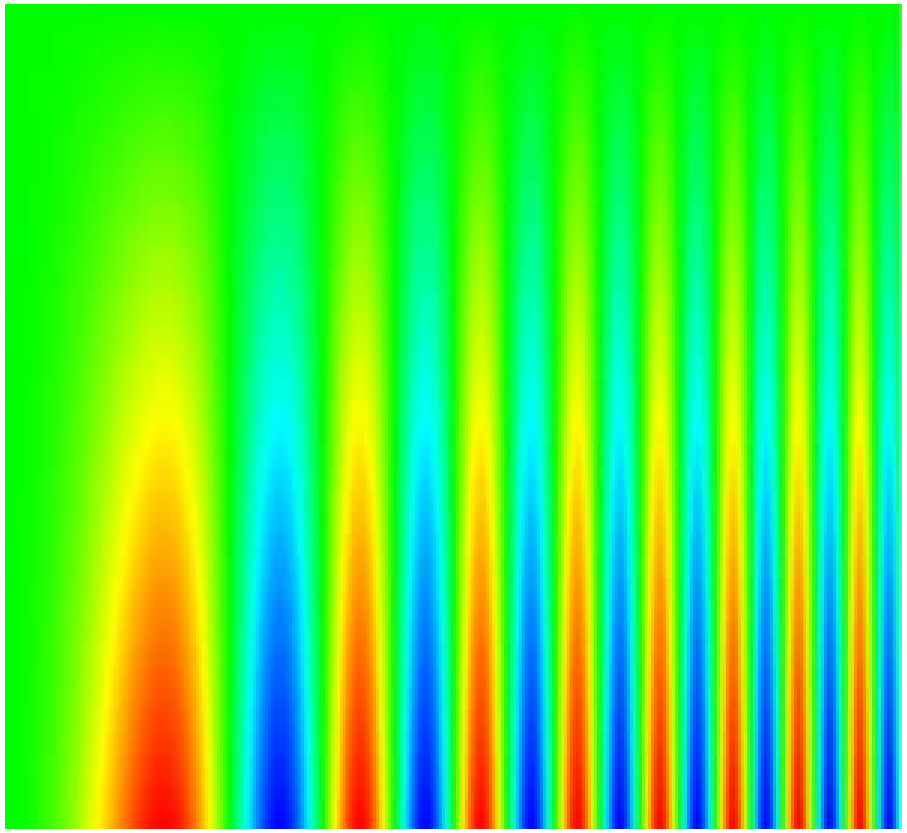


sequential colormap

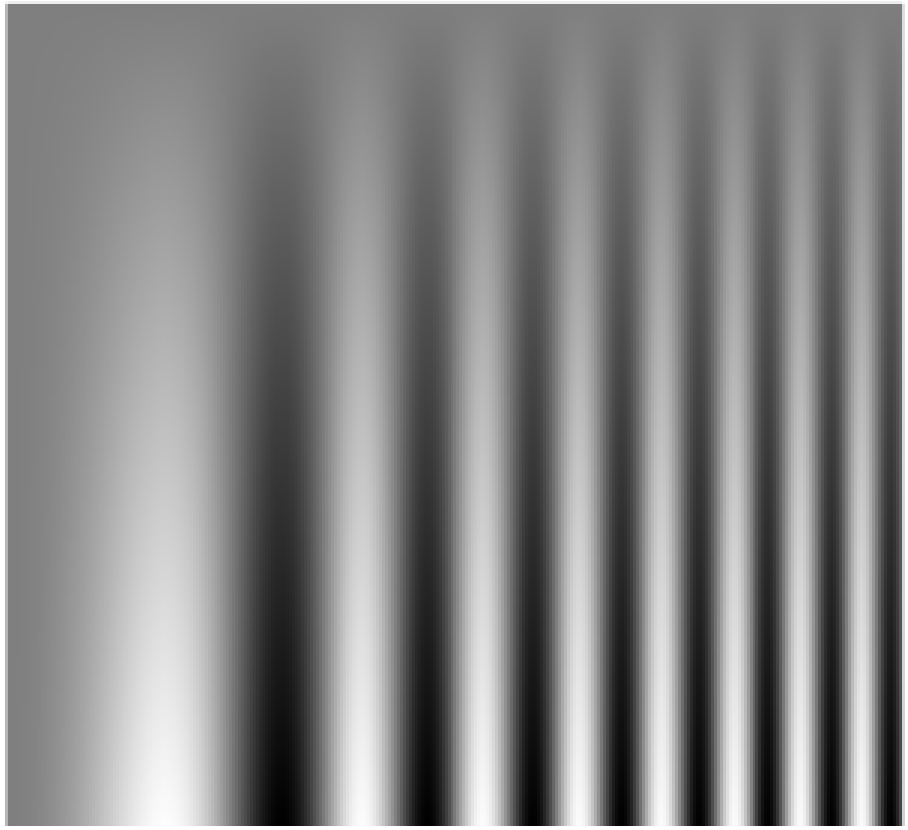
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Rainbow colormaps hide information

- Variation is hidden in low contrast regions
  - example from [Moreland 09]



with a rainbow colormap



with a grayscale colormap

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Grayscale colormaps

- Grayscale colormap
  - instead of changing the hue, only changes the luminance



- Known problems
  - dull and boring
  - interferes with surface shading
    - unsuitable for 3D rendering
  - prone to of all kinds of luminance illusions
    - crispening effects, Chevreul illusion, induced grating …

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Isoluminant colormaps

- Maintain a constant (perceptual) luminance
  - this implies a less effective visualization, though
- Compatible with 3D rendering and shading
- Often look dull and ugly



  - Isoluminant green to red
    - popular because use a pair of opponent colors
    - unclear for color-blind people

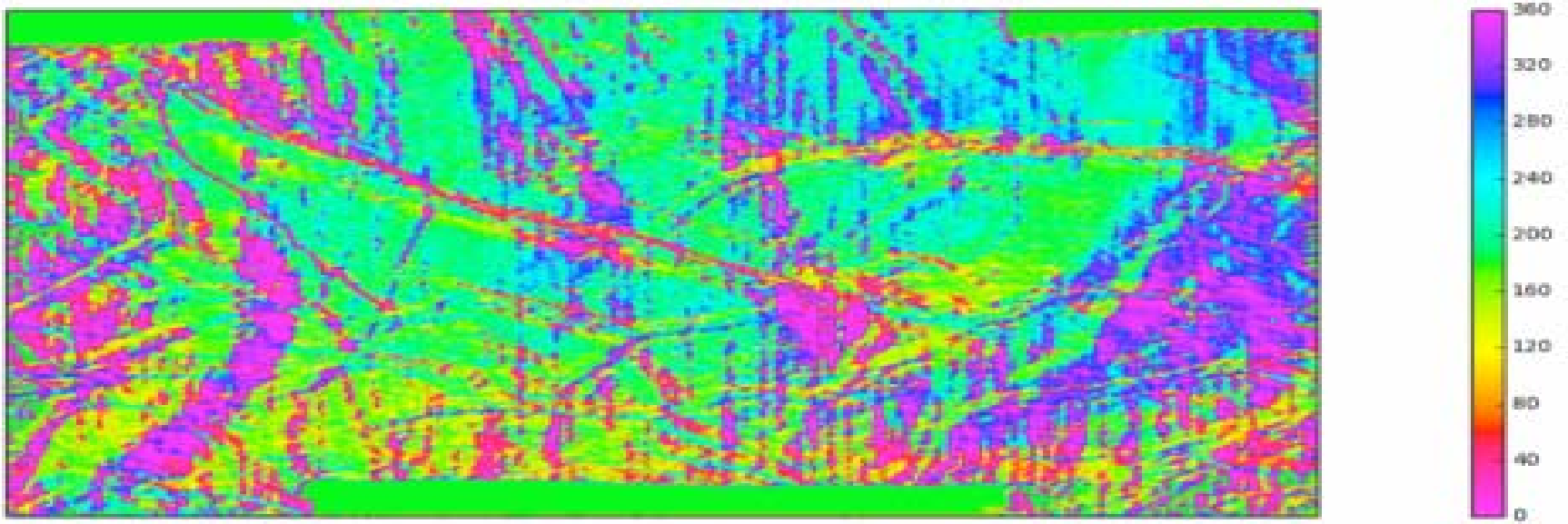


  - Isoluminant cyan to mauve
    - mauve is pale purple (224, 176, 255)
    - easier to see by color-blind people

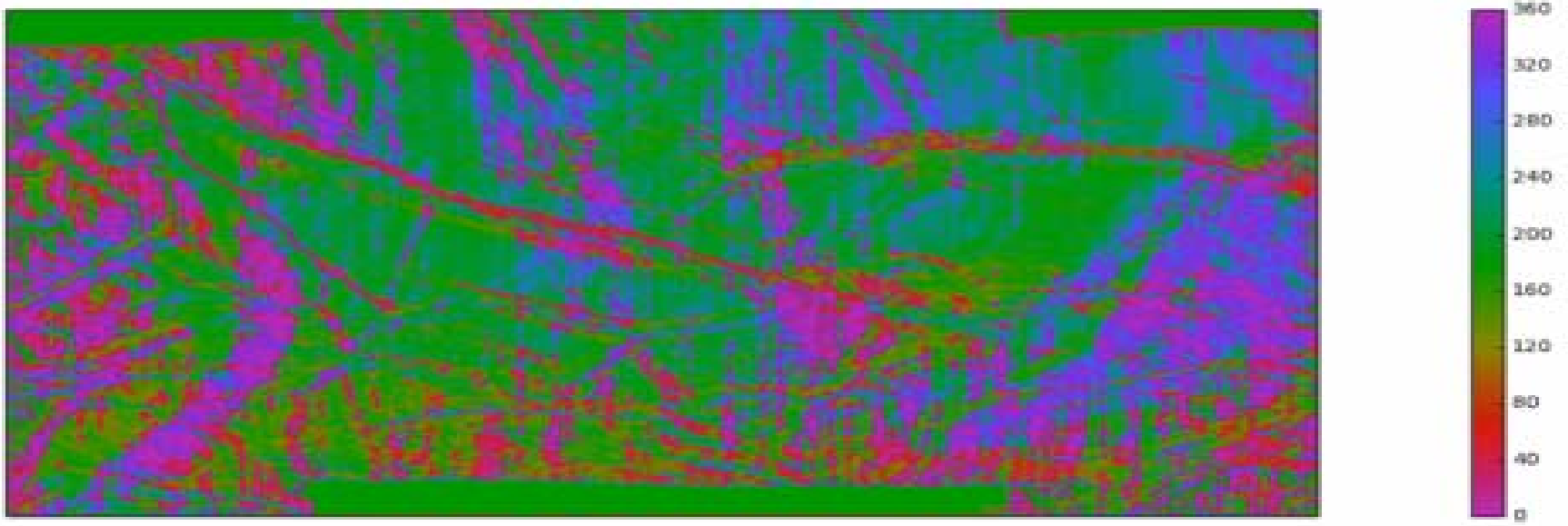
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Isoluminant colormap example

conventional
HSV colormap



isoluminant
colormap



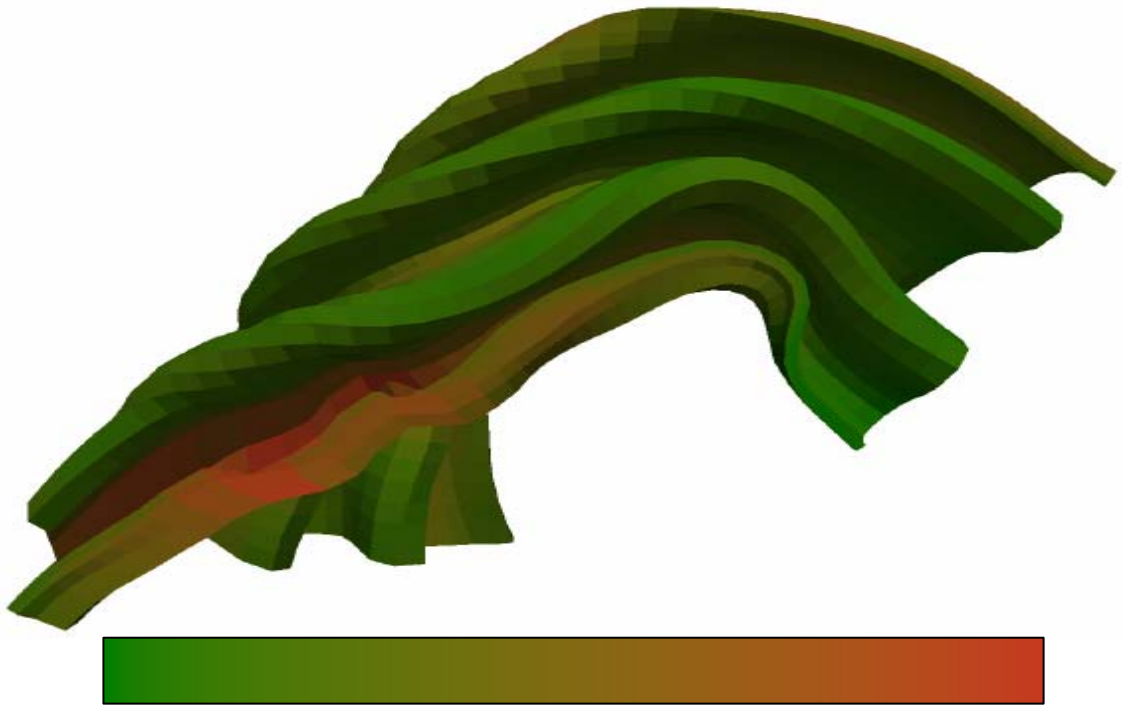
From https://mycarta.wordpress.com/2015/06/17/reinventing-the-color-wheel-part-2/

Isoluminant colormap example

- Minimize shading interference in 3D rendering
  - example from [Moreland 09]



Grayscale colormap

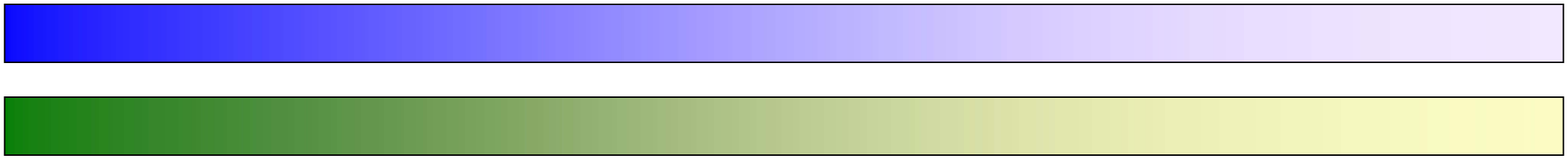


Isoluminant green to red colormap

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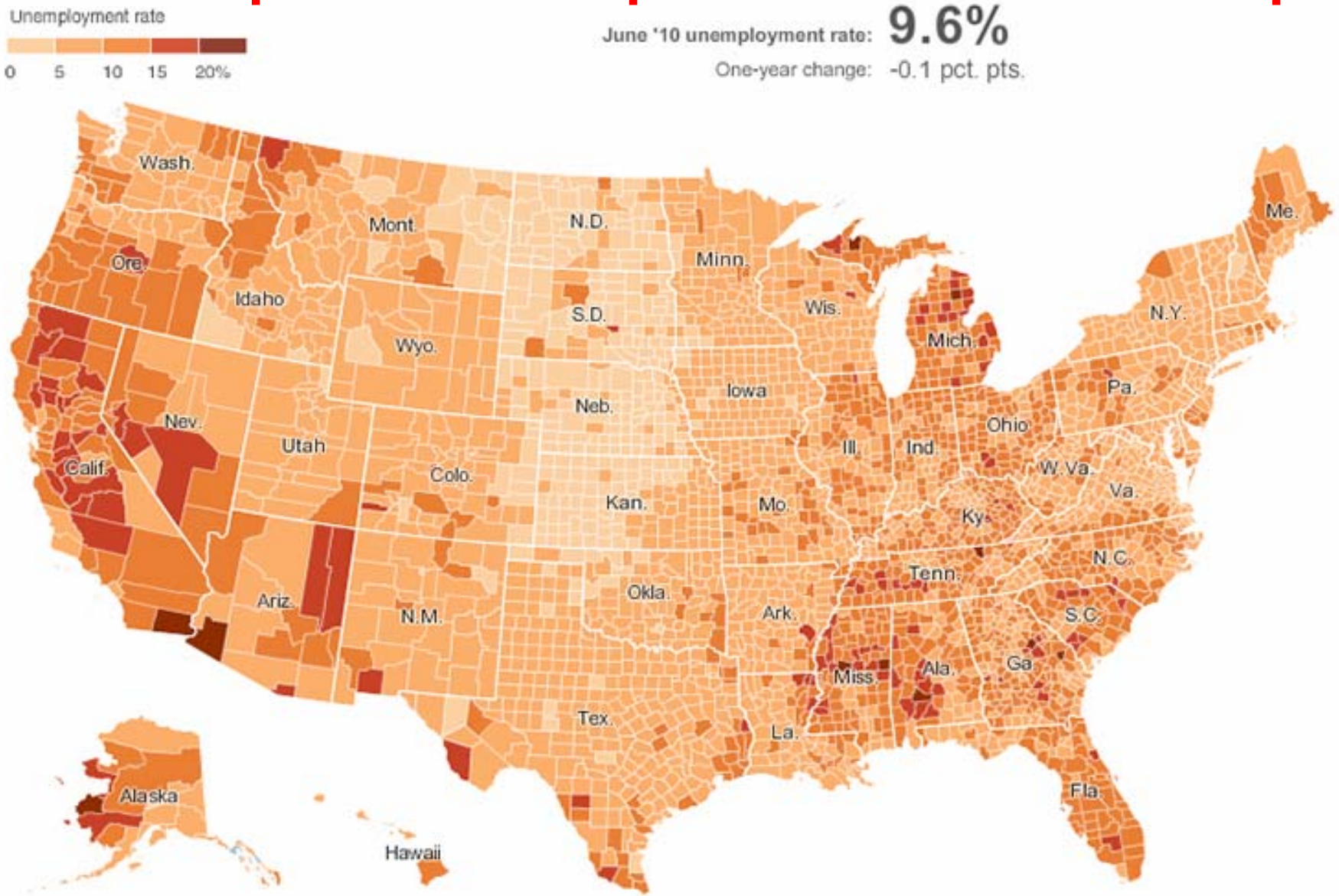
Sequential colormaps

- Also known as “ordinal” or “saturation” colormaps
- Nearly monocromatic
- Start from a heavily saturated color
- Terminates in a color close to white
- Luminance is increased as saturation is decreased



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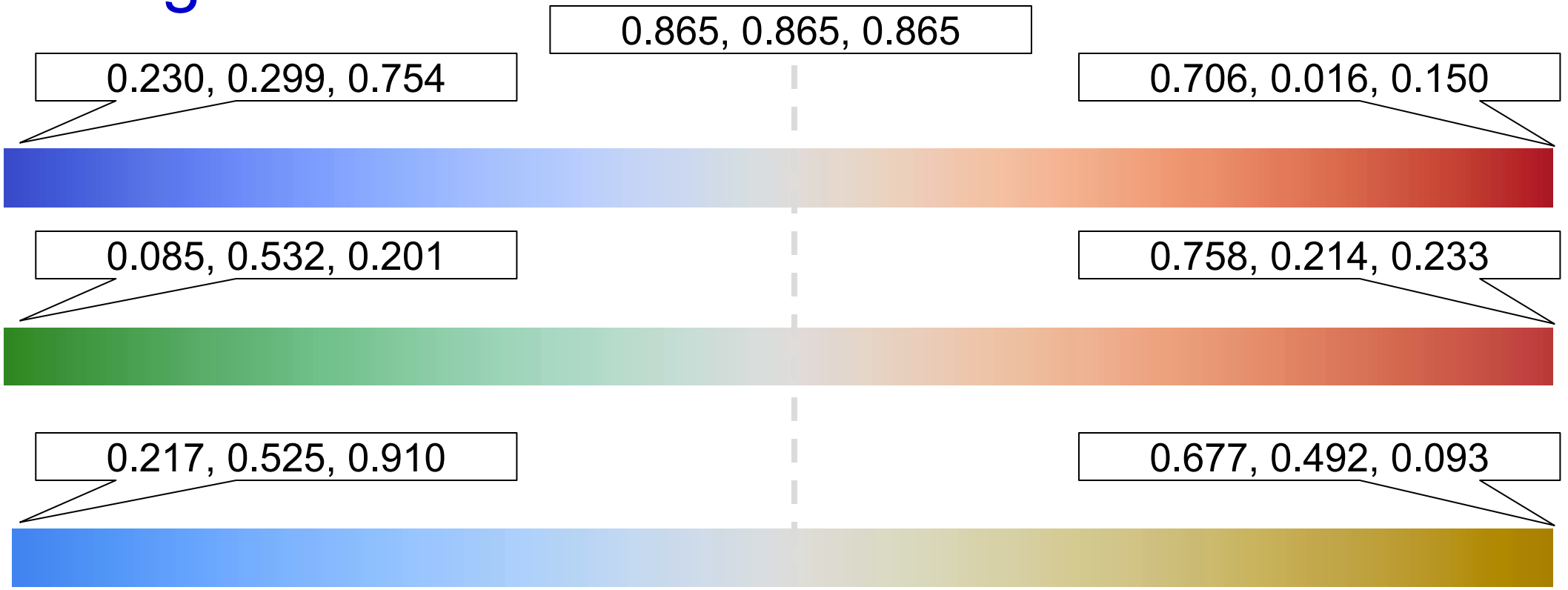
Example of sequential colormap



From http://www.joshuastevens.net/cartography/make-a-bivariate-choropleth-map/

Diverging colormaps

- Also called “bipolar” or “double-ended”
- Transition from one color to another by passing through an unsaturated color



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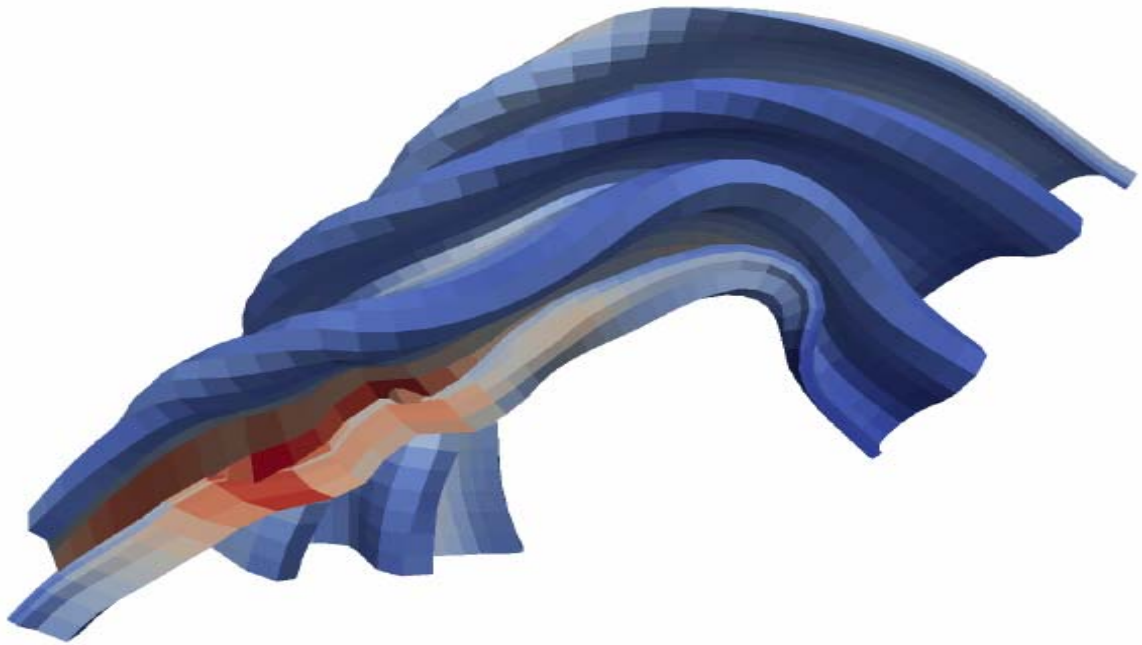
Diverging colormap example

- Shading interference
  - example from [Moreland 09]





Isoluminant green to red colormap



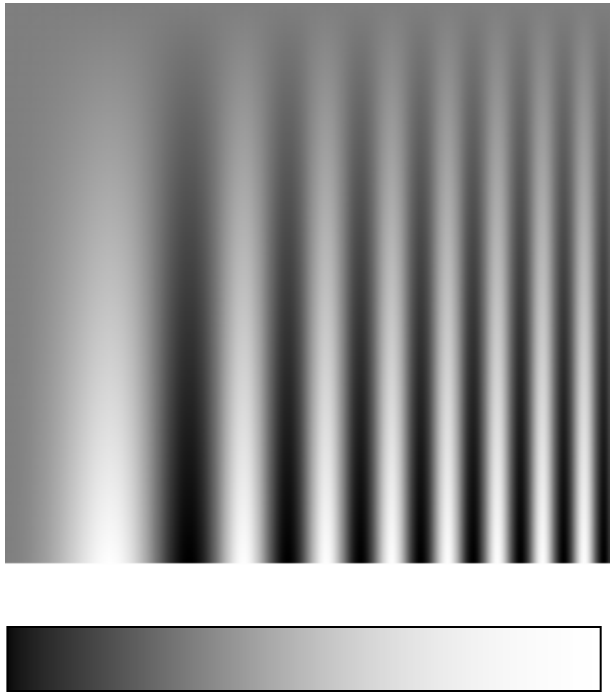


Diverging blue to red colormap

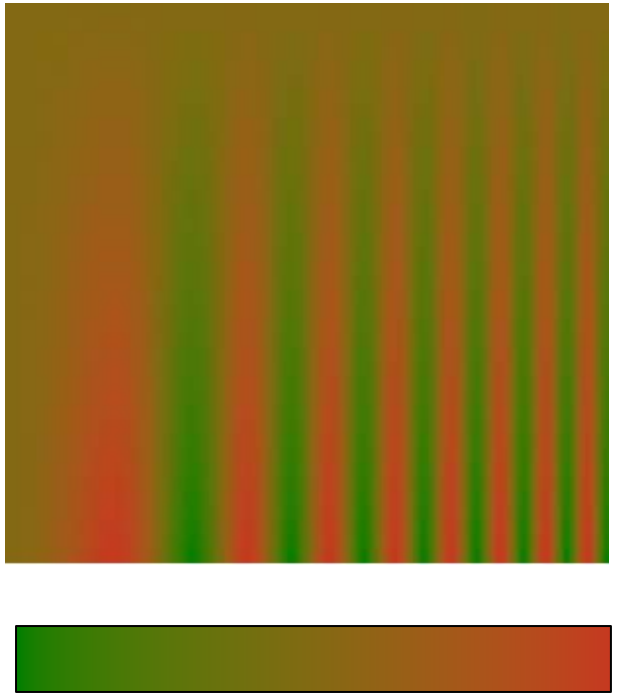
copyright ©2026 m.patrignani

Diverging colormap example

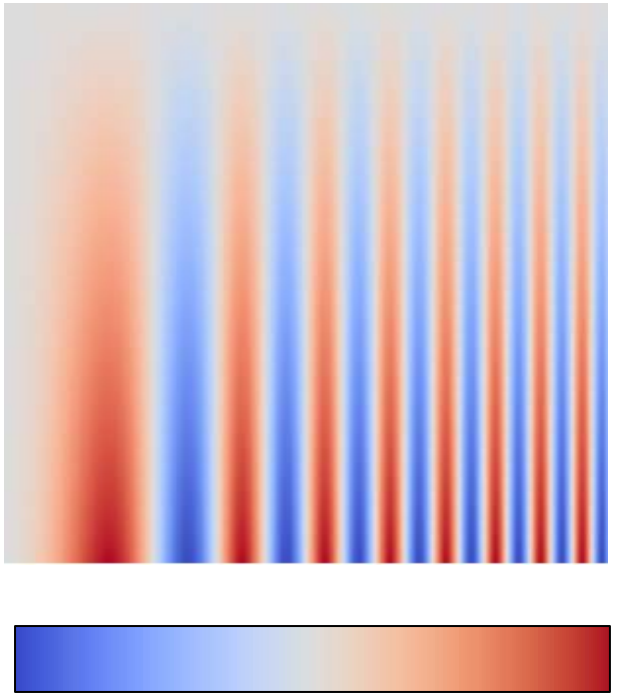
- Low constrast regions
  - example from [Moreland 09]



Grayscale colormap



Isoluminant colormap



Diverging colormap

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Wrap up: perception of colors

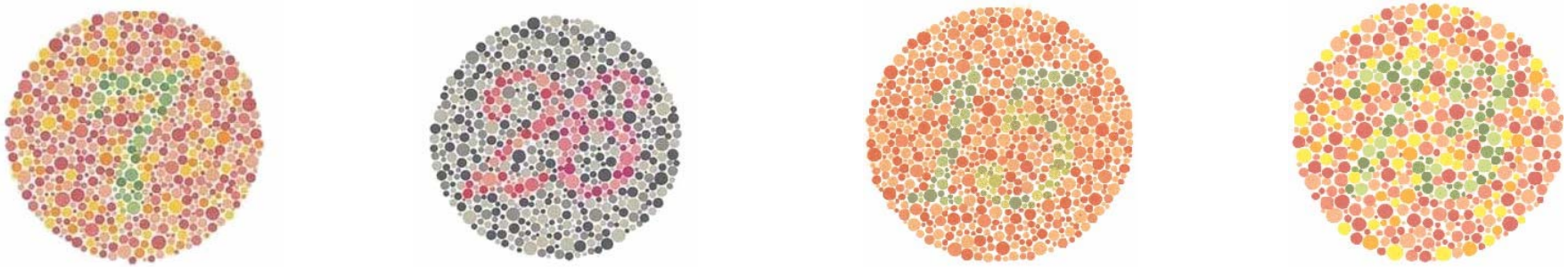
- Human perception of colors is affected by severe limitations
  - only three kinds of cones (trichromacy)
  - lenses cause chromatic aberrations
  - color perception is subject to
    - luminance issues
      - rods do not perceive colors
      - not all colors are perceived equally well
    - context
    - cones' exaustion
    - cultural issues

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Wrap up: never rely on colors only

- Approximately 1 in 12 men and 1 in 200 women are affected by color blindness

Ishihara Color Test



- Get it right in black and white [Stone 10]
  - ensure that the main content is readable on a black and white print out or monochrome display
  - essentially
    - encode the most important attributes with the luminance channel
    - consider the hue and saturation channels as secondary sources of information

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Wrap up: don’t use saturated colors

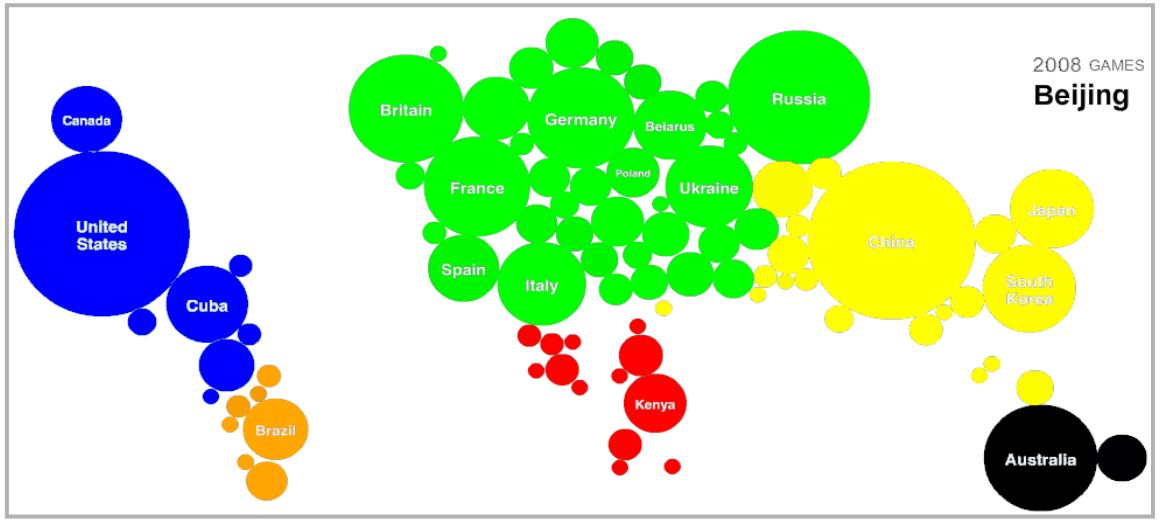
- They cause chromatic aberration
  - they are usually not pleasant and not relaxing
- They have sometimes poor contrast
- They are more prone to cones’ exaustion
- Use pastel colors instead, chosen from color families



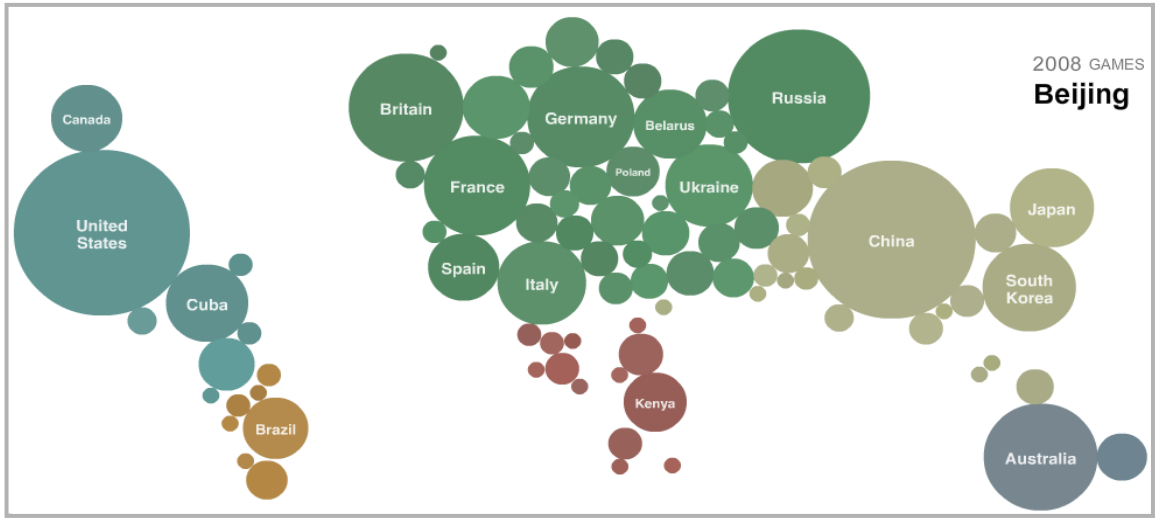
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Wrap up: don’t use saturated colors

- A negative example



saturated colors



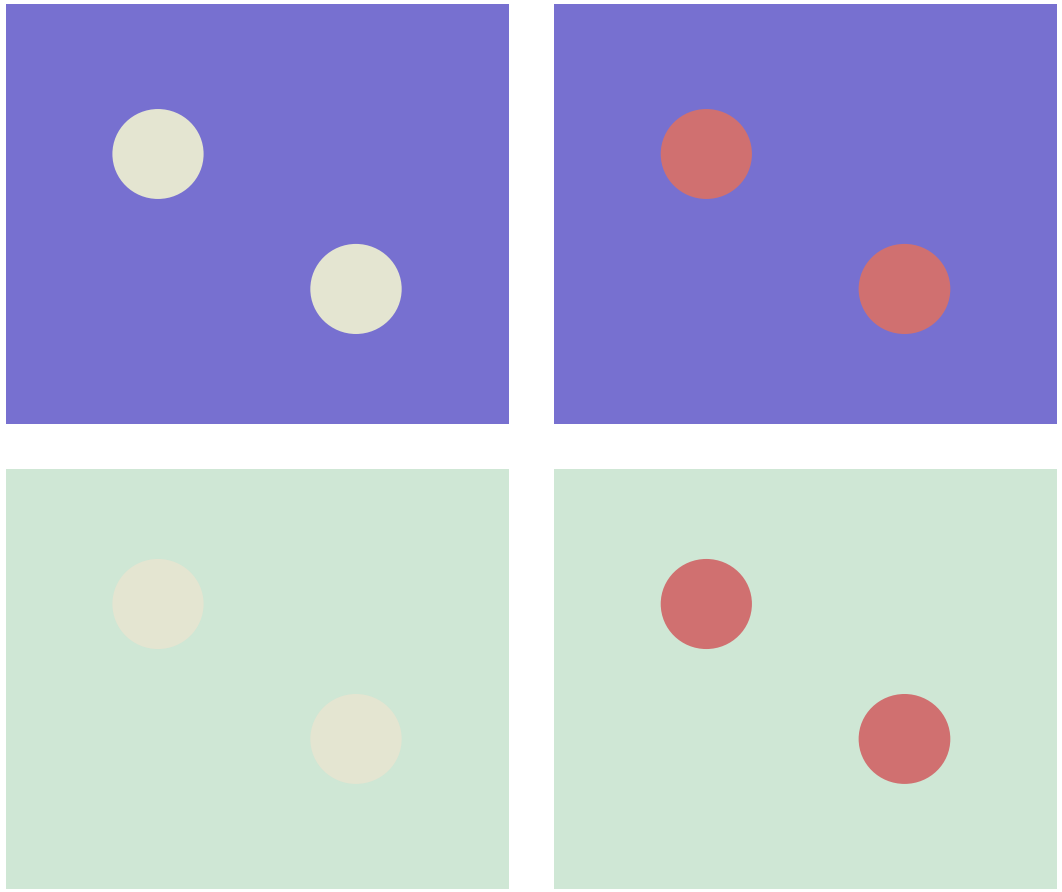
desaturated colors

http://www.nytimes.com/interactive/2008/08/04/sports/olympics/20080804_MEDALCOUNT_MAP.html

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Wrap up: exploit luminance

- Luminance allows you to stress objects with respect to their background
  - colors with high luminance (yellow, green) are strongly perceived if their background has low luminance (blue, red)
  - also the reverse is true



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Wrap up: chosing colormaps

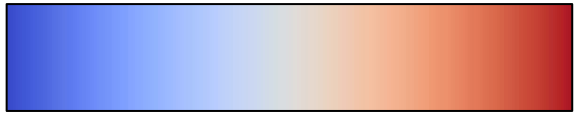
- Never use rainbow colormaps



  - they introduce artifacts
  - they hide information
- Use sequential colormaps instead



- Use divergent colormaps if needed



  - put the middle, unsaturated color, where a significant transition is in the data
- Use isoluminant colormaps only if you have a 3D object



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Resources

- Further resources can be found at
  - http://www.kennethmoreland.com/color-maps/
    - documentation
    - excel RGB values generator
  - http://colorbrewer.org
    - color picker

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Bibliography and further readings

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